PHINMA University of Iloilo
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StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention & CyberLedger: Secure Transaction and Audit Monitoring System

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StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention & CyberLedger: Secure Transaction and Audit Monitoring System

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CHAPTER I

Background of the Study

Have you ever wondered what happens behind the register when no one is watching? Behind every smooth transaction and satisfied customer lies a hidden vulnerability that many business owners rarely question their own employees. In the blink of an eye, a trusted team member with unrestricted system access can alter prices, process fake refunds, or redirect inventory, walking away with assets worth thousands while the system records it as a legitimate operation. This unsettling reality has cost businesses worldwide billions of dollars in losses, yet it remains one of the most overlooked security gaps in modern commerce.

Every day, small and medium-sized café businesses lose thousands of pesos to inventory shrinkage and internal fraud, yet many continue to operate without a single digital safeguard to track who accessed what, when, and why. This critical vulnerability represents not merely a financial leakage but a systemic governance failure that undermines operational transparency and erodes stakeholder confidence. The food service industry, particularly cafés and restaurants, faces persistent challenges related to inventory management and internal fraud. Global and local studies indicate that employee theft accounts for a significant portion of inventory shrinkage, often reaching as high as 75% of total losses in restaurants. In the Philippines, where micro, small, and medium enterprises (MSMEs) dominate the sector, these issues are amplified by limited resources, high staff turnover, and reliance on outdated manual or semi-automated processes. Cash-intensive operations and perishable goods further heighten vulnerability to misappropriation, unauthorized discounts, and unrecorded wastage.

The café industry stands as a dynamic pillar of modern hospitality in the Philippines and across the globe, reflecting evolving consumer lifestyles centered on convenience, community, and experiential consumption. Local coffee culture has flourished dramatically, with the proliferation of independent cafés and specialty shops transforming urban neighborhoods in Manila, Cebu, and beyond into vibrant social and work hubs. This growth mirrors worldwide trends, where the café market continues to expand robustly amid urbanization, rising disposable incomes, and demand for premium beverages. Yet beneath the aroma of freshly brewed coffee and the warmth of customer interactions lies a sector grappling with operational complexities inherent to high-volume, low-margin environments.

Cafés handle perishable goods, complex supply chains, and a workforce often comprising students, part-timers, and seasonal staff, creating fertile ground for inefficiencies that traditional management approaches struggle to contain. The proliferation of food delivery platforms, digital loyalty programs, and contactless payment options has further accelerated the pace at which consumers expect to be served, placing additional pressure on café systems that were designed for a simpler era of commerce. Understanding the roots of these operational vulnerabilities requires examining the interplay of human behavior, technological limitations, and structural business factors in the Philippine café landscape.

The Philippines has witnessed a remarkable transformation in its coffee culture over the past two decades. What began as a predominantly instant coffee-drinking nation has evolved into a sophisticated market for specialty brews, artisanal preparation methods, and third-wave coffee experiences. The Bangko Sentral ng Pilipinas and the Philippine Statistics Authority have consistently reported the growth of the food and beverage sector as one of the most resilient contributors to the country's GDP, even through periods of economic disruption such as the COVID-19 pandemic.

In cities like Iloilo, Cebu, Davao, and Metro Manila, independent cafés have emerged as anchors of neighborhood commerce. They serve not merely as places to consume beverages but as social institutions where professionals work remotely, students study for examinations, families gather for weekend brunches, and friends reconnect over shared meals. This multifunctional role elevates the café from a simple food service establishment to a community hub with deep social and economic significance. The Iloilo City Government's efforts to develop the Mandurriao district and surrounding commercial corridors have further intensified competition among café operators, creating an environment where operational excellence is no longer optional it is existential.

Yet despite this cultural and economic growth, the operational infrastructure supporting many Philippine cafés remains underdeveloped relative to the demands placed upon it. A 2022 report by the Department of Trade and Industry noted that over 85% of food and beverage MSMEs in the Philippines rely on manual or semi-manual processes for inventory tracking, with fewer than 30% having adopted any form of integrated point-of-sale technology. This gap between operational complexity and technological sophistication creates the conditions under which losses accumulate invisibly and fraud flourishes undetected.

The post-pandemic recovery period has further complicated this picture. The return of dine-in customers, combined with the expansion of delivery and takeaway channels, has multiplied transaction volumes for cafés that have limited capacity to process them efficiently. Supply chain disruptions, which became chronic during the pandemic, have made real-time inventory visibility more critical than ever. The volatility of coffee bean prices on international commodity markets, driven by climate events in key producing nations like Brazil and Vietnam, has compressed café margins to the point where even small inefficiencies can determine monthly profitability. In this environment, the adoption of sophisticated yet accessible point-of-sale technology is not a luxury reserved for large chains it is a survival imperative for independent operators.

Central to these challenges is the management of inventory, a critical yet notoriously fragile aspect of café operations. Beans, syrups, milk, pastries, and ingredients must be precisely tracked to prevent both stockouts during peak hours and costly overstocking that leads to waste. Manual processes or basic spreadsheets frequently result in inaccuracies, with discrepancies compounding over time to inflate food costs significantly. In fast-paced Philippine café settings, staff may inadvertently or deliberately mishandle stock ranging from unrecorded giveaways to outright theft, leading to shrinkage rates that drain precious resources.

The literature on food service inventory management identifies several structural factors that make cafés particularly vulnerable to stock-related losses. First, the perishability of café ingredients creates time pressure that incentivizes shortcuts in tracking. A barista who notices that milk is approaching its expiration date may use it in an off-menu drink for a regular customer without recording the transaction, effectively creating an untracked inventory deduction. Over time, hundreds of such informal accommodations create a growing divergence between recorded and actual stock levels. Second, the recipe complexity inherent to specialty coffee menus means that a single beverage may require precise quantities of four to seven different ingredients. Without automated recipe-based deduction, every transaction creates the possibility of inaccurate inventory records.

Third, the physical layout of café kitchens and service areas often creates multiple points of inventory access behind the bar, in dry storage, in walk-in coolers, and at prep stations that are difficult to monitor simultaneously. Fourth, the high turnover characteristic of the café industry means that ownership and management must continuously onboard new staff to inventory protocols, creating recurring windows of vulnerability when new employees may not fully understand or adhere to tracking procedures. The cumulative effect of these structural factors is a persistent, industry-wide inventory accuracy problem that manual systems are fundamentally incapable of solving.

Real-time visibility remains elusive for many small operators, who rely on periodic physical counts that disrupt service and provide only retrospective insights. By the time a discrepancy is identified through a physical count, days or weeks of untracked losses have already occurred. This retrospective approach to inventory management means that corrective action always lags behind the problem, allowing systematic issues to compound before they are addressed. The absence of real-time Inventory data also prevents cafés from making informed decisions about menu design, portion control, and ingredient procurement decisions that collectively determine whether a café operates profitably or absorbs unnecessary costs.

Compounding these inventory woes is the pervasive issue of internal fraud, a silent epidemic disproportionately affecting small businesses like independent cafés and family-run establishments. In the Philippines, where MSMEs form the backbone of the economy, employee-related losses represent a particularly acute threat due to resource constraints that limit sophisticated oversight. The fast-paced, cash-heavy nature of transactions, combined with high staff turnover and limited supervision during busy shifts, creates opportunities for exploitation.

The Association of Certified Fraud Examiners (ACFE) in its global fraud study consistently finds that small businesses suffer disproportionately from occupational fraud compared to larger organizations. The median loss from fraud in small businesses exceeds that of large enterprises, primarily because small operators lack the internal control infrastructure, dedicated audit functions, separation of duties, and regular reconciliation processes that large businesses take for granted. For cafés in particular, the combination of cash transactions, perishable inventory, and high employee autonomy creates what fraud researchers describe as an environment of elevated opportunity.

The fraud triangle, a conceptual framework developed by criminologist Donald Cressey, identifies three elements that must converge for occupational fraud to occur: pressure, opportunity, and rationalization. In the Philippine café context, all three elements are frequently present. Pressure may arise from financial difficulties faced by low-wage café workers, who earn salaries that barely keep pace with rising urban living costs. Opportunity exists wherever internal controls are absent or weak shared system logins, undifferentiated access levels, and infrequent management oversight create abundant occasions for fraud. Rationalization is enabled by the perception that café owners, particularly those who appear prosperous, can absorb small losses without significant harm or that taking food or beverages is less serious than theft of cash.

A barista might ring up a sale but pocket the cash, or a manager could manipulate refunds to siphon funds. These acts erode trust within teams and impose heavy financial burdens that small operators can ill afford. More insidiously, internal fraud that goes undetected tends to escalate over time. What begins as an employee taking occasional undocumented breaks for food typically progresses to systematic cash theft, inventory manipulation, or collusion with suppliers, as the absence of consequences reinforces the behavior. The longer fraud persists undetected, the greater the cumulative loss and the more entrenched the behavior becomes.

For a café operator in Mandurriao like Dulcis Café, operating in a competitive commercial environment with thin margins, the financial impact of even moderate internal fraud can be decisive. Research from the Philippine Institute for Development Studies indicates that MSME failure rates in the food service sector are significantly correlated with undetected internal loss, yet few operators have the tools or knowledge to identify and address the problem systematically. The implementation of technology solutions like role-based access control is not merely a technical upgrade it is a structural intervention that addresses the opportunity dimension of the fraud triangle directly.

Traditional point-of-sale (POS) systems, while foundational for transaction processing, often fall short in addressing these multifaceted risks. Many legacy or basic POS setups prioritize speed and simplicity over security and analytics, lacking integrated mechanisms for granular control or anomaly detection. Staff may share generic logins, enabling untraceable actions, while inventory modules if present fail to sync in real time with sales, leading to persistent mismatches between digital records and physical stock.

The Philippine market for POS technology is characterized by significant fragmentation. At one end of the spectrum, large café chains like local franchise operations of international brands deploy enterprise-grade systems with sophisticated inventory management, customer relationship management integration, and multi-store reporting capabilities. At the other end, the majority of independent cafés use either basic cash registers, simple spreadsheet-based tracking, or low-cost POS applications that provide minimal security or analytical capability. The middle ground of affordable, feature-rich systems designed for the specific operational context of Philippine MSMEs is remarkably underserved.

Cybersecurity vulnerabilities exacerbate the problem; POS terminals handling sensitive payment data become prime targets for malware, skimming, or breaches that expose customer information and invite regulatory penalties. In an era of contactless payments and data privacy regulations like the Data Privacy Act of 2012 (Republic Act 10173) in the Philippines, inadequate encryption or weak network protections can result in devastating incidents, eroding consumer trust and inviting legal repercussions. The National Privacy Commission has increasingly pursued enforcement actions against businesses of all sizes that fail to protect customer data, and the reputational damage from a public data breach can be fatal for a small café that depends on community loyalty.

For cafés operating on thin margins, the cost of even a single breach or prolonged fraud episode can prove existential. Yet the investment required to implement truly robust POS security has historically been beyond the reach of most small operators, who lack both the capital and the technical expertise to deploy and maintain enterprise-grade systems. This is the fundamental gap that StockSecure POS is designed to address, bringing sophisticated security, inventory management, and audit capabilities within reach of the small café operator at a price point and complexity level appropriate for MSME deployment.

This is where the integration of Role-Based Access Control (RBAC) emerges as a transformative approach within POS frameworks. RBAC assigns permissions strictly according to user roles such as cashier, inventory manager, supervisor, or auditor, ensuring individuals access only the data and functions necessary for their responsibilities. This principle of least privilege minimizes insider threats by preventing a single employee from performing conflicting actions, like initiating and approving voids or accessing full financial reports.

The theoretical foundations of RBAC are well-established in computer security literature, dating to foundational work by Ferraiolo and Kuhn at the National Institute of Standards and Technology in the early 1990s. The NIST RBAC model identifies four key components: roles (which capture job functions), permissions (which define access rights), users (who are assigned to roles), and sessions (which are active instances of user-role assignments). In its simplest form, RBAC allows organizations to manage access control at the role level rather than the individual user level, dramatically simplifying administration while improving security consistency.

In a Philippine café setting, a barista might process sales and deduct inventory but lack authority to issue refunds or alter stock levels, while managers gain oversight dashboards without needing broad system access. Such segmentation not only deters fraud but streamlines compliance with audit requirements, as every action generates traceable logs. The practical effect of RBAC implementation in small business contexts has been well-documented in international research. Studies from the Journal of Information Systems Security and related publications consistently show that RBAC adoption reduces unauthorized access incidents by 60 to 80 percent compared to role-undifferentiated access systems, while simultaneously reducing the time required for security administration by eliminating the need to manage individual user permissions.

When paired with advanced inventory monitoring that automates deductions upon sales and triggers alerts for anomalies, RBAC creates a cohesive ecosystem for proactive management. The synergy between access control and inventory monitoring is particularly powerful in the café context because it simultaneously addresses two distinct fraud vectors: the unauthorized action vector (prevented by RBAC role restrictions) and the undetected action vector (addressed by comprehensive audit logging and anomaly detection). The proposed StockSecure POS system embodies this integration, designed specifically to fortify inventory oversight and curb internal threats through intelligent role definitions tailored to hospitality workflows.

Complementing the RBAC framework is CyberLedger, a secure transaction and audit monitoring layer that leverages immutable logging, real-time analytics, and encryption to safeguard every interaction. In environments prone to high transaction volumes, this component ensures that data flows are encrypted end-to-end, with automated reconciliation between sales, inventory, and banking records. It addresses external cyber risks by incorporating features like anomaly detection for unusual patterns such as sudden spikes in voids during off-peak hours that might signal manipulation.

The concept of immutable logging draws from blockchain technology principles, where each record is cryptographically linked to its predecessor, making retroactive alteration detectable. While CyberLedger does not implement a full distributed blockchain architecture, which would be disproportionate for a single-establishment café system, it applies the core principle of cryptographic chaining to create an audit trail that cannot be silently modified. This design choice reflects a broader trend in business technology toward applying blockchain-inspired integrity mechanisms in concentrated, resource-appropriate implementations that deliver security benefits without the overhead of full decentralization.

For a café, where daily operations blend creativity with precision, these systems represent more than technological upgrades; they are guardians of sustainability, enabling owners to focus on customer experience rather than constant firefighting. The dual architecture of StockSecure POS combining RBAC for access control with CyberLedger for audit integrity creates a layered security posture that is greater than the sum of its parts. RBAC reduces the number of unauthorized actions that occur; CyberLedger ensures that actions that do occur are fully documented and verifiable. Together, they address both the prevention and detection dimensions of fraud management.

Real-life situations illustrate the stakes vividly: consider a mid-sized urban café in Metro Manila that, after adopting basic digital tracking, uncovered systematic under-ringing by staff that had drained tens of thousands of pesos monthly for nearly a year before detection. Or consider a family-owned establishment whose basic POS system was compromised by malware, exposing customer payment data and resulting in temporary closure, significant fines, and lasting reputational damage. These are not hypothetical scenarios they represent the documented experiences of businesses whose operators, with access to systems like StockSecure POS, might have detected and prevented these losses far earlier.

The necessity for rigorous research into such integrated solutions becomes evident when examining the unique position of MSMEs in the Philippine café sector. Unlike large chains with dedicated IT departments, independent operators often operate with constrained budgets and technical expertise, relying on off-the-shelf software that inadequately addresses their nuanced needs. This research gap is particularly acute for emerging technologies like RBAC-enhanced POS, which require customization for the dynamic, human-centric environment of local hospitality.

The Philippine government has recognized the critical role of MSMEs in economic development, designating them as priority beneficiaries of digitalization programs under the Philippine Digital Economy Framework and the MSME Development Plan.

The Department of Information and Communications Technology has initiated programs to support technology adoption among small businesses, yet the adoption rate for sophisticated operational systems remains low. Survey data from the Philippine Chamber of Commerce consistently identifies technical complexity and upfront cost as the primary barriers to digital adoption, followed by concerns about reliability in environments with unstable internet connectivity.

StockSecure POS addresses these barriers directly through its offline-first architecture, which ensures continued operation during internet outages, and through its design philosophy of making sophisticated security features accessible through intuitive interfaces that do not require technical expertise to operate. The system is designed to be deployed and managed by café operators themselves, without requiring ongoing support from IT professionals, making it genuinely accessible to the MSME market that needs it most.

Empirical benefits of RBAC extend beyond security to operational efficiency: simplified user provisioning, reduced administrative overhead, and enhanced productivity as staff navigate intuitive, role-aligned interfaces. In Philippine cafés, this translates to faster training for new hires, fewer errors during shifts, and data-driven decision-making for menu optimization and supplier negotiations. The operational efficiency gains alone independent of the fraud prevention benefits can justify the investment in integrated POS technology for most small café operators.

Furthermore, the study addresses broader societal implications. In an economic climate where small businesses drive employment and innovation in the Philippines, fortifying them against fraud and cyber threats preserves jobs and community vitality. Cafés, in particular, serve diverse demographics, from students seeking affordable study spots to professionals craving quality brews, making their stability essential for social cohesion. Real-world parallels abound: establishments that suffered major inventory losses due to undetected theft reported not only financial hits but also strained team morale and customer dissatisfaction from inconsistent service.

The social contract between a café and its community extends beyond the commercial transaction. When a neighborhood café closes due to preventable financial losses, the community loses not just a business but a gathering place, a source of employment for local youth, and a node in the informal social network that sustains neighborhood cohesion. The deployment of technology that helps cafés survive and thrive is, in this sense, an investment in community infrastructure as much as in individual business success.

Conversely, those adopting advanced monitoring tools have reported profit uplifts through waste reduction and improved forecasting. This research, grounded in a student's perspective, emphasizes scalability for resource-limited settings, incorporating user feedback loops to ensure the system remains accessible rather than overly complex. The goal is not to import enterprise-scale complexity into the café context but to distill the essential protective and analytical benefits of enterprise systems into a form that small operators can realistically deploy and maintain.

The integration of RBAC within POS architectures aligns with modern cybersecurity best practices that prioritize zero-trust principles. No user or device is inherently trusted; verification occurs continuously based on context and role. For inventory monitoring, this means automated cross-checks between point-of-sale deductions, supplier deliveries, and waste logs, flagging deviations instantly. CyberLedger's audit capabilities generate comprehensive reports suitable for tax authorities or internal reviews, reducing the manual labor traditionally associated with reconciliation.

The cumulative effect of these challenges, inventory volatility, fraud exposure, and cyber risks, creates a compelling case for why dedicated investigation and development are imperative. Without intervention, cafés risk perpetuating cycles of loss that undermine viability, especially as external factors like climate impacts on coffee supply or economic pressures intensify cost challenges. This thesis project, originating from a student's academic journey, embodies a proactive response: harnessing computer engineering principles to deliver StockSecure POS and CyberLedger as practical, integrated solutions tailored to the Philippine café landscape.

Statement of the Problem

Dulcis Cafe operates in Mandurriao, Iloilo City, a competitive food and beverage landscape where customer expectations rise daily and profit margins shrink just as fast. Mandurriao is one of the most commercially active districts of Iloilo City, characterized by a mix of residential neighborhoods, commercial establishments, educational institutions, and professional offices that collectively generate consistent demand for café services throughout the day. The district's proximity to the Iloilo Business Park has further elevated its commercial profile, attracting a clientele that expects digital payment options, efficient service, and consistently available menu items.

Yet beneath the daily rhythm of service, Dulcis Cafe is weighed down by a web of interconnected operational failures that no single malfunctioning process can explain in isolation. Order processing delays drive customers to competitors during peak hours. Manual inventory practices simultaneously cause stockouts of popular items and spoilage of perishable goods. Security gaps leave the business vulnerable to internal fraud with no means of accountability. Labor-intensive administrative workflows consume the owner's time without producing reliable financial or operational insights.

Each problem feeds into the next inaccurate inventory records mask theft, poor sales data prevents informed purchasing, manual reconciliation hides cash flow leaks, and the cumulative time burden leaves no room for the strategic thinking needed to grow. The absence of an integrated, secure, and role-appropriate point-of-sale system means Dulcis Cafe absorbs costs that a digital solution would prevent, loses revenue that better service would capture, and takes risks that proper access controls and audit trails would eliminate.

The following subsections document the specific operational, security, and strategic problems that define the current state of the business.

Customer-Facing Service and Order Processing Problems

Order Processing Delays During Peak Hours

Order processing delays during peak hours present a significant challenge for cafés like Dulci Cafe, particularly in a bustling area like Mandurriao, Iloilo City. These delays not only frustrate customers but also result in lost business opportunities. During peak periods such as the morning rush when students and professionals are looking for quick service or the weekends when families gather for leisurely meals, the current ordering setup struggles to keep pace. This setup relies heavily on handwritten tickets, disconnected spreadsheets, and manual routing between order entry, kitchen display, and payment processing. Such disjointed handoffs create bottlenecks that directly increase customer wait times, leading to dissatisfaction and potential loss of clientele.

The implications of these delays extend beyond mere inconvenience. Research from the National Restaurant Association indicates that nearly 60% of small restaurants close within their first year, with operational inefficiencies that degrade the customer experience being among the primary drivers of this failure. In the Philippine context, where café culture encourages lingering and socializing, customers may tolerate moderate wait times for quality beverages; however, this tolerance has its limits. Particularly during morning rush periods, customers face time constraints as they head to work or school. Each additional minute a customer waits beyond their expectation is a minute during which a competitor's door becomes more inviting. In Mandurriao's competitive café landscape, where options abound, retaining customers becomes increasingly challenging.

The inefficiency of handwritten order tickets contributes significantly to these operational challenges. Illegible handwriting can lead to preparation errors, requiring remakes that consume additional ingredients and staff time. This not only impacts the café's bottom line but also further delays service, creating a cycle of frustration for both employees and customers. The absence of order sequencing in manual systems exacerbates the issue; busy periods can produce chaotic preparation queues where high-complexity orders may be started before simpler ones that arrived earlier. This disorganization not only leads to longer wait times but also increases the likelihood of mistakes, ultimately degrading the overall customer experience.

Moreover, the lack of a kitchen display system means that preparation staff must rely on verbal communication or the handoff of physical tickets. This reliance creates additional opportunities for miscommunication and error. When orders are communicated verbally, especially in a noisy kitchen environment, important details can easily be overlooked. A kitchen display system would streamline this process by visually presenting orders in real-time, allowing staff to prioritize tasks effectively and ensuring that all orders are prepared accurately and efficiently. Such a system can significantly reduce the chances of errors that arise from miscommunication, ultimately enhancing the speed and quality of service.

The integration of StockSecure POS at Dulci Cafe aims to address these inefficiencies head-on. By automating the order processing workflow, the system eliminates the reliance on handwritten tickets and manual routing. With a user-friendly interface, baristas can quickly enter orders, which are then transmitted directly to the kitchen display system. This real-time communication ensures that preparation staff receive accurate and legible orders, reducing the risk of errors and improving overall efficiency. Additionally, the system allows for order sequencing, ensuring that items are prepared in the correct order, which is especially important during busy periods when multiple orders are being processed simultaneously.

Furthermore, the implementation of StockSecure POS will enable the café to better manage peak periods through features such as automated low-stock alerts and inventory tracking. By having real-time visibility into stock levels, management can proactively reorder supplies before they run out, ensuring that popular items remain available during busy times. This capability not only enhances customer satisfaction but also contributes to smoother operations, as staff can focus on providing excellent service rather than scrambling to fulfill orders due to stock shortages.

In addition to improving efficiency, the StockSecure POS system will also enhance the overall customer experience at Dulci Cafe. With faster order processing times, customers will be less likely to experience long wait times during peak hours. This improvement is crucial for retaining customers who may otherwise be tempted to visit competing cafés. In a competitive market where customer loyalty is hard-won, providing a seamless and efficient service can set Dulci Cafe apart from its competitors.

Moreover, the implementation of StockSecure POS will provide valuable data insights that can inform future business decisions. By analyzing sales performance and customer preferences, management can identify trends and make data-driven decisions regarding menu offerings, pricing strategies, and promotional campaigns. This level of insight will empower Dulci Cafe to adapt to changing customer demands and optimize its operations for long-term success.

Overall, the integration of StockSecure POS at Dulci Cafe addresses the critical issue of order processing delays during peak hours. By streamlining operations, reducing reliance on manual processes, and enhancing communication, the system will significantly improve the café's ability to serve customers efficiently and effectively. For BSIT students at Phinma UI, studying this project offers invaluable insights into the challenges faced by small businesses and the role of technology in driving operational improvements. As they engage with the development and implementation of StockSecure POS, students gain practical experience that will prepare them for successful careers in the technology sector, where problem-solving and innovation are essential for success.

Inefficient Payment Handling

The absence of an integrated payment system at Dulci Cafe presents significant operational challenges that affect both staff efficiency and customer satisfaction. Without a unified approach to handling payments, staff are forced to calculate totals manually and manage various payment methods such as cash, GCash, bank transfers, and card payments through separate channels. This fragmentation leads to a host of issues, including miscalculated bills, delayed change, and time-consuming end-of-day reconciliation processes that consume hours the owner simply cannot spare. In a high-volume microenterprise like Dulci Cafe, where every second counts, each billing error not only erodes customer trust but also requires additional staff time to resolve, further compounding operational inefficiencies.

The proliferation of payment methods in the Philippine market has added a new layer of complexity for café operators. The increasing adoption of digital payment solutions such as GCash, PayMaya, and contactless card payments, alongside traditional cash transactions, means that cafés must now manage multiple payment channels simultaneously. For a café relying on manual processes, this translates into maintaining separate records for each payment method, which can be cumbersome and prone to errors. The need to perform complex reconciliations at the end of each shift can lead to significant discrepancies, creating a challenging environment for staff who must ensure that all transactions are accounted for accurately.

The opportunity for errors and discrepancies in this fragmented payment system is substantial. Staff may inadvertently miscalculate totals, leading to incorrect bills that frustrate customers and damage the café's reputation. Additionally, the time required for reconciliation often extends shift closing procedures by an hour or more. This delay not only incurs additional staff overtime costs but also takes valuable time away from the owner, who may have other pressing responsibilities to attend to. The cumulative effect of these inefficiencies can significantly impact the café's profitability, making it essential to address the payment processing challenges head-on.

Implementing an integrated payment system through StockSecure POS will streamline the payment process at Dulci Cafe, enhancing both efficiency and accuracy. By consolidating all payment methods into a single platform, staff will no longer need to calculate totals manually or juggle multiple records for different payment channels. Instead, the system will automatically calculate totals and process transactions in real-time, reducing the risk of errors and ensuring that customers receive accurate bills without delay. This streamlined approach will not only improve the customer experience but also alleviate the burden on staff, allowing them to focus on providing exceptional service.

Furthermore, the integrated payment system will simplify end-of-day reconciliation processes significantly. With all transactions recorded in a single platform, the café owner will have a clear and accurate overview of daily sales across all payment methods. This visibility will enable quicker and more efficient reconciliation, eliminating the need for complex manual calculations and reducing the time spent on closing procedures. By freeing up valuable time for both staff and management, the café can operate more smoothly and efficiently, ultimately enhancing overall productivity.

The adoption of an integrated payment system also positions Dulci Cafe to better respond to evolving consumer preferences in the payment landscape. As customers increasingly favor digital payment options for their convenience and speed, having a robust system in place will ensure that the café remains competitive. By offering a seamless payment experience that accommodates various payment methods, Dulci Cafe can attract and retain a broader customer base, fostering loyalty and encouraging repeat business.

For BSIT students at Phinma UI, the study of StockSecure POS and its integrated payment system provides critical insights into the challenges faced by microenterprises in managing payment processes. They learn about the importance of creating solutions that enhance operational efficiency and improve customer satisfaction. By engaging with the development and implementation of an integrated payment system, students gain practical experience that prepares them for future careers in software development, financial technology, and operational management. This knowledge will be invaluable as they navigate the complexities of modern business environments, where technology plays a pivotal role in driving success.

In conclusion, the integration of a payment system within StockSecure POS addresses the significant challenges posed by fragmented payment processes at Dulci Cafe. By streamlining payment processing, enhancing accuracy, and simplifying reconciliation, the system will improve operational efficiency and customer satisfaction.
For BSIT students, studying this project offers a unique opportunity to engage with real-world applications of technology that address pressing business challenges, equipping them with the skills and insights necessary to thrive in their future careers. As they contribute to the development of innovative solutions like StockSecure POS, students are empowered to make meaningful impacts in the world of business and technology.

Absence of Customer Preference Tracking

Dulci Cafe currently faces a critical challenge in its ability to capture customer identities, order histories, and visit frequency. The absence of a systematic approach to record customer interactions means that the café cannot maintain a usable customer database, track purchase patterns, or offer personalized promotions. This limitation not only hinders the café's ability to foster customer loyalty but also puts it at a competitive disadvantage against other establishments in Iloilo's vibrant coffee scene. Many of these competitors have already implemented POS-integrated loyalty features that automatically track customer spending and reward repeat visits capabilities that Dulci Cafe cannot match without a foundational data-capture system in place.

The significance of loyalty programs in the café sector has grown substantially in recent years, driven by changing consumer preferences and heightened competition. Research from the Philippine Retailers Association reveals that loyalty program participants spend an average of 67% more per visit than non-participants and visit approximately twice as frequently. For Dulci Cafe, whose profitability hinges on converting occasional visitors into regular patrons, the inability to offer and track loyalty benefits represents a substantial missed revenue opportunity. Without a structured system to capture customer data, the café cannot effectively implement loyalty programs that encourage repeat business and enhance customer retention.

Moreover, the customer data captured through loyalty programs provides invaluable insights that can inform various aspects of the café's operations. By analyzing purchase patterns and visit frequency, management can make more informed decisions regarding menu development, targeted promotions, and inventory planning. For instance, understanding which items are most popular among loyal customers can guide menu adjustments to better cater to their preferences. Additionally, the ability to offer personalized promotions based on individual purchase histories can enhance the overall customer experience, fostering a sense of connection and loyalty to the café. These capabilities compound in value over time as the customer database grows, creating a feedback loop that drives further engagement and profitability.

Implementing a robust data-capture system through the StockSecure POS project will enable Dulci Cafe to address these challenges effectively. By integrating customer relationship management (CRM) features into the POS system, the café can systematically capture customer identities and track their interactions. This integration will allow staff to create customer profiles that include order histories, preferences, and visit frequencies, enabling personalized service and targeted marketing efforts. As a result, Dulci Cafe will be well-positioned to compete with other establishments that already leverage such technology to enhance customer loyalty.

For BSIT students at Phinma UI, studying the implementation of a data-capture system within StockSecure POS offers critical insights into the role of technology in driving customer engagement and loyalty. They learn how to design and develop systems that effectively gather and analyze customer data, empowering businesses to make data-driven decisions. This experience not only enhances their technical skills but also prepares them for future careers in fields such as marketing, data analysis, and customer relationship management, where understanding consumer behavior is paramount.

Furthermore, the integration of a customer loyalty program within StockSecure POS will enable Dulci Cafe to differentiate itself in a competitive market. By offering rewards for repeat visits and tracking customer spending, the café can create a compelling value proposition that encourages customers to choose Dulci over competing establishments. This shift in strategy will not only enhance customer satisfaction but also drive revenue growth, as loyal customers are more likely to spend more and visit more frequently.

In conclusion, the lack of a systematic way to capture customer identities and order histories at Dulci Cafe represents a significant barrier to building customer loyalty and enhancing profitability. By implementing a data-capture system through the StockSecure POS project, the café can unlock valuable insights that drive personalized marketing, menu development, and inventory planning. For BSIT students, engaging with this project provides a unique opportunity to understand the importance of customer data in business operations and equips them with the skills necessary to contribute to the success of future enterprises. As Dulci Cafe embraces technology to enhance its operations, it positions itself for long-term growth and success in Iloilo's competitive café landscape.

Absence of Portion Control and Recipe Costing

High-value ingredients such as coffee beans, specialty syrups, and fresh milk are essential components of any café menu, particularly for establishments like Dulci Cafe that pride themselves on quality offerings. However, the current lack of standardized measurement protocols for these ingredients poses a significant risk to the café's profitability. Without a systematic approach to portion control, the variance between the theoretical cost of goods sold (COGS), what should have been used according to recipes, and the actual COGS, what was actually used, goes entirely undetected. This discrepancy can lead to substantial financial losses, especially in a business where margins are already thin.

Restaurants that rely on manual systems often experience shrinkage rates of one to five percent of total revenue due to a combination of theft, waste, and measurement errors. In the context of a café operating on narrow profit margins, losing even five percent of revenue to invisible waste can mean the difference between profit and loss. For Dulci Cafe, where every peso counts, implementing standardized measurement protocols is not just a best practice; it is a necessity for sustaining operations and ensuring long-term viability.

The economic significance of portion control in specialty beverage preparation cannot be overstated. Coffee beans, which typically represent the highest per-unit cost ingredient on most café menus, are particularly susceptible to variance in portioning if not carefully measured. For instance, if a barista consistently uses 20 grams of coffee per espresso shot instead of the recipe-standard 18 grams, they are effectively increasing the coffee cost of every espresso-based drink by 11 percent. This seemingly minor deviation in measurement may appear insignificant on an individual basis, but when multiplied across a high volume of sales, the cumulative effect can result in a material cost difference that directly impacts profitability.

To illustrate, consider a scenario where Dulci Cafe sells 100 espresso-based drinks in a single day. If each drink is made with the excess coffee, the additional two grams per shot translates to an extra 200 grams of coffee consumed. Given that the cost of high-quality coffee beans can be substantial, this excess usage compounds into a significant financial loss over time. For a café that aims to thrive in a competitive market, such inefficiencies can quickly erode profit margins, making it imperative to adopt measures that ensure accurate portion control.

Implementing a robust system through StockSecure POS that standardizes measurements for all high-value ingredients will address these challenges head-on. By integrating automated dispensing systems or digital scales that provide real-time feedback on ingredient usage, Dulci Cafe can ensure that baristas adhere to established recipes and portion sizes. This technology not only minimizes the risk of human error but also promotes consistency in beverage preparation, enhancing the overall quality of the café's offerings.

For BSIT students at Phinma UI, the study of portion control and its impact on profitability provides valuable insights into the intricacies of managing food and beverage operations. They learn how to design systems that incorporate measurement protocols and track ingredient usage, empowering businesses to make informed decisions based on accurate data. This knowledge is particularly relevant in the food service industry, where operational efficiency and cost control are critical for success. By engaging with the development and implementation of StockSecure POS, students gain practical experience that prepares them for future careers in hospitality management, food science, and culinary entrepreneurship.

Moreover, the integration of portion control measures will enable Dulci Cafe to optimize its inventory management and reduce waste. By accurately tracking ingredient usage, the café can identify patterns in consumption and adjust ordering practices accordingly. This proactive approach to inventory management not only minimizes spoilage and waste but also ensures that the café maintains sufficient stock levels of high-value ingredients. As a result, Dulci Cafe can operate more efficiently, ultimately enhancing its bottom line.

In conclusion, the absence of standardized measurement protocols for high-value ingredients at Dulci Cafe poses a significant threat to profitability. By implementing robust portion control measures through the StockSecure POS system, the café can mitigate the risks associated with measurement errors, theft, and waste. For BSIT students, studying this project offers critical insights into the importance of operational efficiency in the food service industry and equips them with the skills necessary to drive positive change in future enterprises. As Dulci Cafe embraces technology to enhance its operations, it positions itself for long-term success in a competitive market, ensuring that every ingredient is utilized effectively and contributes to the café's overall profitability.

Inventory and Supply Chain Breakdown

Real-Time Inventory Management Deficiencies

Dulci Cafe's reliance on periodic manual inventory counts presents significant challenges in managing its stock of perishable goods, including fresh milk, coffee beans, specialty syrups, and baked pastries. These ingredients require precise tracking due to their short shelf lives and high replacement costs. Unfortunately, the manual counting process is prone to human error and quickly becomes outdated, leading to potential overstocking or stockouts. Industry data indicates that food waste in food service businesses can exceed ten to twenty-five percent of total inventory value. For a small café operating on thin margins, this level of waste is not merely an operational nuisance; it represents a direct and material drain on profitability.

The specific characteristics of café inventory underscore the importance of real-time tracking. Unlike retail merchandise, which can sit on a shelf indefinitely, café ingredients have finite shelf lives that vary significantly. For instance, fresh pastries and cut fruit may only last a few days, while dairy products and syrups might last several weeks. Dry ingredients and packaged goods can last for months, but even these items require careful monitoring to avoid spoilage. Effectively managing this perishable inventory necessitates continuous visibility into current stock levels, consumption rates, and days-to-expiration for each product category. Manual inventory systems, which typically provide only a snapshot of stock levels at the moment of the count, are structurally incapable of delivering the continuous visibility required for effective perishable inventory management.

Implementing a real-time inventory tracking system through StockSecure POS will be crucial for Dulci Cafe. By integrating inventory management features into the POS system, the café can achieve continuous visibility into its stock levels. This integration allows for automatic updates as ingredients are used or received, providing accurate data on current inventory and days-to-expiration. This system can also generate alerts for items that are nearing expiration, allowing staff to prioritize their use in daily specials or promotions. Such proactive measures can significantly reduce waste and enhance the café's ability to serve customers with fresh ingredients, ultimately improving customer satisfaction and loyalty.

Furthermore, the implementation of a real-time inventory system will facilitate better forecasting and ordering practices. By analyzing historical usage data, Dulci Cafe can identify trends in ingredient consumption, allowing for more accurate ordering that aligns with customer demand. For example, if the café notices a spike in the sale of a particular syrup during weekends, it can adjust its orders accordingly to ensure it never runs out of that popular item. This level of responsiveness not only minimizes waste but also maximizes the café's ability to capitalize on peak sales periods, driving revenue growth.

The integration of real-time inventory management will also empower staff to make informed decisions on ingredient usage, leading to improved portion control. With accurate tracking of ingredient levels, baristas can be trained to use precise measurements when preparing beverages, ensuring consistency in quality while minimizing waste. For example, if the system indicates that only a limited quantity of a specialty syrup remains, staff can adjust their recipes or promote the use of alternative ingredients, thereby ensuring that the café continues to operate smoothly without compromising on quality. This approach fosters a culture of accountability among staff, as they become more aware of the impact of their actions on the café's overall profitability.

Moreover, the impact of real-time inventory tracking extends beyond just waste reduction and cost savings; it can also enhance the café's ability to innovate. With a clearer understanding of ingredient usage and customer preferences, Dulci Cafe can experiment with new recipes and seasonal offerings with confidence. If certain ingredients are consistently underutilized or nearing expiration, staff can create limited-time promotions that highlight these items, encouraging customers to try something new while minimizing waste. This not only diversifies the menu but also keeps the customer experience fresh and engaging, fostering a sense of excitement and loyalty among patrons.

Additionally, the insights gained from a robust inventory management system can inform marketing strategies. For example, if data reveals that certain products are particularly popular among specific customer segments, targeted marketing campaigns can be developed to promote those items. By leveraging customer data alongside inventory insights, Dulci Cafe can create personalized promotions that resonate with its audience, driving both sales and customer engagement. This data-driven approach allows the café to adapt quickly to market trends, ensuring that its offerings remain relevant and appealing to its customer base.

For BSIT students at Phinma UI, studying the implementation of a real-time inventory tracking system within StockSecure POS provides valuable insights into the importance of technology in managing perishable goods. They learn how to design and develop systems that enhance operational efficiency and reduce waste, equipping them with skills that are highly relevant in the food service industry. By engaging with this project, students gain practical experience that prepares them for future careers in supply chain management, operations, and data analysis. As they witness firsthand the transformative impact of technology on business processes, students are inspired to think critically about how they can leerage similar solutions in their future endeavors.

In conclusion, Dulci Cafe's reliance on manual inventory counts poses significant risks to its ability to manage perishable goods effectively. By implementing a real-time inventory tracking system through StockSecure POS, the café can enhance its operational efficiency, minimize waste, and ultimately improve profitability. This technological advancement will not only streamline inventory management but also foster a culture of accountability and precision among staff.

For BSIT students, this project offers a unique opportunity to understand the critical role of technology in inventory management and equips them with the skills necessary to drive improvements in future enterprises. As Dulci Cafe embraces technology to optimize its operations, it positions itself for long-term success in a competitive market, ensuring that every ingredient is utilized effectively and contributes to the café's overall profitability. The potential for growth and innovation through effective inventory management is immense, and as Dulci Cafe adapts to these changes, it can set a benchmark for other small businesses in the region, demonstrating how technology can transform operations and enhance customer experiences.

Supplier Coordination and Procurement Inefficiencies

Without automated low-stock alerts or consumption-based forecasting, Dulci Cafe is forced into reactive buying patterns that can severely impact its financial health. This approach often leads to emergency purchases at premium prices when stock runs out, resulting in inflated costs that eat into the café's already thin profit margins. Additionally, the lack of proactive inventory management can lead to excess orders, pushing perishable stock past its shelf life. This cycle of financial waste not only strains the café's resources but also prevents it from establishing a stable cost structure. Studies on Philippine food service micro, small, and medium enterprises (MSMEs) confirm that disjointed inventory-to-procurement workflows disrupt both finances and menu consistency, making it difficult for owners to maintain control over their operational costs.

The supply chain challenges facing Philippine café operators are further compounded by the fragmented nature of the local supplier market. Unlike large chain operators that can negotiate long-term supply agreements with major distributors, independent cafés typically rely on multiple small suppliers who operate informally and unpredictably. This fragmentation can make it difficult for café owners to establish reliable supply chains, as delivery schedules may shift based on supplier capacity, seasonal availability, and informal relationship dynamics. Such unpredictability can lead to unexpected shortages or surpluses, complicating inventory management and hindering the café's ability to serve customers consistently.

Without systematic tracking of procurement history and consumption patterns, café operators like Dulci Cafe cannot effectively plan their orders or hold suppliers accountable for delivery commitments. This lack of visibility can lead to a reactive approach to procurement, where decisions are made based solely on immediate needs rather than informed forecasting. Consequently, the café may find itself in a constant state of flux, struggling to balance stock levels and maintain menu offerings that meet customer expectations. Additionally, the inability to analyze past purchasing behavior means that the café misses opportunities to negotiate better terms with suppliers or identify more reliable sources for key ingredients.

Implementing a comprehensive inventory management system through StockSecure POS can address these challenges by providing real-time insights into stock levels and consumption trends. By integrating procurement history with inventory data, Dulci Cafe can develop a more strategic approach to ordering that minimizes waste and stabilizes costs. Automated low-stock alerts will notify staff when ingredient levels fall below predefined thresholds, allowing for timely reordering and reducing the risk of emergency purchases. Furthermore, consumption-based forecasting can help the café anticipate demand fluctuations, enabling it to adjust orders based on historical usage patterns and seasonal trends.

In addition to improving procurement practices, a robust inventory management system can enhance supplier relationships. By tracking delivery performance and maintaining records of procurement history, Dulci Cafe can hold suppliers accountable for their commitments. This level of transparency fosters better communication and collaboration between the café and its suppliers, ultimately leading to more reliable service and improved product quality. As the café establishes a reputation for consistency and accountability, it may also find opportunities to negotiate better pricing or terms with suppliers, further enhancing its financial position.

For BSIT students at Phinma UI, studying the implementation of an integrated inventory management and procurement system within StockSecure POS provides valuable insights into the complexities of supply chain management in the food service industry. They learn how to design systems that facilitate effective inventory control and procurement strategies, equipping them with skills that are highly relevant in today's business environment. By engaging with this project, students gain practical experience that prepares them for future careers in logistics, operations, and supply chain management, where understanding the intricacies of procurement and inventory is essential for success.

In conclusion, Dulci Cafe's current reliance on reactive buying patterns and fragmented supplier relationships poses significant challenges to its operational efficiency and financial stability. By implementing a comprehensive inventory management system through StockSecure POS, the café can transition to a proactive approach that minimizes waste and enhances supplier accountability. This technological advancement will not only streamline procurement processes but also empower staff to make informed decisions that stabilize costs and improve menu consistency. For BSIT students, this project offers a unique opportunity to understand the critical role of technology in supply chain management and equips them with the skills necessary to drive improvements in future enterprises. As Dulci Cafe embraces these changes, it positions itself for long-term success in a competitive market, ensuring that it can meet customer demand while maintaining profitability and operational efficiency.

Absence of Automated Alerts and Exception Reporting

When milk spoilage exceeds usable volume, when a specific item's sales drop sharply, or when a staff member processes an unusual number of voids, Dulci Cafe's current system lacks any mechanism to flag these critical events. There is no dashboard, no notification system, and no threshold alarm to alert the owner or staff of these anomalies. As a result, significant cumulative losses can occur before the owner even becomes aware of them. This reactive approach to management means that by the time patterns are detected through manual review, the damage has already been done, often resulting in substantial financial implications for the café.

The compounding nature of these small, undetected leaks makes them more dangerous than a single large loss. For instance, if milk spoilage goes unnoticed for several days, the café not only incurs the cost of the spoiled product but also loses potential revenue that could have been generated from sales of beverages requiring that milk. Similarly, if sales of a particular item drop sharply due to changing customer preferences or inadequate marketing, the café may continue to order that item based on historical data, leading to excess inventory that ultimately spoils. Each of these scenarios represents a slow erosion of profitability that escapes attention until it becomes entrenched, making it increasingly difficult for the owner to identify and rectify the underlying issues.

Moreover, the lack of a systematic approach to monitoring and analyzing key performance indicators (KPIs) means that the café is unable to make data-driven decisions that could mitigate these risks. Without real-time insights into inventory levels, sales trends, and operational anomalies, the café misses opportunities to adjust its offerings, optimize its stock, and respond to customer demands effectively. This absence of visibility can lead to a cycle of inefficiency where the café is constantly reacting to problems rather than proactively managing its operations. For example, if the café had a dashboard that highlighted declining sales for specific items, management could quickly investigate and implement strategies to boost sales, such as promotional offers or menu adjustments.

The absence of alerts for unusual voids processed by staff can also point to deeper issues within the café's operations. High void rates may indicate problems such as staff training deficiencies, customer dissatisfaction, or even potential theft. When these voids go unmonitored, the café not only loses revenue from those transactions but also fails to address the root causes of the issue. By implementing a system that flags unusual activity, management can investigate the reasons behind the voids, whether they stem from legitimate customer complaints or operational inefficiencies. This proactive approach allows the café to improve service quality, enhance customer satisfaction, and ultimately protect its bottom line.

Furthermore, the cumulative effect of these undetected issues can create a culture of complacency among staff. When employees see that there are no immediate consequences for spoilage, voids, or inventory discrepancies, they may not feel motivated to adhere to best practices in inventory management and customer service. This lack of accountability can further exacerbate the problems facing the café, leading to a cycle of waste and inefficiency that becomes increasingly difficult to break. By establishing a system that monitors performance and provides feedback, Dulci Cafe can foster a culture of accountability and continuous improvement, empowering staff to take ownership of their roles and contribute to the café's success.

Additionally, the lack of visibility into operational metrics can hinder the café's ability to identify trends that may require strategic adjustments. For instance, if sales of certain seasonal items are declining, management may miss the opportunity to pivot their marketing efforts or introduce new promotions in a timely manner. This oversight can lead to missed sales opportunities and an inability to adapt to changing customer preferences. Furthermore, without a comprehensive understanding of how different variables affect sales and inventory, the café may struggle to develop effective pricing strategies or promotional campaigns that resonate with its target audience.

The financial implications of these undetected leaks extend beyond immediate losses; they can also affect the café's long-term viability. Consistent financial strain from unaddressed issues can lead to cash flow problems, making it difficult for the café to invest in necessary improvements or innovations. Over time, this can hinder the café's ability to compete effectively in a crowded market, as it may lack the resources to enhance its offerings or improve customer experiences. The cumulative impact of small inefficiencies can create a significant barrier to growth, trapping the café in a cycle of reactive management that stifles its potential.

In addition, the emotional toll on the café owner and staff cannot be overlooked. Constantly dealing with the fallout from undetected issues can lead to frustration and burnout among employees, which can impact morale and productivity. When staff members feel overwhelmed by operational challenges and lack the tools to address them effectively, it can create a negative work environment that affects customer service. A disengaged staff may not deliver the level of service that customers expect, leading to further declines in sales and customer loyalty.

Implementing a comprehensive monitoring and alert system through StockSecure POS can fundamentally transform how Dulci Cafe manages its operations. With real-time dashboards that display key metrics such as inventory levels, sales trends, and void rates, the café can gain immediate insights into its performance. Automated alerts can notify management of significant changes or anomalies, allowing for timely intervention before small issues escalate into larger problems. This proactive approach not only enhances operational efficiency but also positions the café to respond swiftly to changing market conditions and customer preferences.

In addition to improving operational efficiency, a robust monitoring system can facilitate better training and development for staff. With real-time data on performance metrics, management can identify areas where employees may need additional support or training. For instance, if certain staff members consistently have high void rates or if specific items are frequently reported as spoiled, targeted training sessions can be organized to address these issues. This tailored approach to staff development not only improves individual performance but also enhances the overall service quality of the café. Staff members will feel more empowered and capable in their roles, contributing to a positive work culture that ultimately benefits the café's bottom line.

Moreover, the integration of a monitoring system can enhance customer engagement and loyalty. By analyzing sales data, Dulci Cafe can better understand customer preferences and tailor its offerings accordingly. For example, if data shows that certain beverages are particularly popular during specific times of day or seasons, management can adjust inventory levels to ensure that these items are always available. Additionally, the café can implement targeted marketing campaigns based on customer buying patterns, such as loyalty programs that reward frequent customers or promotions that highlight popular items. By leveraging data to create a more personalized customer experience, Dulci Cafe can foster stronger relationships with its patrons, encouraging repeat visits and boosting overall sales.

Another critical aspect of implementing a monitoring system is the potential for enhanced financial forecasting and budgeting. By having access to real-time data on inventory costs, sales trends, and operational expenses, Dulci Cafe can develop more accurate financial projections. This capability allows the café to make informed decisions regarding pricing strategies, menu changes, and capital investments. For instance, if the café identifies a trend of increasing costs for certain ingredients, it can proactively adjust prices or seek alternative suppliers to mitigate the impact on profitability. This level of financial agility is crucial for small businesses operating in competitive markets, where even minor fluctuations can significantly affect the bottom line.

Additionally, the ability to track and analyze operational metrics can lead to improved supplier relationships. With a detailed history of procurement and inventory management, Dulci Cafe can negotiate better terms with suppliers based on performance metrics. For example, if a supplier consistently delivers late or provides subpar products, the café can address these issues directly and seek alternatives if necessary. Conversely, if a supplier consistently meets or exceeds expectations, the café may be able to negotiate discounts or favorable terms based on the volume of business being conducted. This strategic approach to supplier management not only enhances the quality of ingredients but also contributes to the overall financial health of the café.

Furthermore, the implementation of a monitoring system can facilitate better collaboration among staff members. With a centralized platform for tracking performance metrics, employees can work together more effectively to address challenges and achieve common goals. For example, if sales data indicates that a particular item is not performing well, staff can collaborate to brainstorm promotional ideas or adjust the way the item is presented to customers. This collaborative approach fosters a sense of teamwork and shared responsibility, creating a more cohesive work environment that enhances both employee satisfaction and customer service.

In summary, the challenges faced by Dulci Cafe due to the absence of a monitoring system are multifaceted and deeply rooted in operational inefficiencies. From the inability to detect spoilage and sales declines to the emotional toll on staff and the financial implications of undetected leaks, the café's current approach to management is fraught with risks. By implementing a comprehensive monitoring and alert system through StockSecure POS, Dulci Cafe can transform its operations, enabling proactive management that minimizes waste, enhances customer engagement, and fosters a culture of accountability and continuous improvement. As the café embraces these changes, it will be better equipped to navigate the complexities of the competitive food service landscape, ensuring that it can maintain profitability while delivering exceptional service to its customers.

Security, Fraud, and Accountability Gaps

Absence of Role-Specific Access Controls

Multiple staff members share broad, undifferentiated system access at Dulci Cafe, creating a significant vulnerability that can lead to various forms of fraud and operational inefficiencies. This lack of access differentiation enables underreporting of sales, manipulation of refund and void transactions, and inflation of wastage logs to conceal theft. In the food service industry, employee theft accounts for approximately twenty-nine percent of all inventory shrinkage, with estimates suggesting it can reach as high as four percent of total restaurant revenue. Such figures underscore the critical need for robust security measures within the café's operational framework. The absence of role-specific permissions, tamper-evident audit logs, and transaction-level traceability means that identifying culprits is nearly impossible, creating an environment where dishonest behavior can flourish. The current system’s lack of accountability invites abuse, as employees may feel emboldened to exploit these vulnerabilities without fear of detection.

The specific fraud vectors enabled by undifferentiated system access in a café context are numerous and well-documented. Cashiers with access to void and refund functions can process transactions for customers, collect payment, and then retroactively void the sale, pocketing the cash while the system shows no net revenue for the transaction. This practice not only results in direct financial losses for the café but also distorts sales data, making it difficult for management to accurately assess performance and make informed decisions. Additionally, staff members with inventory adjustment access can inflate wastage records to conceal the theft of ingredients or finished products. For instance, if an employee consistently reports excessive wastage of high-value items like specialty coffee beans or pastries, it may indicate an attempt to cover up theft. Without individual user accountability, which requires that each employee log in with unique credentials and that their actions be individually tracked, attributing suspicious activity to specific individuals becomes nearly impossible, even when patterns of loss are evident.

The ramifications of these vulnerabilities extend beyond immediate financial losses; they can also undermine the overall integrity of the café's operations. When employees perceive that theft or manipulation of the system is possible without consequence, it can foster a culture of dishonesty and disengagement. This environment not only affects the morale of honest employees but can also lead to increased turnover, as dedicated staff may become disillusioned by the lack of accountability among their peers. Furthermore, the financial strain caused by employee theft can hinder the café's ability to invest in improvements, such as upgrading equipment or enhancing the customer experience. The longer these vulnerabilities remain unaddressed, the more entrenched the issues become, creating a cycle of inefficiency and distrust that can be challenging to break.

Moreover, the absence of robust auditing mechanisms means that management lacks the necessary tools to conduct thorough investigations when discrepancies arise. Without tamper-evident audit logs, it is difficult to determine when and how losses occurred, making it challenging to implement corrective actions. For example, if management notices a sudden spike in wastage reports or voided transactions, the lack of detailed logs prevents them from tracing the issue back to specific employees or shifts. This lack of transparency can lead to frustration among management, as they are left to speculate on the causes of losses without concrete evidence. Consequently, the café may struggle to establish effective loss prevention strategies, as management remains in the dark about the specific behaviors that contribute to inventory shrinkage.

Implementing a system with role-specific permissions and robust tracking capabilities can significantly mitigate these risks. By ensuring that each employee has access only to the functions necessary for their role, the café can reduce opportunities for fraud. For instance, cashiers should have limited access to void and refund functions, with these actions requiring managerial approval or additional verification. Similarly, staff responsible for inventory management should have their adjustments logged and monitored closely, with any significant changes requiring justification. This level of oversight not only helps to deter dishonest behavior but also fosters a sense of accountability among employees, as they understand that their actions are being monitored and evaluated.

Additionally, integrating tamper-evident audit logs into the system can enhance transparency and provide management with valuable insights into operational activities. These logs can track each employee's actions within the system, documenting changes made to sales, refunds, and inventory adjustments. By analyzing these logs, management can identify patterns of suspicious behavior, such as repeated voids or excessive wastage reports, and take appropriate action. This proactive approach to loss prevention empowers management to address issues before they escalate, ultimately protecting the café's bottom line.

The implementation of unique login credentials for each employee is another critical step in establishing accountability. By requiring staff to log in with their individual credentials, the café can ensure that actions taken within the system are attributed to specific individuals. This level of traceability not only helps to deter fraudulent behavior but also allows management to hold employees accountable for their actions. For instance, if a particular employee consistently engages in suspicious activity, management can address the issue directly, providing coaching or disciplinary action as necessary. This focus on accountability can foster a culture of integrity within the café, encouraging employees to take pride in their work and adhere to best practices.

The introduction of a comprehensive training program is essential to complement the technological enhancements in the café's operations. Employees should be educated on the importance of maintaining accurate records, the implications of theft and fraud, and the café's policies regarding accountability. By fostering an environment of transparency and trust, management can empower employees to take ownership of their roles and contribute to the café's success. Additionally, ongoing training can help staff develop the skills necessary to recognize and report suspicious behavior, further strengthening the café's defenses against fraud.

In conclusion, the vulnerabilities created by undifferentiated system access at Dulci Cafe pose significant risks to its operational integrity and financial health. The potential for employee theft and manipulation of sales data can lead to substantial losses, undermining the café's ability to thrive in a competitive market. By implementing a system with role-specific permissions, tamper-evident audit logs, and unique login credentials, the café can establish a robust framework for accountability that deters fraudulent behavior and promotes transparency. Coupled with comprehensive training programs, these enhancements can foster a culture of integrity and responsibility among staff, ensuring that Dulci Cafe can operate effectively and profitably while delivering exceptional service to its customers.

Inaccurate Sales Recording and Financial Reporting

Inaccurate sales recording and financial reporting at Dulci Cafe stem from a reliance on manual tallying through basic cash registers and handwritten receipts, a practice that is inherently fraught with challenges. This system is highly susceptible to errors, omissions, and inconsistencies, which can significantly impact the café's financial health and operational efficiency. When staff members manually record sales, there is a greater likelihood of miscalculations, transcription errors, or even overlooking transactions altogether. For example, if a cashier accidentally omits a sale during the manual entry process or miscalculates the total, the café's reported revenue will be inaccurate, leading to a distorted view of its financial performance. Such inaccuracies can mask financial leaks that erode profitability, making it difficult for the owner to identify and address issues in a timely manner.

The consequences of relying on manual sales recording extend beyond individual transactions; they can have a cascading effect on the café's overall financial reporting. Owners typically receive delayed reports that are often inaccurate by the time they reach review. This delay can be particularly problematic during peak business hours when timely decision-making is crucial. If management is unaware of real-time sales performance, they may miss opportunities to adjust staffing levels, optimize inventory, or implement promotional strategies that could enhance revenue. The inability to access accurate and timely data can lead to reactive rather than proactive management, hindering the café's ability to respond effectively to changing market conditions or customer preferences.

The data the owner needs to make informed decisions such as daily revenue by category, payment-method breakdown, and tax-liability summaries requires hours of manual assembly and is still unreliable when produced. This labor-intensive process not only consumes valuable time but also increases the risk of human error. For instance, if staff members are tasked with compiling daily sales reports from handwritten receipts, they may inadvertently misinterpret or misrecord figures, resulting in discrepancies in the final report. These inaccuracies can have serious implications for financial planning and forecasting, as the café may overestimate or underestimate its revenue and expenses, leading to poor budgeting decisions.

Moreover, the lack of a centralized system for tracking sales data exacerbates these issues. Without an integrated point-of-sale (POS) system, each transaction must be manually recorded and reconciled, creating additional opportunities for errors. In a bustling café environment, where speed and efficiency are paramount, the potential for mistakes increases significantly. Staff may be distracted by customer interactions or overwhelmed by high volumes of orders, further contributing to inaccuracies in sales reporting. As a result, the café's financial statements may not accurately reflect its true performance, complicating efforts to assess profitability and make informed strategic decisions.

The implications of inaccurate sales recording extend to tax reporting as well. When financial data is unreliable, calculating tax liabilities becomes a complex and error-prone process. Inaccurate revenue figures can lead to misreporting of taxes owed, which can have serious legal consequences for the café. If the café underreports its income due to inaccuracies in sales recording, it may face penalties from tax authorities if discrepancies are discovered during audits. Conversely, overreporting income can result in higher tax burdens than necessary, further straining the café's finances. The lack of reliable data can create a cycle of stress and uncertainty for the owner, who must navigate the complexities of tax compliance without accurate financial insights.

In addition to the financial ramifications, the reliance on manual processes can hinder the café's ability to leverage data for strategic decision-making. In today's competitive landscape, businesses that can analyze and interpret data effectively are better positioned to succeed. However, without accurate sales data, Dulci Cafe may struggle to identify trends, assess customer preferences, or evaluate the effectiveness of promotional campaigns. For example, if the café launches a new menu item but lacks reliable data on sales performance, it may be difficult to determine whether the item is resonating with customers or if adjustments are needed. This inability to derive actionable insights from sales data can stifle innovation and limit the café's growth potential.

Furthermore, the inefficiencies associated with manual sales recording can lead to increased operational costs. Owners may find themselves spending excessive amounts of time reconciling accounts, investigating discrepancies, and addressing issues related to inaccurate reporting. This time could be better spent on strategic initiatives, such as enhancing customer service, developing new products, or expanding marketing efforts. The cumulative effect of these inefficiencies can hinder the café's ability to operate at peak performance, ultimately impacting its profitability and long-term viability.

To address these challenges, Dulci Cafe must consider transitioning to a more advanced point-of-sale (POS) system that automates sales recording and financial reporting. An integrated POS system can streamline the sales process, reducing the likelihood of errors and providing real-time insights into sales performance. With automated data entry and reporting capabilities, the café can eliminate the need for manual tallying and significantly reduce the risk of inaccuracies. Additionally, an advanced POS system can generate comprehensive reports that provide the owner with the critical financial data needed for informed decision-making, including daily revenue by category, payment-method breakdowns, and tax liability summaries.

Implementing a modern POS system would not only enhance the accuracy of sales recording but also improve operational efficiency. Staff members would spend less time on manual data entry and reconciliation, allowing them to focus on providing exceptional customer service. With real-time access to sales data, management can make informed decisions on staffing, inventory management, and marketing strategies. The ability to quickly analyze sales trends and customer preferences can empower the café to adapt its offerings in response to changing demands, ultimately driving revenue growth.

Moreover, a robust POS system can facilitate better inventory management by integrating sales data with inventory levels. This integration allows the café to track ingredient usage in real-time, helping to prevent overstocking or stockouts. By analyzing sales patterns, the café can optimize its purchasing decisions, ensuring that it has the right amount of inventory on hand to meet customer demand without incurring unnecessary waste. This level of operational efficiency not only enhances profitability but also contributes to a more sustainable business model.

Additionally, an advanced POS system can improve the café's ability to comply with tax regulations. By automating the calculation of tax liabilities based on accurate sales data, the café can minimize the risk of errors and ensure compliance with tax reporting requirements. This capability can alleviate the stress associated with tax season, allowing the owner to focus on other aspects of the business without the constant worry of potential audits or penalties.

In summary, the reliance on manual tallying and handwritten receipts at Dulci Cafe has created significant challenges related to inaccurate sales recording and financial reporting. The susceptibility to errors, delays in reporting, and labor-intensive data assembly all contribute to a distorted view of the café's financial performance. By transitioning to a modern point-of-sale system, the café can enhance the accuracy of its sales recording, streamline financial reporting, and ultimately improve operational efficiency. This shift will empower the café to make informed decisions, optimize inventory management, and navigate the complexities of tax compliance with greater confidence. As the café embraces these changes, it will be better positioned to thrive in a competitive market, ensuring that it can deliver exceptional value to its customers while maintaining financial stability.

Regulatory Compliance and Data Security Vulnerabilities

Regulatory compliance and data security vulnerabilities present significant challenges for Dulci Cafe, particularly in the context of maintaining accurate records and adhering to legal requirements. The café's reliance on paper records and locally stored digital files exposes it to various risks, including fire, water damage, accidental misplacement, and hardware failure. These vulnerabilities can undermine the integrity of critical business documents and lead to substantial operational disruptions. Without an automatic backup system in place, the café faces the possibility of losing essential records, which can have dire consequences for its ability to comply with regulatory obligations. When the end of the quarter arrives, generating accurate reports for Bureau of Internal Revenue (BIR) requirements, health inspections, and business permit renewals becomes a laborious, error-prone exercise. The time spent scrambling for records during these critical periods is time that the owner cannot reinvest in the business, ultimately hindering growth and operational efficiency.

The BIR's mandatory electronic invoicing and reporting requirements, which are progressively being extended to cover more micro, small, and medium enterprises (MSMEs), impose additional compliance obligations on small café operators like Dulci Cafe. These regulations require businesses to generate and submit specific documentation formats that reflect their sales and tax liabilities accurately. However, without digital systems capable of producing the required reports, the café is at a disadvantage. The manual processes currently in place not only increase the likelihood of errors in reporting but also make it challenging to meet the deadlines set by regulatory authorities. Non-compliance with BIR regulations exposes the café to audit risk, penalties, and, in serious cases, the suspension of operations. For a café already operating under financial pressure, the risk of a BIR enforcement action is an existential threat that integrated POS technology can meaningfully reduce.

The consequences of non-compliance can be severe, ranging from financial penalties to reputational damage. If Dulci Cafe fails to submit accurate reports or meet the BIR's requirements, it may face fines that further strain its already limited resources. Additionally, the stress and anxiety associated with potential audits can divert the owner's attention away from core business operations. The time and effort spent addressing compliance issues could be better spent on strategic initiatives, such as enhancing customer service, developing new menu items, or expanding marketing efforts. The cumulative effect of these compliance burdens can create a cycle of inefficiency that hinders the café's ability to thrive in a competitive market.

Furthermore, the lack of secure data storage solutions increases the risk of data breaches and unauthorized access to sensitive information. Paper records are particularly vulnerable to theft or loss, while locally stored digital files can be compromised if hardware failures occur. This lack of data security can put the café at risk of losing confidential information, including customer data and financial records. In an era where data privacy regulations are becoming increasingly stringent, the café must prioritize the protection of its data to avoid potential legal repercussions. A breach of customer information not only damages the café's reputation but can also lead to costly legal battles and fines.

Implementing an integrated point-of-sale (POS) system can significantly enhance Dulci Cafe's ability to meet regulatory compliance requirements and improve data security. A modern POS system can automate the generation of reports required by the BIR, ensuring that the café can produce accurate documentation quickly and efficiently. By digitizing sales and financial records, the café can minimize the risk of errors associated with manual data entry and streamline the reporting process. This automation not only saves time but also reduces the likelihood of discrepancies that could lead to compliance issues.

Moreover, a cloud-based POS system can provide the café with robust data security features, including automatic backups and encryption. By storing data in the cloud, Dulci Cafe can protect its records from physical threats such as fire or water damage, as well as hardware failures. Cloud-based solutions offer the added benefit of remote access, allowing the owner to retrieve important documents and reports from any location, which can be particularly valuable during audits or inspections. This flexibility ensures that the café can maintain compliance even in challenging circumstances, reducing the stress associated with regulatory obligations.

Additionally, an integrated POS system can facilitate better inventory management, which is crucial for compliance with health regulations. By tracking ingredient usage and expiration dates in real-time, the café can ensure that it adheres to food safety standards and minimizes the risk of serving expired products. This proactive approach to inventory management not only enhances customer safety but also helps the café avoid potential fines associated with health code violations.

Furthermore, the implementation of an integrated POS system can improve communication and collaboration among staff members. With a centralized platform for managing sales, inventory, and reporting, employees can work together more effectively to ensure compliance with regulatory requirements. For example, staff responsible for inventory management can easily access sales data to identify trends and adjust orders accordingly, helping the café maintain optimal stock levels while adhering to health regulations. This level of coordination fosters a culture of accountability and responsibility, empowering staff to contribute to the café's success.

In addition to enhancing operational efficiency and compliance, the integration of technology can also improve customer trust and loyalty. In today's digital age, consumers are increasingly concerned about how their data is handled and protected. By adopting secure, modern systems for managing transactions and customer information, Dulci Cafe can demonstrate its commitment to data security and privacy. This proactive stance can enhance the café's reputation in the community, attracting customers who prioritize businesses that prioritize their data protection. As customers become more aware of data security issues, the café's investment in technology can serve as a competitive advantage, setting it apart from competitors who may still rely on outdated practices.

Moreover, the transition to a digital system can facilitate better analytics and insights into customer behavior and preferences. With accurate sales data readily available, Dulci Cafe can analyze purchasing patterns, identify popular menu items, and tailor marketing efforts accordingly. This data-driven approach can lead to more effective promotional strategies, helping the café maximize its revenue potential. For instance, if sales data shows that a particular beverage is consistently popular among customers, the café can consider introducing seasonal variations or special promotions to capitalize on that demand. By leveraging technology to gain insights into customer preferences, the café can enhance its offerings and improve overall customer satisfaction.

Additionally, the implementation of an integrated POS system can streamline the onboarding process for new employees. With standardized procedures for entering sales, managing inventory, and generating reports, new staff members can be trained more efficiently. This consistency not only reduces the likelihood of errors during the training period but also helps to establish a culture of accountability from the outset. When new employees understand the importance of accurate record-keeping and compliance with regulations, they are more likely to adopt best practices and contribute positively to the café's operations.

Finally, the adoption of an integrated POS system can provide Dulci Cafe with a scalable solution that can grow alongside the business. As the café expands its operations, whether through additional locations or increased customer volume, the technology can adapt to meet the evolving needs of the business. This scalability ensures that the café can maintain compliance and operational efficiency even as it faces new challenges and opportunities in the marketplace. Investing in a robust POS system not only addresses current vulnerabilities but also positions the café for long-term success in an increasingly competitive environment.

Administrative Time Drain and Workforce Management Gaps

Administrative Overload and Time Poverty

The challenges faced by Dulci Cafe in terms of administrative overload and the resistance to digital adoption are not unique to this establishment; they reflect broader trends observed across the micro, small, and medium enterprise (MSME) landscape in the Philippines. Many small business owners find themselves juggling multiple roles, from managing finances to overseeing daily operations, which can lead to a significant strain on their time and resources. This situation is exacerbated by the complexities of regulatory compliance, inventory management, and customer service, all of which require careful attention and meticulous record-keeping. As a result, owners often become entrenched in routine tasks, leaving little room for strategic planning or innovation.

The impact of administrative overload extends beyond mere inefficiency; it can also lead to burnout among owners and staff. When individuals are constantly bogged down by repetitive tasks, their motivation and morale can suffer. This can create a negative feedback loop, where low employee engagement further exacerbates operational inefficiencies.

In contrast, businesses that successfully implement digital solutions often report higher levels of employee satisfaction, as staff members are empowered to focus on more meaningful work that contributes to the café's success. By reducing the burden of manual processes, Dulci Cafe can cultivate a more motivated workforce that is eager to contribute to the café's growth and success.

Moreover, the failure to adopt digital systems can hinder the café's ability to compete effectively in an increasingly technology-driven marketplace. Consumers today expect businesses to provide seamless experiences, whether through online ordering, loyalty programs, or personalized promotions. Cafés that rely on outdated practices may struggle to meet these expectations, resulting in lost business opportunities. For instance, without a robust POS system, Dulci Cafe may miss out on the chance to implement a customer loyalty program that rewards repeat visits or to analyze sales data that could inform targeted marketing efforts. In a competitive environment, the ability to leverage technology effectively can be a key differentiator that sets successful establishments apart from their competitors.

Furthermore, the transition to digital systems can enhance the café's ability to respond to changing consumer preferences and market trends. In the food service industry, consumer tastes can shift rapidly, and businesses must be agile enough to adapt. A modern POS system can provide real-time insights into sales trends, allowing Dulci Cafe to identify emerging preferences and adjust its offerings accordingly. For example, if data reveals a growing interest in plant-based options, the café can quickly introduce new menu items to cater to this demand. This level of responsiveness not only helps the café stay relevant but also positions it as a forward-thinking establishment that is attuned to its customers' needs.

In addition to enhancing operational efficiency and customer responsiveness, digital systems can also improve financial management. Accurate financial reporting is essential for making informed business decisions, but manual processes often lead to discrepancies and delays. With an integrated POS system, Dulci Cafe can automate financial reporting, ensuring that owners have access to up-to-date information on revenue, expenses, and profit margins. This transparency allows for more effective budgeting and forecasting, enabling the café to allocate resources more strategically and identify areas for improvement. By gaining a clearer understanding of its financial health, the café can make informed decisions that drive growth and sustainability.

Moreover, the integration of technology can facilitate better supplier management and negotiation. With accurate inventory tracking and sales data, Dulci Cafe can optimize its purchasing decisions, ensuring that it maintains optimal stock levels while minimizing waste. This level of insight can empower the café to negotiate more favorable terms with suppliers, as it can provide data-driven justifications for bulk purchases or discounts based on sales trends. By fostering stronger relationships with suppliers, the café can enhance its operational efficiency and potentially reduce costs, contributing to improved profitability.

As the café embraces digital transformation, it can also enhance its marketing efforts through data-driven strategies. A modern POS system can capture valuable customer data, including purchasing habits and preferences. This information can be leveraged to create targeted marketing campaigns that resonate with specific customer segments. For instance, if the café identifies a group of customers who frequently purchase certain items, it can tailor promotions or special offers to encourage repeat visits. By utilizing data analytics, Dulci Cafe can maximize the effectiveness of its marketing initiatives, ultimately driving customer engagement and loyalty.

Additionally, the transition to digital systems can improve the café's ability to gather and respond to customer feedback. In today's competitive landscape, customer reviews and feedback play a crucial role in shaping a business's reputation. By implementing systems that facilitate customer feedback collection, such as digital surveys or loyalty programs, Dulci Cafe can gain valuable insights into customer satisfaction and areas for improvement. This proactive approach to gathering feedback not only demonstrates the café's commitment to customer service but also provides actionable data that can inform operational changes and enhance the overall customer experience.

The potential benefits of adopting an integrated POS system extend beyond immediate operational improvements; they can also contribute to the long-term sustainability of Dulci Cafe. In an environment where small businesses face numerous challenges, including economic fluctuations and changing consumer behaviors, the ability to adapt and innovate is paramount. By investing in technology that streamlines operations, enhances customer engagement, and provides valuable insights, the café can position itself for success in an ever-evolving market. The transition to digital systems is not merely a response to current challenges; it is a strategic investment in the future of the business.

In conclusion, the administrative overload faced by Dulci Cafe represents a significant barrier to growth and innovation. By recognizing the structural challenges that impede digital adoption, the café can take proactive steps to break free from the cycle of inefficiency. The implementation of StockSecure POS offers a pathway to streamline operations, improve compliance, and enhance customer engagement. As the café embraces digital transformation, it can unlock new opportunities for growth, drive profitability, and ultimately create a more sustainable business model. In a rapidly changing landscape, the ability to leverage technology effectively will be a key determinant of success for Dulci Cafe and similar establishments within the Philippine MSME sector.

Evidence-Free Employee Scheduling

Evidence-Free Employee Scheduling and the Integrated Solution of StockSecure POS and CyberLedger at Dulce Cafe

Employee scheduling in the Philippine food and beverage industry has long operated on intuition rather than evidence. Managers at establishments like Dulce Cafe, a mid-sized café operating in a competitive Metro Manila market, typically build weekly rosters based on historical habit, staff seniority, or simple guesswork about which hours will be busiest. This approach, while familiar and easy to administer, carries significant hidden costs. When scheduling is divorced from actual transaction data, the result is a recurring mismatch between labor supply and customer demand. During slow mid-afternoon hours, a café may have three or four staff members on the floor when one or two would suffice, each earning wages that cut directly into already thin margins. Conversely, during the morning rush between 7:00 and 9:00 AM or the after-work surge between 5:00 and 7:00 PM, the same establishment may be understaffed, forcing baristas to split attention between taking orders, preparing drinks, handling payments, and restocking supplies. Service slows, wait times lengthen, and customers who expect the quick, cheerful experience that defines a neighborhood café leave frustrated or, worse, do not return.

For Dulce Cafe, where the average transaction value hovers around PHP 180 to PHP 250 and repeat patronage is the backbone of revenue, every instance of understaffing erodes the goodwill that independent cafés depend on to survive alongside larger chains. Overstaffing, meanwhile, quietly drains the bottom line: if even one extra crew member is scheduled per slow shift, the cumulative labor waste over a month can reach PHP 8,000 to PHP 12,000, a meaningful sum for a small business operating on net profit margins of five to ten percent. The problem is compounded by the fact that most scheduling decisions are made manually on paper or basic spreadsheets, with no mechanism to review historical transaction counts, peak hour patterns, or the labor-cost-to-revenue ratio of each shift. The manager's best guess becomes the schedule, and the schedule becomes the status quo, repeated week after week without interrogation. Breaking this cycle requires a fundamental shift from intuition-based to evidence-based scheduling, and that shift depends on having the right data infrastructure in place.

The StockSecure POS system, with its integrated role-based access control architecture and real-time transaction logging, provides exactly that foundation. By capturing every sale, void, refund, and inventory adjustment at the point of service, StockSecure generates a precise digital record of when customers arrive, what they order, and how long each transaction takes. Over weeks and months, this data reveals the invisible rhythm of the business: the predictable spikes before lunch, the dead periods after three o'clock, the weekend patterns that differ sharply from weekday ones.

For Dulce Cafe, deploying StockSecure means that scheduling can finally be grounded in the empirical reality of customer behavior rather than managerial instinct, transforming labor from a fixed cost into a variable cost that flexes with actual demand. The result is a café that is neither overstaffed nor understaffed but precisely staffed, with every peso of labor expenditure justified by the transaction volume it supports. This is the first and most consequential layer of the integrated solution that StockSecure and CyberLedger together bring to Dulce Cafe, and it sets the stage for a broader transformation in how the business manages its operations from the front counter to the back office.

Integrating transaction volume data into the scheduling process is only the beginning of what StockSecure enables at Dulce Cafe. The system's core architecture is built around role-based access control, or RBAC, which means that every user who interacts with the POS operates within a strictly defined permission tier. A cashier sees only the functions needed to ring up sales and process payments. A barista can mark orders as complete and update modifiers but cannot access pricing menus or refund transactions. A shift supervisor can void items within a preset daily limit and run end-of-day reports but cannot alter user permissions or modify inventory cost bases.

The store manager and owner hold the highest tiers, with access to employee management, inventory adjustments, system configuration, and full audit logs. This layered permission structure serves a dual purpose that directly addresses the scheduling problem and extends well beyond it. First, it ensures that when evidence-based scheduling is implemented, the staff members assigned to each shift have precisely the access they need to perform their roles and nothing more, reducing the risk of unauthorized actions during understaffed periods when oversight is thin. Second, RBAC functions as a powerful internal fraud prevention mechanism. In the Philippine café sector, where establishments like Dulce Cafe typically employ eight to twelve staff members across morning, mid, and closing shifts, the risk of internal theft through unauthorized discounts, phantom refunds, or inventory skimming is significant and often underreported.

Employees who handle cash and inventory daily operate in an environment of high trust and low supervision, especially during closing shifts when the manager is absent. A cashier who can process refunds without supervisory approval, for example, could theoretically ring up a fake refund after a cash sale and pocket the difference. A barista who can adjust inventory counts could conceal the theft of premium coffee beans or milk. StockSecure closes these vulnerabilities by enforcing that every sensitive action requires the appropriate access level, logged with the identity of the user who performed it. When a void or discount exceeds a configurable threshold, the system escalates approval to the next RBAC tier, ensuring that no single employee can manipulate a transaction from end to end without a second set of eyes.

For Dulce Cafe, implementing RBAC means that the same system that captures the transaction data for smarter scheduling also enforces operational discipline across every shift. The morning cashier cannot access closing procedures, the closing barista cannot modify inventory, and the weekend relief supervisor cannot alter system settings. This granular control does not just prevent fraud; it creates a culture of accountability where every action is traced to a named user, and every exception is flagged for review. When combined with the scheduling insights derived from StockSecure's transaction logs, RBAC ensures that the right people are not only scheduled at the right times but also empowered with the right permissions, neither over-privileged nor under-equipped for the demands of their shift.

Inventory monitoring represents the third pillar of the StockSecure solution at Dulce Cafe, and it connects directly to both the scheduling and fraud prevention dimensions already discussed. In a typical Philippine café, inventory management is often a weekly or biweekly ritual: the manager prints a count sheet, walks the storage area tallying bags of coffee beans, cases of milk, syrups, cups, and lids, then manually compares the counts against sales records to calculate usage. This process is labor-intensive, prone to transcription errors, and updated too infrequently to catch problems early. By the time a discrepancy surfaces between physical inventory and system records, the underlying cause may be days or weeks old, making it impossible to determine whether the loss resulted from theft, spoilage, misportioning, or simple recording error.

StockSecure transforms this paradigm by integrating inventory tracking directly into every transaction. When a barista rings up a latte, the system automatically decrements the appropriate quantity of espresso beans, milk, and cup from the digital inventory. When a pastry is sold, its corresponding ingredient or finished-good count adjusts in real time. This perpetual inventory model means that at any moment, the manager can view the system's expected stock levels and compare them against physical counts to identify shrinkage. For Dulce Cafe, where perishable ingredients like dairy and baked goods represent a substantial portion of the cost of goods sold, real-time inventory visibility enables proactive rather than reactive management. If milk usage is running twenty percent higher than expected for the current transaction volume, the system flags the discrepancy immediately rather than waiting for the weekly count.

The manager can then investigate: Are baristas over-pouring? Is there a training issue with standard recipes? Or is product being consumed for staff meals without being recorded? Each of these scenarios has a different operational remedy, and catching them early prevents small losses from compounding into significant margin erosion over time. The scheduling connection is equally important. When StockSecure tracks inventory depletion rates alongside transaction volumes, it generates data that informs not only how many staff members to schedule but also what skills and tasks those staff members should prioritize during each shift. A mid-morning rush that generates high volumes of espresso-based drinks requires a different allocation of barista attention than an afternoon lull dominated by drip coffee and pastry sales. By correlating inventory movement patterns with staffing levels, the system can recommend shift compositions that match labor skills to predicted demand, ensuring that the café is never short-handed on the espresso machine during peak latte hours or overstaffed on the register during a slow period dominated by grab-and-go transactions. The inventory module of StockSecure thus completes a virtuous cycle: transaction data informs scheduling, scheduling ensures adequate coverage for anticipated demand, proper coverage reduces execution errors and waste, and tighter inventory control feeds back into more accurate cost forecasting and pricing decisions.

Internal fraud prevention at Dulce Cafe extends beyond RBAC permissions and into the domain of transaction pattern analysis, where StockSecure and CyberLedger together create a defense-in-depth architecture that protects revenue from both external and internal threats. While RBAC prevents unauthorized actions at the point of sale, determined employees can still find ways to exploit system gaps if no mechanism exists to detect anomalous patterns after the fact. Consider the scenario of a cashier who processes a legitimate cash transaction, then moments later issues a refund or void for that same transaction and pockets the cash. If the system only logs each action individually, this kind of sweethearting or fraudulent refund scheme can go unnoticed for months, especially if the cashier is careful to keep the amounts small and spread them across different customer names. StockSecure counters this by maintaining a complete, immutable transaction log that includes the timestamp, user identity, terminal, register drawer activity, and a cryptographic hash of the transaction data.

This log is not merely stored on the local POS terminal but is continuously transmitted to the CyberLedger audit monitoring system, which functions as a separate, tamper-evident record of every financial event at Dulce Cafe. CyberLedger receives each transaction as it occurs, applies a time-stamped cryptographic seal, and stores it in a sequential ledger that cannot be altered retroactively without breaking the cryptographic chain. This means that even if someone gains physical access to the POS terminal and attempts to modify or delete local transaction records, the original data survives intact within CyberLedger. The system's audit monitoring component then applies automated rules to detect suspicious patterns: multiple voids by the same user in a short time window, refunds that exceed a certain percentage of daily sales, transactions initiated outside normal operating hours, or register drawers that are opened without a corresponding sale. Each anomaly triggers an alert that is sent to the manager or owner, complete with the relevant transaction IDs, timestamps, and user details, enabling swift investigation.

For Dulce Cafe, where the owner may not be on-site during every shift, this remote monitoring capability is invaluable. The owner can review the day's exception report from a mobile dashboard after closing, confident that any irregular activity has been captured and flagged rather than papered over by the closing crew. The deterrent effect of this system is equally important: when employees know that every transaction is independently recorded in a cryptographic ledger that management can review remotely, the perceived risk of engaging in fraudulent behavior rises sharply, and the incidence of such behavior falls. CyberLedger thus acts as both a detective control, catching fraud after it occurs, and a preventive control, discouraging fraud before it is attempted. Integrated with StockSecure's RBAC layer, it creates a closed loop where access is controlled at the point of sale, monitored in real time, and audited retroactively with cryptographic certainty.

The CyberLedger component of the integrated system at Dulce Cafe deserves particular attention, as it represents the most technologically advanced layer of the solution and the one with the farthest-reaching implications for trust, compliance, and strategic decision-making. CyberLedger is a secure transaction and audit monitoring system built on distributed ledger principles, although for a single-establishment deployment like Dulce Cafe, the ledger can operate in a centralized or multi-node configuration depending on the owner's preference for redundancy and transparency. What matters is the immutability guarantee: once a transaction is committed to the CyberLedger, it cannot be silently altered or deleted. This guarantee extends beyond simple sales records to encompass every event that StockSecure generates, including inventory adjustments, user permission changes, schedule modifications, price updates, and cash management actions such as opening floats, cash pickups, and end-of-day reconciliations.

For the owner of Dulce Cafe, this means that the financial record is no longer a collection of files that can be edited after the fact or lost in a hard drive failure. It is a permanent, verifiable chain of events that reconstructs the complete operational history of the business from the moment the system was deployed. The practical value of this capability becomes apparent in several scenarios. When the business files its monthly taxes with the Bureau of Internal Revenue, the owner can export a CyberLedger-verified report that matches the POS sales data exactly, eliminating the fear of discrepancies during a BIR audit. When a dispute arises with a supplier over delivery quantities or pricing, the inventory adjustment records in CyberLedger provide an authoritative timestamped reference for what was received and when. When the owner considers selling the business or bringing in an investor, the CyberLedger audit trail serves as a due diligence asset, demonstrating that the café's financial records are complete, consistent, and tamper-proof.

On an operational level, the audit monitoring features of CyberLedger empower the owner to track key performance indicators with confidence. Labor cost percentage, inventory shrinkage rate, average transaction value, table turnover time, and sales per labor hour can all be computed from the immutable ledger data, giving the owner a reliable baseline for measuring the impact of scheduling changes, menu adjustments, and promotional campaigns. When evidence-based scheduling is first implemented at Dulce Cafe, the owner can compare post-implementation labor metrics against the CyberLedger-verified pre-implementation baseline to quantify the improvement with precision, rather than relying on anecdotal impressions of whether the new schedule is working. This creates a culture of continuous improvement where decisions are evaluated by their measured outcomes, and the ledger provides the single source of truth that makes those measurements meaningful. CyberLedger thus transforms from a security tool into a strategic asset, one that gives the owner of a small café the same kind of financial visibility and control that large restaurant chains achieve through dedicated finance teams and enterprise resource planning systems.

The integration of scheduling data, RBAC, inventory monitoring, and cryptographic audit logging creates synergies that none of these components could achieve in isolation. At Dulce Cafe, the StockSecure POS system functions as the operational hub, capturing granular transaction data that feeds into three distinct but interconnected analytical streams. The first stream is the scheduling engine, which uses historical transaction volumes by day of week, time of day, and season to generate predictive staffing recommendations. The second stream is the inventory management module, which tracks real-time stock levels and usage rates, flagging discrepancies that may indicate training issues, recipe deviations, or theft.

The third stream is the RBAC and fraud detection layer, which monitors user behavior for anomalies and enforces permission boundaries that align with each employee's role and shift responsibilities. All three streams feed their data into CyberLedger, which seals each record cryptographically and applies cross-stream analytics that would be impossible if the data remained siloed. For example, if the scheduling engine predicts a high-volume Saturday morning and recommends four staff members, but the actual transaction volume falls short by forty percent, CyberLedger can flag the discrepancy and correlate it with inventory usage data to determine whether the overstaffing led to excessive waste through overproduction or whether the forecast model itself needs recalibration. Conversely, if a scheduled low-volume Tuesday afternoon sees an unexpected surge in transactions that the skeleton crew of two cannot handle, the system records the resulting slowdown in service speed, the increase in voided or corrected orders from rushed staff, and any inventory discrepancies from hurried portioning.

These correlated data points become the evidence base for refining the next week's schedule, creating a closed-loop learning system that becomes more accurate with every cycle. The economic impact at Dulce Cafe is measurable across multiple dimensions. Labor costs, which typically consume twenty-five to thirty-five percent of revenue in the Philippine restaurant sector, can be reduced by an estimated eight to fifteen percent through the elimination of overstaffed shifts without sacrificing service quality. Inventory shrinkage, which industry benchmarks place at two to five percent of revenue for cafes and quick-service operations, can be cut by half or more through real-time monitoring and rapid discrepancy detection.

Fraud losses, which are notoriously difficult to quantify but are estimated to affect as many as thirty to forty percent of small food-service businesses in some form, are reduced to near zero through the combined deterrent and detective effects of RBAC and CyberLedger's immutable audit trail. For a café generating PHP 80,000 to PHP 120,000 in monthly revenue, these improvements translate into PHP 15,000 to PHP 30,000 in recovered or avoided costs per month, a sum that can mean the difference between merely surviving and genuinely thriving in the competitive Philippine café market.

Beyond the immediate operational and financial benefits, the integrated StockSecure and CyberLedger system positions Dulce Cafe for long-term strategic growth in ways that traditional POS systems cannot match. The evidence-based scheduling capability, powered by transaction data and validated through CyberLedger's audit trail, gives the owner a rigorous framework for making hiring decisions, expansion plans, and menu pricing adjustments. When the owner considers opening a second Dulce Cafe location, the scheduling models, inventory benchmarks, and fraud prevention protocols developed at the first branch become a replicable operational playbook rather than a set of undocumented habits that the manager carries in their head. The new location can be equipped with StockSecure from day one, configured with the same RBAC tiers and CyberLedger integration, and staffed according to scheduling algorithms that have been refined using real data from the original café. This repeatability is the hallmark of a scalable business model, and it is enabled by the data infrastructure that StockSecure and CyberLedger provide. On the compliance front, the system offers protections that are particularly valuable in the Philippine regulatory environment.

The Bureau of Internal Revenue has increasingly emphasized digital receipt issuance and real-time sales reporting, and the comprehensive, tamper-evident records generated by CyberLedger far exceed the minimum requirements for BIR accreditation. When the café undergoes a BIR audit or a local government inspection, the owner can present a complete, cryptographically verified history of every transaction, every inventory adjustment, and every cash movement, eliminating the anxiety and uncertainty that accompanies manual recordkeeping. For the employees of Dulce Cafe, the system creates a fairer and more transparent workplace. Scheduling is based on data rather than favoritism, performance is measured by objective metrics such as transaction throughput and order accuracy rather than subjective impressions, and the RBAC structure ensures that each staff member has the access they need to do their job well without being placed in positions where they might be tempted or pressured to circumvent controls.

The combination of StockSecure POS with its role-based access control and inventory monitoring, together with CyberLedger's secure transaction and audit monitoring, thus addresses the full spectrum of operational challenges that face a modern Philippine café. It solves the scheduling problem by replacing intuition with evidence. It solves the fraud problem by replacing trust with verification. It solves the inventory problem by replacing periodic counts with real-time visibility. And it solves the strategic problem by replacing undocumented practices with an immutable record that supports learning, scaling, and continuous improvement.

For Dulce Cafe, the adoption of this integrated system represents not merely a technology upgrade but a fundamental shift in how the business understands itself, shifting from a café run on habit and guesswork to a café run on data, accountability, and cryptographic certainty. The evidence-free scheduling that once cost the business in wasted labor, lost sales, and unaddressed theft gives way to a precision operation where every staffing decision is anchored in transaction reality, every inventory movement is tracked, every financial action is logged, and every improvement is measured. This is the promise of the StockSecure and CyberLedger integration, and it is a promise that transforms a small café from a vulnerable enterprise into a resilient, scalable, and confident business

Limited Business Intelligence and Strategic Visibility

Without automated reporting, the operational reality of a small café like Dulcis in Mandurriao's competitive commercial district becomes a landscape of blind spots and reactive decisions rather than strategic foresight. Identifying bestsellers requires the owner to manually cross-reference handwritten receipts against inventory sheets, a process so labor-intensive that it is performed sporadically at best and never comprehensively. Seasonal patterns in customer traffic, menu preferences, and ingredient availability remain invisible because nobody has the time to compile months of paper records into a coherent trend analysis. Promotion effectiveness is evaluated anecdotally a customer mentions they came for the discount, or the owner notices a slight uptick in a particular item, but no controlled comparison exists between promotional periods and baseline performance. This means every decision about what to feature, what to discount, and when to run a campaign is made without reliable evidence about what actually works. The compounding effect across dozens of weekly decisions is a slow but steady erosion of potential revenue that the owner never sees, never measures, and therefore never corrects. Each missed insight is individually small, but their accumulation over months and years represents a structural disadvantage that no amount of hard work or intuition can fully overcome.

The owner of Dulcis Cafe cannot answer fundamental strategic questions about the business without spending hours digging through paper receipts, reconciling delivery invoices, and relying on imperfect mental recall of events that happened weeks or months earlier. Which menu items are most profitable when ingredient costs, prep time, and waste are all factored in, not just the selling price? No single source of truth exists for this calculation; the answer has to be pieced together from supplier invoices, production notes, and sales slips, and by the time it is assembled, the ingredient prices have already changed. Which time slots generate the highest margins when staffing costs and traffic volumes are weighed against each other? The morning rush might sell more units while the afternoon lull yields higher per-ticket averages, but without time-stamped sales data disaggregated by staffing shifts, this remains a guess. Is the customer base growing or shrinking in terms of repeat visitation rates, not just gross foot traffic that could reflect one-time promotional visitors? The owner sees familiar faces and has a general sense of whether business is up or down but cannot quantify retention, churn, or the lifetime value of different customer segments. These are not esoteric analytics questions suited only for corporate chains they are the basic operational intelligence needed to run any competitive food-service business, and Dulcis is operating without them.

This puts Dulcis Cafe at a competitive disadvantage in a dynamic market where data-driven competitors adjust their menus, pricing, and promotions in near real time based on actual performance signals rather than gut feelings. Mandurriao's commercial scene has seen a proliferation of cafés and quick-service concepts over the past several years, many of which are operated by younger entrepreneurs who have grown up with point-of-sale systems, inventory management apps, and cloud-based reporting dashboards as natural parts of running a business. These competitors know, with a few taps on a tablet, their top-ten items by contribution margin, the hour of the week when labor costs peak relative to revenue, and which promotional channels deliver the highest return on marketing spend.

They can identify a declining sales trend in a specific category within days of its onset and adjust their menu positioning, pricing, or sourcing accordingly. Dulcis, by contrast, operates on a weeks-to-months lag in detecting any trend, and even then, the detection is qualitative and imprecise. The gap between intuition-driven and data-driven operations in a market with thin margins and high customer churn is not merely an efficiency difference it is the difference between adapting to the market and being overtaken by it.

The strategic importance of business intelligence for small operators has grown substantially as café competition has intensified across urban commercial corridors throughout the Philippines. What was once a luxury reserved for large chains with dedicated analytics departments has become increasingly accessible through affordable, cloud-based point-of-sale systems that bundle reporting capabilities into their monthly subscription fees. Yet the barrier is not primarily technological or financial it is cognitive and behavioral. Many small operators do not realize what they are missing because they have never had access to operational metrics presented in a decision-ready format. They have normalized the absence of data.

They make pricing, menu, and staffing decisions the same way their parents and grandparents ran small food businesses, not realizing that the competitive landscape has shifted beneath their feet. In a market where a competitor can identify a decline in afternoon-traffic revenue within 48 hours and launch a targeted promotion within the same week, the operator who discovers the same decline four weeks later through a manual receipt audit has already lost the revenue and the customer relationships that went elsewhere in the interim.

The ability to identify high-margin items, optimize menu mix, and time promotional campaigns based on traffic pattern data can meaningfully differentiate a café from its competitors on dimensions that directly affect profitability and customer loyalty. High-margin items are not always the ones with the highest selling prices they are the ones where the gap between selling price and fully loaded cost (ingredients, labor, packaging, and waste) is widest, and identifying them requires data that cuts across sales, inventory, and production records. A pasta dish priced at 320 pesos with a 45 percent food cost is less profitable than a 180-peso pastry with an 18 percent food cost that turns over five times faster during the same period, but this comparison is invisible without a system that tracks both cost percentages and velocity. Menu mix optimization means understanding not just which items sell well individually but also how items perform in combination, whether certain drinks drive food add-ons, whether the lunch special cannibalizes higher-margin à la carte sales, or whether the dessert menu is underperforming because of poor placement or because customers are simply too full after the main course. These are analytical questions, not intuitive ones, and the answers vary by day part, season, and customer demographic.

Yet generating these insights from manual records is so labor-intensive that most small operators simply do not have access to them, not because the data is unavailable but because the effort required to extract it exceeds the available hours in a day already consumed by operational firefighting. The owner of Dulcis is not sitting idle she is managing suppliers, supervising staff, handling customer complaints, cleaning equipment, arranging deliveries, reconciling cash drawers, and covering shifts when someone calls in sick. The idea of spending three additional hours per week building a spreadsheet that cross-references sales by time slot, item profitability, and promotion timing is simply unrealistic, even if she recognizes the value of doing so. The opportunity cost is too high, and the immediate operational demands always win. This is the fundamental paradox of business intelligence for small operators: the businesses that need data-driven insights the most are precisely those with the least organizational capacity to generate them through manual effort. The very conditions that make automated reporting valuable thin margins, high competition, limited staff are the same conditions that make manual analysis impossible to sustain as a regular practice.

The result is that operational decisions at Dulcis Cafe, such as which items to feature on the display counter, how to price new menu additions, when to run promotions, and how many staff to schedule for each shift, are made based on intuition and anecdote rather than data, leading to systematically suboptimal outcomes across every dimension of business performance. Featured items are chosen because the owner likes them or because a supplier offered a deal, not because analysis shows they drive traffic or improve margin mix. Pricing decisions are reactive, set to match a competitor across the street or adjusted upward when a supplier raises prices rather than strategic, informed by customer willingness-to-pay signals and cross-elasticity effects that could be derived from transaction data.

Promotions are run on gut feel: a slow Tuesday prompts a midweek discount, but nobody knows whether the discount generated incremental revenue or merely shifted purchases from Wednesday. Staff scheduling follows historical routines rather than predicted traffic patterns, so the café is either overstaffed during quiet periods, burning margin through excess labor costs, or understaffed during rushes, burning margin through lost sales and slower service. Each of these decisions, taken individually, might shift weekly profit by a small percentage. Taken together across dozens of decisions made every month, the cumulative drag on revenue and profitability is substantial, and it compounds as the market grows more competitive and margins grow tighter.

The compounded effect of all identified problems at Dulcis exceeds the sum of their individual costs because the problems do not operate independently they interact with and amplify each other in ways that are invisible to the operator who only sees each symptom in isolation. Revenue leakage from service delays caused by chronic understaffing during peak hours does not merely cost the individual transactions that walk out the door; it also drives down per-ticket averages as hurried customers skip add-ons and desserts, reduces repeat visitation as frustrated customers choose faster alternatives on subsequent visits, and increases employee stress and turnover as staff struggle to keep up with demand during understaffed shifts. Inventory waste from poor demand forecasting is not just the cost of spoiled ingredients; it cascades into stock-outs of popular items on high-traffic days that disappoint customers and depress sales and into emergency supplier orders that carry premium pricing and disrupt the cost structure of affected menu items. Fraud or reporting inaccuracies that go undetected for months erode trust in the financial picture of the business, making it impossible to distinguish between a genuine decline in customer demand and a leak in the revenue stream that needs plugging. Each problem feeds into the others, creating a web of interconnected constraints that collectively reduce the business's effective capacity far more than any single issue would suggest.

Poor data visibility prevents strategic correction because the owner cannot distinguish between problems that require operational fixes (a supplier issue, a training gap, a process breakdown) and problems that require strategic repositioning (a shift in customer preferences, new competitive offerings, a change in the neighborhood's demographic profile). Every setback looks the same through the lens of manual record-keeping: revenue is down, costs are up, and the reason is unclear. The owner responds by trying harder at everything rather than targeting the specific intervention that would have the highest return. Compliance risks threaten operational continuity in ways that rarely appear in day-to-day management but can manifest suddenly and severely a health inspection that flags undocumented temperature logs, a tax filing that reveals inconsistencies between reported revenue and bank deposits, or a supplier audit that exposes gaps in receiving documentation. These are low-probability, high-impact events that build up slowly over time through the accumulated friction of manual record-keeping, and they strike hardest at businesses that lack the systematic documentation that automated systems provide as a byproduct of normal operations. The owner's time, meanwhile, is consumed by firefighting rather than growth. Every hour spent tracking down a discrepancy in the cash drawer or reconciling a supplier invoice is an hour not spent on menu development, customer experience improvements, or strategic planning that could move the business forward.

Rising ingredient costs, staff turnover, and competitive pressure compound these vulnerabilities further as external headwinds that the business is structurally unprepared to absorb or respond to. A 10 percent increase in flour prices might reduce gross margin by a small percentage across the entire bakery line, but without item-level costing data the owner cannot determine whether to absorb the increase, pass it on selectively, or reformulate affected items. Staff turnover endemic in the food-service industry costs far more than the recruitment and training expense; it costs institutional knowledge about customer preferences, operational routines, and supplier relationships that accumulates slowly in a manual operation and disappears entirely when an experienced staff member leaves. Each new hire cycle resets the operational clock, and without systematized processes documented and enforced through a reporting infrastructure, the business's effective capability fluctuates with every staffing change. Competitive pressure from new entrants and from existing competitors who are upgrading their own systems means that the baseline for acceptable operational performance is rising steadily. Dulcis is not merely falling behind its current competitors, it is falling behind the accelerating trajectory of what customers will expect and what competitors will deliver in the next twelve to twenty-four months. For Dulcis Cafe, these are not isolated inconveniences or temporary growing pains. They are interconnected, self-reinforcing constraints that, taken together, pose an existential threat to the business's sustainability in Mandurriao's bustling commercial scene.

Objectives of the Study

General Objective

The general objective of this study is to develop and implement an integrated point-of-sale system specifically designed for Dulci Cafe, a thriving cafe business located in Mandurriao, Iloilo City. Named StockSecure POS, the system integrates three core components: Role-Based Access Control (RBAC), real-time inventory management, and CyberLedger, a dedicated module for secure transaction monitoring and audit tracking. This comprehensive solution aims to significantly enhance operational efficiency, strengthen internal controls against fraud, and provide accurate, real-time data to support informed decision-making. By addressing the limitations of traditional manual processes and basic POS systems commonly used in local cafes, the proposed platform seeks to modernize daily operations while ensuring security, accuracy, and accountability across all levels of the business.

The first major objective is to **design and develop a role-based access control module** that precisely defines user permissions according to staff roles at Dulci Cafe. Baristas receive access limited to order entry and ticket management. Cashiers handle payment processing and receipt generation without being able to modify menu prices or access financial summaries. Supervisors gain shift-management privileges, including the ability to review transaction logs and approve refunds within defined thresholds. Managers hold full system oversight, encompassing user administration, report generation, and configuration control. Each role carries only the permissions its function requires, and every action is tagged to the authenticated user who performed it. This structure eliminates the broad, undifferentiated access that currently enables underreporting, unauthorized voids, and inventory manipulation, replacing it with a permission model that makes fraud far more difficult to commit and far easier to trace when it occurs.

The second major objective is to **create a real-time inventory management system** that automatically adjusts stock levels for all ingredients, beverages, and supplies with every transaction or adjustment. When a barista rings up a latte, the system deducts the configured quantities of coffee beans, milk, and syrup from active inventory counts. When a delivery arrives, the manager records receipt quantities, and the system updates available stock immediately. Reorder points trigger proactive low-stock alerts so the owner can place orders before an item runs out, eliminating the reactive emergency purchases that currently inflate food costs. The system also tracks waste and spoilage entries separately from sales deductions, providing visibility into where and why inventory variance occurs. All inventory movements are time-stamped and attributed to the user who performed them, creating a complete chain of custody for every ingredient that enters and leaves the cafe.

The third major objective is to **integrate CyberLedger** as a secure, tamper-evident digital ledger that records and monitors all financial transactions, refunds, voids, and audit trails with cryptographic protection. Each financial event receives a unique, sequentially linked entry that cannot be altered retroactively without breaking the chain. This gives management a reliable, verifiable record of every peso that moves through the business. Automated anomaly detection scans for suspicious patterns, an unusual concentration of voids from one cashier, a sudden spike in wastage entries, and discounts applied outside authorized hours and surfaces real-time alerts so the owner can investigate before small leaks compound into significant losses. CyberLedger transforms audit from a periodic, reactive exercise into a continuous, proactive capability. Management can perform random spot checks or scheduled audits at any time, confident that the underlying data has not been tampered with. This level of traceability directly addresses the accountability gap that currently leaves Dulci Cafe vulnerable to internal fraud.

Beyond the core modules, the study aims to deliver measurable operational improvements through business intelligence and process integration. The system generates actionable reports on sales performance by menu item and time period, inventory turnover rates, staff productivity metrics, and payment-method breakdowns that currently require hours of manual assembly and are still unreliable. Employee scheduling can be informed by transaction volume patterns rather than habit, aligning labor cost with actual revenue. Customer purchase history, captured automatically through the system, enables loyalty features that competing establishments already offer. The study also assesses how these integrated features reduce operational losses, enhance staff accountability, and support strategic decision-making by the cafe owner. Through the successful deployment of StockSecure POS with RBAC and CyberLedger, this study aims to set a new benchmark for technology adoption among small cafe businesses in Iloilo promoting sustainable growth, operational resilience, and long-term profitability while offering a scalable model for other establishments facing the same challenges across the region.

The StockSecure POS project also emphasizes the importance of user-centered design in software development. As students engage with the system, they learn to prioritize the needs and experiences of the end-users, baristas, cashiers, and managers when developing features and functionalities. This focus on user experience not only enhances the usability of the system but also fosters a positive work environment. By understanding how to create intuitive interfaces that reduce cognitive load and streamline operations, students are better equipped to design technology solutions that resonate with users. This knowledge is invaluable as they prepare for careers in software development, where user satisfaction is a key determinant of success.

Moreover, the experience gained from implementing StockSecure POS provides students with insights into the iterative nature of software development. Throughout the project, they learn the importance of gathering feedback from users and stakeholders to refine and improve the system continuously. This iterative process is essential in ensuring that the final product meets the evolving needs of the café and its employees. Students discover that effective communication with users is crucial for identifying pain points and areas for improvement, thereby enhancing their ability to deliver solutions that truly add value. This experience prepares them for dynamic work environments where adaptability and responsiveness to feedback are critical.

The project also highlights the significance of data security and privacy in the digital age. As students work with sensitive financial information and customer data, they gain a deeper understanding of the ethical responsibilities associated with handling such information. They learn about best practices for securing data, including encryption, access control, and compliance with relevant regulations. This awareness of data security is particularly relevant in today's landscape, where businesses face increasing scrutiny regarding their data handling practices. By understanding the implications of data privacy, students are better prepared to navigate the challenges associated with managing information in their future careers.

Additionally, the collaborative nature of the StockSecure POS project fosters a sense of community among BSIT students. Working together to develop a comprehensive solution encourages peer learning and knowledge sharing, as students bring diverse perspectives and skills to the table. This collaborative environment not only enhances their technical abilities but also builds essential interpersonal skills that are critical for success in the workforce. By engaging in teamwork and collaboration, students learn to appreciate the value of diverse viewpoints and the importance of collective problem-solving in achieving common goals.

Finally, the StockSecure POS project serves as a model for future initiatives aimed at modernizing small businesses in Iloilo and beyond. As students reflect on their experiences, they are inspired to consider how technology can address challenges faced by various industries. The skills and insights gained from this project empower them to pursue entrepreneurial ventures or contribute to technology-driven solutions in their communities. By applying the lessons learned from StockSecure POS, students are positioned to make a positive impact in the business landscape, driving innovation and promoting sustainable growth in the region. This forward-thinking mindset will enable them to become catalysts for change, shaping the future of technology adoption in small businesses and contributing to the overall economic development of their communities.

Specific Objectives

The primary goal of this study is to develop a **comprehensive, intuitive, and fully functional web-based point of sale system** tailored specifically for both frontline employees and administrators. This platform will feature a clean, modern interface with streamlined navigation, allowing staff to process transactions quickly and accurately while administrators manage backend operations with ease. By replacing outdated manual systems with this digital solution, the project aims to enhance overall workflow productivity, reduce training time for new employees, and create a more professional environment that improves both staff morale and customer perception of the business. The system will incorporate responsive design principles that ensure seamless performance across desktops, tablets, and mobile devices, enabling staff to move fluidly between different stations without friction. Intuitive touch-friendly controls, customizable shortcuts, and guided workflows will minimize cognitive load during busy periods, while contextual help prompts and interactive tutorials will accelerate onboarding for new hires, allowing them to become productive team members within hours rather than days.

A central component of the initiative involves building a sophisticated digital order processing system that completely eliminates common human errors such as miscommunication, incorrect pricing, or forgotten items, while dramatically increasing the speed and efficiency of customer service. This system will automate the entire order lifecycle from initial entry at the counter or table to preparation, payment, and fulfillment, incorporating smart validation rules and real-time synchronization between different departments. As a result, service times will decrease significantly, customer satisfaction will rise, and operational bottlenecks will be minimized, leading to higher throughput during peak hours. Orders will be instantly routed to the appropriate kitchen or preparation stations with visual indicators for urgency and special requests, while integrated kitchen display systems will update in real time to reflect modifications or cancellations. Payment processing will support multiple methods including cash, cards, mobile wallets, and contactless options, with automatic tax calculations, discounts, and split-bill functionality handled seamlessly to create a frictionless checkout experience that leaves customers impressed with the speed and professionalism of service.

The project will also establish a **robust real-time inventory tracking system** that continuously monitors stock levels, product usage, and movement across all areas of the business. Advanced algorithms will provide instant alerts for low-stock situations, predict potential shortages based on historical consumption patterns, and generate automated reorder suggestions. This level of visibility ensures that inventory remains optimally balanced, waste is reduced, overstocking is avoided, and purchasing decisions are always data-driven rather than based on guesswork. Every item sold or transferred will automatically deduct from inventory counts across multiple locations or storage areas, while barcode and RFID integration will enable quick scanning during receiving, sales, and stocktaking processes. The system will analyze trends such as daily, weekly, and seasonal demand variations to offer intelligent forecasting, helping managers maintain just the right amount of stock to satisfy customer demand without tying up excessive capital in storage.

To protect sensitive business and customer information, a comprehensive role-based access control system will be implemented, meticulously defining and enforcing permissions according to each user's position and responsibilities. Employees will only have access to the tools and data relevant to their roles, while managers and administrators enjoy broader visibility and control. This layered security approach not only prevents unauthorized access but also reduces the risk of internal data breaches, ensures regulatory compliance, and fosters a culture of accountability throughout the organization. Permissions will be granular and configurable, allowing administrators to create custom roles such as cashier, kitchen supervisor, inventory clerk, or financial auditor, with each role inheriting specific capabilities like transaction processing, report viewing, or system configuration. Multi-factor authentication, session timeouts, and IP-based access restrictions will add additional layers of protection to safeguard against both external threats and potential insider risks.

An **advanced audit logging system** will be integrated to record every user action within the platform with precise timestamps, user identification, and detailed descriptions of changes made. Whether it involves processing a sale, modifying inventory records, or accessing customer information, all activities will be securely captured. This creates a transparent and tamper-proof trail that business owners can review during investigations, performance evaluations, or compliance audits, significantly strengthening fraud prevention measures and internal governance. Logs will capture not only what was changed but also the previous and new values, providing complete before-and-after context that makes it easy to reconstruct events or identify patterns of suspicious behavior. Searchable and filterable log interfaces will allow quick retrieval of specific events, while automated anomaly detection will flag unusual activities such as unusually large discounts or frequent voids for immediate managerial review.

The entire solution will rest upon a meticulously designed secure database architecture engineered to safely store all sales transactions, user activities, financial records, and operational data. Utilizing industry-standard encryption protocols, regular backup mechanisms, and multi-layered security defenses, this database will safeguard information against unauthorized access, data corruption, or system failures. Its reliable structure will also enable fast querying and reporting, ensuring that historical data remains readily available for analysis and long-term business intelligence. Data will be stored using normalized relational structures combined with strategic indexing to maintain high performance even as transaction volumes grow over time, while encrypted connections and field-level encryption will protect sensitive fields such as customer payment details and personal information.

Business owners will gain powerful financial oversight tools that allow them to monitor sales trends, track expenses, analyze revenue streams, and generate comprehensive profit reports in real time. These features will transform complex raw data into clear visual dashboards and actionable insights, empowering leaders to make informed strategic decisions, identify cost-saving opportunities, optimize pricing strategies, and drive sustainable business growth with confidence. Interactive charts, graphs, and heat maps will illustrate performance across different time periods, product categories, and staff members, while comparative analytics will highlight year-over-year growth or underperforming areas that require attention. Custom report builders will let owners create tailored financial summaries that can be exported in multiple formats for meetings or tax preparation.

In addition, the system will incorporate a highly responsive **real-time stock monitoring subsystem** that provides business owners and managers with instant visibility into current inventory levels across all storage locations or branches. This capability will include predictive analytics for seasonal demand fluctuations and automatic notifications for critical stock thresholds. By maintaining precise control over inventory, the business can avoid lost sales due to stockouts, minimize carrying costs, and ensure consistent product availability that supports customer loyalty. Mobile-friendly alerts delivered via email, SMS, or in-app notifications will keep decision-makers informed even when they are away from the premises, while integration with supplier portals could eventually enable one-click reordering based on intelligent recommendations.

At the core of the security infrastructure lies the specially designed CyberLedger subsystem, which functions as a secure transaction and audit monitoring solution. CyberLedger will automatically record, encrypt, and preserve a complete, unalterable record of all financial transactions and user activities, establishing strong data integrity while creating a detailed forensic audit trail. This enables effective fraud detection, supports accountability across all levels of the organization and provides verifiable evidence for internal reviews or external audits when necessary. Using blockchain-inspired immutability techniques combined with traditional cryptographic signing, CyberLedger ensures that once data is written, it cannot be altered without leaving a clear trace, creating an extremely high level of trust in the system's records.

Furthermore, CyberLedger will feature **intelligent data lifecycle management** that automatically archives operational logs after a 30-day active retention period, followed by the secure deletion of active records to optimize system performance and storage usage. Encrypted backup copies will be maintained separately for extended tracking and retrieval purposes, ensuring compliance with data protection standards while balancing operational efficiency with the need for historical accountability. This automated approach prevents the system from becoming bogged down with years of accumulated data while still preserving the ability to retrieve older records when required for legal, financial, or investigative purposes.

To enhance long-term reliability, the system will include **built-in disaster recovery and redundancy features** that protect against hardware failures, cyberattacks, or unexpected data loss. Regular automated integrity checks and secure cloud synchronization options will guarantee continuous availability and quick restoration of services. This forward-thinking approach ensures the platform remains resilient, scalable, and capable of supporting business expansion without compromising security or performance. Failover mechanisms will allow the system to switch to backup servers with minimal downtime, while point-in-time recovery options will enable restoration to any previous healthy state if corruption or ransomware is ever detected.

Finally, the integrated solution will offer extensive customization options and user feedback mechanisms, allowing the system to evolve according to the specific needs of the business over time. Regular performance analytics, user activity summaries, and improvement recommendations will help administrators fine-tune operations continuously. This holistic design not only addresses immediate operational challenges but also positions the business for future growth through technology-driven efficiency, security, and data-informed decision-making. Users will be able to suggest interface improvements, request new features, or report issues directly through the platform, creating a continuous improvement loop that keeps the system aligned with changing business requirements and industry best practices.

Scope and Delimitation

The primary focus of this study is the development of StockSecure POS, a robust, fully offline web-based point-of-sale system tailored specifically for small to medium-sized café businesses. In an era where technology is rapidly evolving, the need for efficient and reliable systems in the café industry has become paramount. The StockSecure POS system aims to address these needs by providing a solution that is not only user-friendly but also capable of operating seamlessly within a café's local area network (LAN). This design ensures that café owners can modernize and streamline their daily operations without the dependency on constant internet connectivity, which can often be unreliable in many locations.

At the heart of StockSecure POS is its integration with CyberLedger, a secure local audit and transaction monitoring module that guarantees complete transparency and traceability of all business activities. This integration is crucial for building trust among café owners and their customers. By maintaining a detailed record of transactions and operational activities, the system fosters accountability and helps deter internal fraud. The research emphasizes the importance of a tamper-proof digital trail, which is vital for compliance with local regulations and for maintaining the integrity of business operations.

The core functionalities of StockSecure POS encompass comprehensive end-to-end order processing. From menu selection and customization to order fulfillment and billing, every step is designed to be intuitive and efficient. The system supports thermal receipt ticket printing, providing customers with detailed transaction breakdowns that enhance their overall experience. Real-time inventory management is another critical feature, allowing café owners to automatically deduct stock as sales occur, receive low-stock notifications, and track ingredients at a granular level for both beverages and food items. This level of detail not only aids in inventory control but also helps in minimizing waste and optimizing ingredient usage.

A sophisticated role-based access control (RBAC) mechanism is implemented within the system, defining clear privileges for different users. Café owners are granted administrative access, allowing them full oversight of the system, while baristas can manage order preparation and inventory updates. Cashiers, on the other hand, are limited to transaction handling and payment recording. This structured access ensures that sensitive information is protected while allowing staff to perform their roles effectively. Comprehensive audit logging captures every user action, including login attempts and modifications, creating a secure environment that supports accountability and transparency.

The payment module of StockSecure POS is designed to accommodate both traditional cash transactions and modern cashless options. For electronic payments, integration with **GCash** allows for the secure capture and recording of official reference numbers, facilitating accurate reconciliation during end-of-day closing. This feature is particularly advantageous for café owners as it simplifies the payment process and enhances customer satisfaction. The inclusion of business analytics features further empowers café owners by providing interactive dashboards and reports that cover various aspects of their operations, such as daily, weekly, and monthly sales performance, expense categorization, and profit margin analysis.

Despite its robust features, this study deliberately excludes several advanced and complex functionalities to maintain a sharp focus on delivering a stable, secure, and immediately implementable system for single-establishment use. For instance, cloud-based data synchronization and real-time remote access are not part of the project scope, as these features could introduce unnecessary complexity and potential security vulnerabilities. Additionally, the system will not integrate with external digital payment gateways beyond basic GCash reference capture, nor will it encompass advanced financial modules like automated tax calculations or payroll processing. This deliberate exclusion allows for a more streamlined development process, ensuring that the core functionalities are thoroughly tested and refined.

Furthermore, the system is strictly delimited to a single-establishment deployment using local database storage, such as MySQL or SQLite, on a dedicated workstation or server within the café premises. This design choice emphasizes the importance of operational resilience, as it enables the system to function effectively even during network outages or power fluctuations. By focusing on a single café environment, the research can delve deeper into the unique challenges faced by café owners, resulting in a high-quality, production-ready POS solution that directly addresses their needs.

In conclusion, by establishing clear delimitations, this research ensures a manageable and in-depth development process that prioritizes security, usability, and operational efficiency in typical café environments. The focused approach allows for thorough testing, refinement, and validation of core system features, ultimately delivering a practical, reliable, and secure solution that café owners can deploy immediately. StockSecure POS not only enhances operational efficiency but also empowers café owners to make informed business decisions, thereby contributing to the overall success and sustainability of their establishments.

1.5 Significance of the Study

The implementation of the StockSecure POS System with its integrated CyberLedger blockchain-inspired audit layer represents a transformative advancement in café management technology. By merging traditional point-of-sale functionalities with modern security and operational innovations, the system delivers substantial value to café owners, managers, employees, customers, and even regulatory bodies. In an industry characterized by tight margins, high staff turnover, and increasing demands for transparency, StockSecure addresses longstanding pain points through real-time monitoring, role-based access control, offline-first architecture, and immutable transaction logging. This holistic approach not only streamlines daily operations but also establishes a new standard for trust and accountability in small-to-medium food service businesses. The significance lies in its ability to bridge the gap between conventional POS systems and enterprise-grade solutions, making sophisticated technology accessible and practical for local café environments.

From an operational perspective, the system significantly enhances efficiency and reduces losses. Real-time inventory tracking prevents stock discrepancies and minimizes wastage, while automated reorder suggestions optimize supply chain management. The offline-first architecture ensures uninterrupted service during internet outages, a common challenge in many locations, allowing transactions to sync seamlessly once connectivity is restored. Role-based access control streamlines staff workflows by granting appropriate permissions, reducing errors caused by unauthorized actions. For café owners, these features translate into better cash flow management, accurate sales analytics, and data-driven decision-making. Managers benefit from comprehensive dashboards that provide instant insights into peak hours, popular items, and staff performance, enabling proactive adjustments that improve overall productivity and customer satisfaction. In a competitive market where operational leaks can determine business survival, StockSecure offers a practical framework for sustainable growth.

Security stands as one of the most critical contributions of this system. The CyberLedger audit layer, inspired by blockchain principles, creates an immutable record of every transaction, making tampering or fraudulent alterations virtually impossible. This feature is particularly significant in an era of rising cybersecurity threats and internal fraud risks within the hospitality sector. Every sale, refund, inventory adjustment, and user action is cryptographically secured and timestamped, providing a reliable audit trail for both internal reviews and external compliance requirements. Business owners gain peace of mind knowing their financial data is protected, while regulatory authorities can easily verify records during inspections. Customers indirectly benefit through increased trust in the establishment's integrity, knowing that their transactions are handled with the highest standards of transparency and accountability. This security dimension elevates the system beyond mere convenience into a strategic risk management tool.

The educational and human resource value of StockSecure further amplifies its significance. The system serves as a practical training platform for employees, particularly in smaller cafés where formal training programs are often limited. Through intuitive interfaces and clear audit logs, new staff members can quickly learn proper procedures, understand inventory management, and appreciate the importance of accurate record-keeping. Managers can use system-generated reports for performance evaluations and targeted coaching, fostering a culture of accountability and continuous improvement. For business education contexts, the platform offers real-world exposure to modern retail technologies, preparing students and young professionals for careers in an increasingly digital hospitality industry. This educational aspect transforms the POS system from a transactional tool into a developmental asset that builds long-term organizational capability.

Ultimately, the StockSecure POS System with CyberLedger contributes to the broader evolution of small business technology adoption. It demonstrates that advanced features like blockchain-inspired auditing are no longer exclusive to large corporations but can be effectively implemented in everyday café settings. By addressing operational, security, and educational needs simultaneously, the system supports economic resilience for small businesses while promoting higher standards of transparency across the food service industry. Its successful deployment could inspire similar innovations in other micro and small enterprises, driving digital transformation at the grassroots level. This study, therefore, holds both immediate practical significance and long-term strategic importance for modernizing traditional business models in a technology-driven economy.

Administrators & Managers

The implementation of StockSecure POS significantly empowers administrators and managers within café businesses by providing them with powerful oversight and control capabilities that enhance their ability to manage day-to-day operations effectively. This system is designed to streamline management processes, allowing leaders to focus on strategic decision-making rather than merely reacting to operational issues. The centralized dashboard serves as a command center, enabling managers to define granular permissions for each role and individual user, thus determining exactly what screens, functions, and data each employee can access or modify. This fine-grained access control minimizes unauthorized actions and reduces the risk of internal misuse, ensuring that sensitive information is protected and that employees can only perform tasks relevant to their roles.

For students in the Bachelor of Science in Information Technology (BSIT) program at Phinma UI, understanding the significance of such access control mechanisms is crucial. They learn how to design systems that incorporate security measures tailored to the specific needs of an organization. This knowledge is not only applicable to café operations but extends to various industries where data integrity and security are paramount. By engaging with these concepts, students develop a strong foundation in security best practices, preparing them for future roles in software development, cybersecurity, and systems administration.

Real-time activity monitoring is another critical feature that enhances managerial oversight. Through StockSecure POS, managers can view live employee actions, such as order processing, inventory adjustments, cash drawer openings, and refunds, directly from their mobile or desktop interface. This visibility enables immediate intervention when suspicious behavior is detected, allowing managers to address potential issues before they escalate. For students, learning about real-time monitoring systems provides insights into how technology can enhance operational transparency and accountability. They gain experience in implementing monitoring solutions that not only improve security but also foster a culture of trust and responsibility among staff.

The integration of the CyberLedger component further strengthens the management capabilities of StockSecure POS. CyberLedger maintains a complete, tamper-evident chronological record of every transaction and system event, creating an immutable audit trail that simplifies fraud detection and internal investigations. This feature is particularly valuable for administrators and managers who need to ensure compliance with regulatory requirements and maintain the integrity of their financial records. Students studying this aspect of the system learn about the importance of audit trails in business operations and how they can be leveraged to enhance accountability and transparency. This understanding is essential for future careers in finance, compliance, and risk management.

With complete, accurate, and real-time data on sales, inventory levels, supplier deliveries, wastage, and financial performance, administrators can make informed, data-driven decisions. The ability to analyze this data allows managers to identify slow-moving items, optimize stock replenishment schedules, and forecast demand more accurately. For BSIT students, this exposure to data analytics and business intelligence tools is invaluable. They learn how to harness data to drive operational improvements and enhance decision-making processes. This skill set is increasingly in demand in today’s data-driven business environment, where organizations rely on insights to remain competitive.

Furthermore, the ability to reduce overstocking or stockouts through effective inventory management contributes to healthier cash flow. Students involved in the StockSecure POS project gain insights into inventory optimization strategies that can significantly impact a café's profitability. By understanding how to analyze sales trends and adjust inventory levels accordingly, students are better prepared to contribute to the operational success of any business they join. This practical experience equips them with the tools necessary to make data-driven decisions that enhance operational efficiency and drive financial performance.

Overall, the StockSecure POS system transforms management from reactive problem-solving to proactive business optimization. For students, this shift in focus is an important lesson in the strategic role of technology in modern business operations. By learning how to implement systems that facilitate proactive decision-making, students develop a mindset that values foresight and innovation. This perspective is essential for future leaders in the technology sector, where the ability to anticipate challenges and seize opportunities can set organizations apart from their competitors.

The collaborative nature of the StockSecure POS project also fosters essential soft skills among BSIT students. Working in teams to develop and implement the system encourages communication, teamwork, and problem-solving. These skills are critical for success in any professional setting, as most projects require input from diverse disciplines and perspectives. By navigating group dynamics and project management, students build a foundation for successful collaboration in their future careers, preparing them for the multifaceted nature of work in the technology sector.

Additionally, the capstone-level focus of the StockSecure POS project encourages students to engage in critical thinking and ethical considerations surrounding technology. They are challenged to reflect on the broader implications of their work, such as how their designs can impact businesses and society. This holistic approach ensures that graduates are not only technically proficient but also socially responsible. They learn to consider the ethical ramifications of their decisions, preparing them to be conscientious professionals who prioritize the welfare of users and stakeholders. As they transition into the workforce, these competencies will distinguish them as leaders who can navigate the complexities of technology with both skill and integrity.

The study of StockSecure POS and CyberLedger provides BSIT students at Phinma University of Iloilo with a comprehensive education that balances theoretical concepts with practical application. The skills and experiences gained through these projects prepare them for diverse career paths in software development, cybersecurity, and enterprise application development. By bridging the gap between academic knowledge and real-world challenges, students emerge from the program equipped to tackle the complexities of the technology landscape. They are armed with the knowledge and skills necessary to succeed in their future endeavors, making them valuable assets to any organization they join.

Ultimately, the StockSecure POS project represents a culmination of learning experiences for BSIT students at Phinma UI, providing them with the tools, skills, and insights necessary to thrive in a competitive and rapidly changing technology landscape. By engaging with real-world challenges and developing innovative solutions, students are empowered to make meaningful contributions to their future workplaces and society at large. The lessons learned from this project will serve them well as they embark on their professional journeys, equipping them to navigate the complexities of the technology sector with confidence and competence.

In conclusion, the implementation of StockSecure POS not only benefits café administrators and managers by enhancing their oversight and control capabilities but also serves as a transformative learning experience for BSIT students. By engaging with the system's features, students develop a comprehensive understanding of how technology can optimize business operations, improve decision-making, and foster a culture of accountability. This knowledge prepares them to become innovative leaders in the technology sector, capable of driving positive change and contributing to the success of their future organizations.

End Users (Employees, Baristas, Cashiers)
For frontline employees such as baristas, cashiers, and service staff, the implementation of StockSecure POS represents a significant advancement in operational efficiency and employee satisfaction. This system is designed to be an intuitive, efficient, and supportive working tool that significantly reduces cognitive load and operational stress. By streamlining the order entry process and providing a user-friendly interface, StockSecure POS enhances the daily work experience of employees, allowing them to focus on delivering exceptional customer service.

The user interface of StockSecure POS is specifically crafted to guide staff through the order entry process seamlessly. With visual product selectors, automated modifiers, and recipe-based inventory deduction, employees can quickly and accurately input orders.

This design minimizes the likelihood of ordering errors, which can lead to customer dissatisfaction and operational delays. As a result, service times are accelerated, allowing cafés to handle high volumes of customers, especially during peak hours. For students at Phinma University of Iloilo studying this system, understanding the importance of user experience in software design is crucial. They learn how intuitive interfaces can enhance productivity and improve the overall service experience for both employees and customers.

Each employee is provided with their own secure, password-protected (or biometric-enabled) account, which comes with personalized access rights tailored to their specific roles. This individualized approach fosters a sense of accountability among staff, as all actions taken within the system are properly attributed to the respective employee. This not only encourages responsible behavior but also promotes professional growth. Students involved in the development of StockSecure POS gain insights into the significance of accountability in the workplace and how technology can facilitate a culture of responsibility. This understanding is particularly relevant for future roles in management and team leadership, where fostering accountability is essential for success.

Real-time stock visibility is another critical feature of StockSecure POS that benefits frontline employees. The system alerts staff when items are running low or are unavailable, which helps prevent disappointing customers with out-of-stock menu items. This proactive approach to inventory management reduces unnecessary preparation time for products that cannot be served, allowing employees to focus on providing quality service rather than managing inventory issues. For BSIT students, learning how to integrate real-time data into operational processes is an invaluable skill. They understand how access to accurate information can enhance decision-making and improve customer satisfaction.

Additional features of StockSecure POS, such as quick order templates and split billing options, further enhance the user experience for frontline employees. These functionalities allow staff to process orders more efficiently, catering to customer preferences and improving overall service quality. The ability to handle complex transactions, such as splitting bills among multiple customers, demonstrates the system's flexibility and adaptability to various service scenarios. By engaging with these features, students learn how technology can be leveraged to meet diverse customer needs, ultimately contributing to a more satisfying dining experience.

The offline functionality of StockSecure POS is particularly beneficial for cafés located in areas with unstable internet connections. This feature ensures that employees can continue serving customers seamlessly, even during internet outages. By minimizing disruptions to service, the system enhances operational resilience and protects the café's revenue stream. For students studying this aspect of the system, understanding the importance of reliability in technology solutions is critical. They learn how to design systems that can operate effectively in various environments, ensuring business continuity regardless of external factors.

Moreover, StockSecure POS minimizes the need for manual inventory counting and paperwork, allowing staff to devote more time to customer interaction and service quality. This reduction in administrative tasks not only alleviates operational stress but also enhances job satisfaction among employees. When staff can focus on building relationships with customers rather than being bogged down by paperwork, the overall atmosphere of the café improves. For BSIT students, this insight into the interplay between technology and employee experience is vital. They learn that effective technology solutions can lead to happier employees, which in turn can foster customer loyalty and repeat business.

As students engage with the StockSecure POS project, they also develop critical soft skills that are essential for their future careers. The collaborative nature of the project encourages teamwork, communication, and problem-solving. Working alongside peers to tackle complex challenges simulates the collaborative environments they will encounter in their professional lives. This experience teaches them how to effectively communicate their ideas, negotiate solutions, and manage conflicts, all of which are vital skills in the tech industry. By navigating group dynamics and project management, students build a foundation for successful collaboration in their future careers.

The capstone-level focus of the StockSecure POS project encourages students to engage in critical thinking and ethical considerations surrounding technology. They are challenged to reflect on the broader implications of their work, such as how their designs can impact businesses and society. This holistic approach ensures that graduates are not only technically proficient but also socially responsible. They learn to consider the ethical ramifications of their decisions, preparing them to be conscientious professionals who prioritize the welfare of users and stakeholders. As they transition into the workforce, these competencies will distinguish them as leaders who can navigate the complexities of technology with both skill and integrity.

The study of StockSecure POS and CyberLedger provides BSIT students at Phinma University of Iloilo with a comprehensive education that balances theoretical concepts with practical application. The skills and experiences gained through these projects prepare them for diverse career paths in software development, cybersecurity, and enterprise application development. By bridging the gap between academic knowledge and real-world challenges, students emerge from the program equipped to tackle the complexities of the technology landscape. They are armed with the knowledge and skills necessary to succeed in their future endeavors, making them valuable assets to any organization they join.

Ultimately, the StockSecure POS project represents a culmination of learning experiences for BSIT students at Phinma UI, providing them with the tools, skills, and insights necessary to thrive in a competitive and rapidly changing technology landscape. By engaging with real-world challenges and developing innovative solutions, students are empowered to make meaningful contributions to their future workplaces and society at large. The lessons learned from this project will serve them well as they embark on their professional journeys, equipping them to navigate the complexities of the technology sector with confidence and competence.

In conclusion, StockSecure POS serves as a transformative tool for frontline employees, enhancing their work experience while simultaneously improving operational efficiency within the café business. For BSIT students, the opportunity to study and engage with such a comprehensive system provides them with invaluable insights into the intersection of technology, employee experience, and customer satisfaction. By understanding how to design and implement user-friendly systems that support staff and streamline operations, students are well-prepared to enter the workforce as knowledgeable, skilled, and responsible professionals ready to contribute to the future of technology in various industries.

Organization (Café Business)

The café business landscape is rapidly evolving, and the integration of robust technology solutions like StockSecure POS, which incorporates Role-Based Access Control (RBAC) and CyberLedger, is transforming traditional operations into modern, efficient, and transparent enterprises. For students in the Bachelor of Science in Information Technology (BSIT) program at Phinma UI, studying this system provides a wealth of knowledge and practical skills essential to their future careers in technology. The benefits of this project extend beyond the immediate application to café businesses; they also significantly enhance students' educational experience, preparing them for the challenges of the workforce.

One of the primary advantages of implementing StockSecure POS in a café setting is its ability to minimize internal fraud. Internal fraud can take many forms, including theft, sweetheart deals, and inventory pilferage, all of which can erode profit margins. Through comprehensive audit logging, role-based restrictions, and real-time alerts, the StockSecure POS system creates a secure operational environment that protects the café's financial health. Students involved in this project learn how to design and implement these security measures, gaining insights into the importance of protecting business assets. This knowledge is invaluable, as it equips them with the skills necessary to create secure systems in any industry, making them highly desirable candidates for future employment.

The implementation of audit logging within StockSecure POS allows café owners to maintain a transparent record of all transactions and user activities. Students gain firsthand experience in developing systems that capture detailed information about user actions, including timestamps, user identifiers, and action outcomes. This exposure not only enhances their technical skills but also instills a sense of responsibility regarding data management and security. Understanding the significance of maintaining accurate records prepares students for roles where accountability and transparency are paramount, particularly in fields such as finance, healthcare, and technology.

Role-Based Access Control (RBAC) is another critical component of StockSecure POS that students explore. By learning how to define user roles and permissions, students understand how to restrict access to sensitive data and functionalities based on an individual's job responsibilities. This practical experience is particularly relevant in today's digital landscape, where data breaches can have devastating consequences. Students who master RBAC are well-equipped to design secure applications that protect both businesses and customers from unauthorized access. This skill set is increasingly sought after by employers, as organizations prioritize the security of their information systems.
In addition to enhancing security, StockSecure POS optimizes inventory utilization, which is vital for the operational efficiency of any café. Students learn how automated stock deduction, low-stock notifications, expiration tracking, and intelligent reordering suggestions can lead to reduced wastage and lower carrying costs. This knowledge is crucial for understanding how technology can streamline operations and improve cash flow. By engaging with these concepts, students develop a keen awareness of how effective inventory management can impact a business's bottom line. This understanding prepares them for roles in supply chain management, logistics, and operations, where inventory control is a key focus.

The ability to improve customer service quality is another significant benefit of implementing StockSecure POS. Through faster order processing, fewer errors, and consistent product availability, cafés can enhance customer satisfaction and encourage repeat business. Students involved in this project learn how to design user-friendly interfaces that facilitate quick and accurate transactions. By focusing on user experience, students develop applications that not only meet business needs but also delight customers. This emphasis on customer-centric design is essential in today's competitive market, where businesses must differentiate themselves through exceptional service.

The integration of modern technology within the café also demonstrates a commitment to sustainable and professional business practices. For students, this aspect of the project highlights the importance of using technology to drive positive change within organizations. By understanding how StockSecure POS helps cafés establish a foundation for scalable growth, students learn about the strategic role of technology in business development. This knowledge is particularly relevant for aspiring entrepreneurs who seek to leverage technology to create innovative solutions and drive growth in their own ventures.
Compliance with tax and regulatory requirements is another critical area where StockSecure POS excels. The system's detailed digital records make it easier for café owners to maintain compliance, reducing the risk of costly penalties or audits. Students gain insights into the importance of regulatory compliance in business operations and how technology can simplify this process. This understanding prepares them for careers in industries where compliance is a significant concern, such as finance, healthcare, and manufacturing. By mastering these concepts, students position themselves as knowledgeable professionals who can navigate the complexities of regulatory environments.

The credibility that comes with implementing a secure and efficient system like StockSecure POS enhances a café's reputation with investors, partners, and financial institutions. Students learn how technology can be leveraged to build trust and foster relationships within the business community. This knowledge is invaluable for those pursuing careers in business development, finance, or entrepreneurship. By understanding the importance of credibility in business, students are better equipped to advocate for their ideas and secure the resources needed to bring them to fruition.
As students engage with the StockSecure POS project, they also develop essential soft skills that are critical for success in the workplace. The collaborative nature of the project encourages teamwork, communication, and problem-solving. By working in groups to tackle complex challenges, students learn how to effectively communicate their ideas, negotiate solutions, and manage conflicts. These skills are vital in any professional setting, as most projects require input from diverse disciplines and perspectives. By navigating group dynamics and project management, students build a foundation for successful collaboration in their future careers.

The capstone-level focus of the StockSecure POS project encourages students to engage in critical thinking and ethical considerations surrounding technology. They are challenged to reflect on the broader implications of their work, such as how their designs can impact businesses and society. This holistic approach ensures that graduates are not only technically proficient but also socially responsible. They learn to consider the ethical ramifications of their decisions, preparing them to be conscientious professionals who prioritize the welfare of users and stakeholders. As they transition into the workforce, these competencies will distinguish them as leaders who can navigate the complexities of technology with both skill and integrity.

The study of StockSecure POS and CyberLedger also provides students with a solid reference framework and empirical foundation for future academic and industry research. The documentation, architecture diagrams, source code patterns, and evaluation results from this project can serve as baseline references, benchmarks, or comparative studies for subsequent research in information systems, cybersecurity, business technology, and digital transformation. This foundational knowledge empowers students to contribute to ongoing discussions and innovations in these fields, positioning them as informed participants in the academic and professional communities.

Students may also investigate a variety of topics stemming from the StockSecure POS project, such as the effectiveness of offline POS systems in unstable internet environments and automated audit retention policies. These areas of inquiry are particularly relevant for

SMEs in the retail and food service sectors, where operational efficiency and security are paramount. By engaging with these research topics, students can explore the practical implications of their work and contribute to the advancement of knowledge in the field. This not only enhances their understanding but also builds their credibility as emerging professionals who can engage with contemporary issues in technology.

Furthermore, the project encourages user experience studies on POS systems for service industry workers. Understanding the end-user perspective is crucial for the success of any technology, and students learn to prioritize usability in their designs. By considering the needs and preferences of service industry workers, students develop applications that are not only functional but also intuitive and user-friendly. This focus on user experience is essential for creating systems that enhance productivity and satisfaction, ultimately leading to better business outcomes.

The integration of inventory management with financial auditing in small business contexts is another critical area that students can explore through this project. By understanding how these two domains intersect, students learn to design solutions that streamline operations and enhance financial transparency. This knowledge is particularly valuable for SMEs, which often face challenges in managing inventory and ensuring accurate financial reporting. By addressing these challenges, students contribute to the long-term sustainability and profitability of businesses, positioning themselves as valuable assets in the workforce.

The practical implications of the StockSecure POS project extend beyond the classroom, as students gain insights into the realities of working in the technology sector. They learn about the challenges faced by SMEs, including limited resources and the need for cost-effective solutions. This understanding fosters a mindset of innovation and resourcefulness, encouraging students to think creatively about how to address real-world problems. By developing solutions that are not only technically sound but also economically viable, students prepare themselves for the demands of the job market, where employers seek individuals who can deliver value in a competitive environment.

Additionally, the project serves as a catalyst for fostering entrepreneurship among students. By engaging with the challenges faced by SMEs, students are inspired to think about how they can leverage their skills to create their own ventures or contribute to startups. The knowledge gained from working on StockSecure POS and CyberLedger equips them with the tools necessary to identify opportunities, develop business plans, and implement technology solutions that drive growth and innovation. This entrepreneurial mindset is essential for navigating the rapidly changing landscape of technology and business, where adaptability and creativity are crucial for success.

Networking and relationship-building are additional benefits of participating in the StockSecure POS project. Through presentations, workshops, and discussions, students have the opportunity to connect with industry professionals and experts in the field. These connections can lead to internships, job opportunities, and mentorship relationships that are invaluable as students transition from academia to the workforce. By fostering a sense of community and collaboration, the BSIT program at Phinma University of Iloilo enhances the overall educational experience, preparing students for the realities of working in a globalized and interconnected world.

Ultimately, the study of StockSecure POS and CyberLedger provides BSIT students at Phinma University of Iloilo with a comprehensive education that balances theoretical concepts with practical application. The skills and experiences gained through these projects prepare them for diverse career paths in software development, cybersecurity, and enterprise application development. By bridging the gap between academic knowledge and real-world challenges, students emerge from the program equipped to tackle the complexities of the technology landscape. They are armed with the knowledge and skills necessary to succeed in their future endeavors, making them valuable assets to any organization they join.

This educational journey not only enhances their technical capabilities but also shapes them into responsible and innovative professionals ready to contribute to the ever-evolving field of information technology. The StockSecure POS project sets a new standard for technology adoption in small-scale retail and food service operations, promoting long-term sustainability, profitability, and professional growth for both students and businesses alike. The integration of RBAC and CyberLedger in this project exemplifies the potential of technology to transform operations, enhance security, and drive efficiency, ultimately benefiting all stakeholders involved. As students reflect on their experiences, they are not only prepared to enter the workforce but also motivated to become leaders and innovators in the technology sector, shaping the future of business and information systems.

In summary, the StockSecure POS project represents a culmination of learning experiences for BSIT students at Phinma UI, providing them with the tools, skills, and insights necessary to thrive in a competitive and rapidly changing technology landscape. By engaging with real-world challenges and developing innovative solutions, students are empowered to make meaningful contributions to their future workplaces and society at large. The lessons learned from this project will serve them well as they embark on their professional journeys, equipping them to navigate the complexities of the technology sector with confidence and competence.

BSIT Students

The Bachelor of Science in Information Technology (BSIT) program at Phinma University of Iloilo emphasizes practical learning experiences that are crucial for bridging theoretical knowledge with real-world applications. One of the standout projects in this curriculum is the development of StockSecure POS, which integrates Role-Based Access Control (RBAC) into a Point of Sale (POS) system. This project serves as a vital educational tool, allowing students to engage deeply with concepts of security, data management, and system design. By working with RBAC, students learn how to implement security measures that restrict access to sensitive information based on user roles, thereby preventing unauthorized access and potential internal fraud. This experience is particularly relevant in today's business environments where data breaches and fraud can have devastating financial and reputational consequences.

Through the StockSecure POS project, students are exposed to essential database security practices, including encryption and secure querying techniques. These skills are critical in protecting sensitive customer and transactional data from cyber threats. Students engage with real-world scenarios where they must consider the implications of data breaches, leading to a strong understanding of the importance of data integrity and confidentiality. The hands-on experience with these technologies not only enhances their technical skills but also instills a sense of responsibility regarding data management and security.

This knowledge is particularly beneficial for students as they prepare to enter an industry where cyber threats are increasingly sophisticated and prevalent. In addition to StockSecure POS, the integration of CyberLedger as a secure transaction and audit monitoring system enriches the educational experience for BSIT students. CyberLedger introduces students to innovative blockchain-inspired ledger systems, which are gaining traction for their ability to provide transparent and immutable records of transactions. This aspect of the project is particularly engaging for students, as it allows them to explore cutting-edge technologies that are reshaping industries such as finance, supply chain, and healthcare. By understanding the principles behind these systems, students gain insights into how to design applications that ensure audit integrity and facilitate secure transactions. This not only broadens their technical skill set but also prepares them for emerging roles in technology that require knowledge of blockchain and distributed ledger technologies.

Moreover, the collaborative nature of the StockSecure POS and CyberLedger projects fosters essential soft skills such as teamwork, communication, and problem-solving. Students work in groups to tackle complex challenges, simulating the collaborative environments they will encounter in their future careers. This experience teaches them how to effectively communicate their ideas, negotiate solutions, and manage conflicts, all of which are vital skills in the tech industry. The ability to work well in teams is increasingly important, as many projects require input from diverse disciplines and perspectives. By navigating group dynamics and project management, students build a foundation for successful collaboration in their professional lives.

The capstone-level focus of these projects encourages students to engage in critical thinking and ethical considerations surrounding technology. They are challenged to reflect on the broader implications of their work, such as how their designs can impact businesses and society. This holistic approach ensures that graduates are not only technically proficient but also socially responsible. They learn to consider the ethical ramifications of their decisions, preparing them to be conscientious professionals who prioritize the welfare of users and stakeholders. As they transition into the workforce, these competencies will distinguish them as leaders who can navigate the complexities of technology with both skill and integrity.

Furthermore, the emphasis on testing strategies and deployment considerations equips students with practical knowledge essential for bringing software applications to market. They learn about various testing methodologies, including unit testing, integration testing, and user acceptance testing, which are crucial for ensuring that applications function correctly and meet user needs. This practical experience prepares students for the rigorous demands of the software development lifecycle, enabling them to contribute effectively to projects from conception through to deployment and maintenance. Understanding these processes is vital for any IT professional, as it allows them to deliver high-quality software solutions that meet business requirements and user expectations.

Ultimately, the study of StockSecure POS and CyberLedger provides BSIT students at Phinma University of Iloilo with a comprehensive education that balances theoretical concepts with practical application. The skills and experiences gained through these projects prepare them for diverse career paths in software development, cybersecurity, and enterprise application development. By bridging the gap between academic knowledge and real-world challenges, students emerge from the program equipped to tackle the complexities of the technology landscape. They are armed with the knowledge and skills necessary to succeed in their future endeavors, making them valuable assets to any organization they join. This educational journey not only enhances their technical capabilities but also shapes them into responsible and innovative professionals ready to contribute to the ever-evolving field of information technology.

Future Researchers

The Bachelor of Science in Information Technology (BSIT) program at Phinma University of Iloilo offers students a unique opportunity to engage in projects that not only enhance their academic understanding but also prepare them for future careers in technology. One such project is the development of StockSecure POS, which integrates Role-Based Access Control (RBAC) into a Point of Sale (POS) system, providing a comprehensive solution for inventory monitoring and internal fraud prevention. This project serves as a significant learning platform for students, allowing them to explore critical aspects of software development, security, and system design. By working on StockSecure POS, students gain hands-on experience that is directly applicable to real-world scenarios, particularly in the context of small and medium enterprises (SMEs) in the retail and food service sectors.

The design and implementation of StockSecure POS allow students to delve into the intricacies of RBAC, a crucial component of modern security practices. Through this project, students learn how to define user roles and permissions, ensuring that access to sensitive data and functionalities is restricted based on the user's role within the organization. This practical experience is invaluable, as it equips students with the skills necessary to implement robust security measures in various applications. The importance of RBAC cannot be overstated, especially in environments where data breaches can lead to significant financial losses and reputational damage. By understanding how to effectively implement RBAC, students position themselves as knowledgeable professionals who can contribute to the security and integrity of information systems.

In addition to RBAC, the StockSecure POS project emphasizes the significance of database security measures, such as encryption and secure querying techniques. Students engage with these concepts through practical applications, learning how to protect sensitive customer and transactional data from unauthorized access. This exposure not only enhances their technical skills but also fosters a deep understanding of the ethical responsibilities that come with handling sensitive information. As they navigate the complexities of data security, students develop a strong foundation in best practices that are essential for any IT professional. This knowledge is particularly relevant in today's digital landscape, where cyber threats are becoming increasingly sophisticated and prevalent.

The integration of CyberLedger as a secure transaction and audit monitoring system further enriches the educational experience for BSIT students. CyberLedger introduces students to innovative blockchain-inspired ledger systems, which are gaining traction for their ability to provide transparent and immutable records of transactions. By exploring the principles behind these systems, students gain insights into how to design applications that ensure audit integrity and facilitate secure transactions. This knowledge is not only technically enriching but also prepares students for emerging roles in technology that require expertise in blockchain and distributed ledger technologies. The hands-on experience with CyberLedger allows students to appreciate the practical implications of these technologies in real-world business scenarios, enhancing their problem-solving capabilities.

Moreover, the collaborative nature of the StockSecure POS and CyberLedger projects fosters essential soft skills such as teamwork, communication, and problem-solving. Working in groups to tackle complex challenges simulates the collaborative environments students will encounter in their future careers. This experience teaches them how to effectively communicate their ideas, negotiate solutions, and manage conflicts, all of which are vital skills in the tech industry. The ability to work well in teams is increasingly important, as many projects require input from diverse disciplines and perspectives. By navigating group dynamics and project management, students build a foundation for successful collaboration in their professional lives.

The capstone-level focus of these projects encourages students to engage in critical thinking and ethical considerations surrounding technology. They are challenged to reflect on the broader implications of their work, such as how their designs can impact businesses and society. This holistic approach ensures that graduates are not only technically proficient but also socially responsible. They learn to consider the ethical ramifications of their decisions, preparing them to be conscientious professionals who prioritize the welfare of users and stakeholders. As they transition into the workforce, these competencies will distinguish them as leaders who can navigate the complexities of technology with both skill and integrity.

The study of StockSecure POS and CyberLedger also provides students with a solid reference framework and empirical foundation for future academic and industry research. The documentation, architecture diagrams, source code patterns, and evaluation results from this project can serve as baseline references, benchmarks, or comparative studies for subsequent research in information systems, cybersecurity, business technology, and digital transformation. This foundational knowledge empowers students to contribute to ongoing discussions and innovations in these fields, positioning them as informed participants in the academic and professional communities.

Researchers may investigate a variety of topics stemming from the StockSecure POS project, such as the effectiveness of offline POS systems in unstable internet environments, automated audit retention policies, and their impact on fraud prevention. These areas of inquiry are particularly relevant for SMEs in the retail and food service sectors, where operational efficiency and security are paramount. By engaging with these research topics, students can explore the practical implications of their work and contribute to the advancement of knowledge in the field. This not only enhances their understanding but also builds their credibility as emerging professionals who can engage with contemporary issues in technology.

Furthermore, the project encourages user experience studies on POS systems for service industry workers. Understanding the end-user perspective is crucial for the success of any technology, and students learn to prioritize usability in their designs. By considering the needs and preferences of service industry workers, students develop applications that are not only functional but also intuitive and user-friendly. This focus on user experience is essential for creating systems that enhance productivity and satisfaction, ultimately leading to better business outcomes.

The integration of inventory management with financial auditing in small business contexts is another critical area that students can explore through this project. By understanding how these two domains intersect, students learn to design solutions that streamline operations and enhance financial transparency. This knowledge is particularly valuable for SMEs, which often face challenges in managing inventory and ensuring accurate financial reporting.By addressing these challenges, students contribute to the long-term sustainability and profitability of businesses, positioning themselves as valuable assets in the workforce.

Ultimately, the study of StockSecure POS and CyberLedger provides BSIT students at Phinma University of Iloilo with a comprehensive education that balances theoretical concepts with practical application. The skills and experiences gained through these projects prepare them for diverse career paths in software development, cybersecurity, and enterprise application development. By bridging the gap between academic knowledge and real-world challenges, students emerge from the program equipped to tackle the complexities of the technology landscape. They are armed with the knowledge and skills necessary to succeed in their future endeavors, making them valuable assets to any organization they join. This educational journey not only enhances their technical capabilities but also shapes them into responsible and innovative professionals ready to contribute to the ever-evolving field of information technology. The StockSecure POS project sets a new standard for technology adoption in small-scale retail and food service operations, promoting long-term sustainability, profitability, and professional growth for both students and businesses alike.

Definition of Terms

In the development and implementation of StockSecure POS, a robust point-of-sale system tailored for small to medium-sized café businesses, it is essential to establish a clear understanding of key terms and concepts that underpin its functionality. This section provides precise definitions of critical components and features integral to the system, ensuring that all stakeholders have a comprehensive grasp of the terminology involved. By elucidating these terms, we aim to enhance clarity and facilitate effective communication among developers, café owners, staff, and users, thereby contributing to the successful deployment and operation of the StockSecure POS system.

“Audit Logging” is the systematic and automatic process of recording comprehensive, detailed information about every single user activity, transaction, system change, and operation that takes place within the StockSecure POS environment. This process is crucial for maintaining transparency and accountability within the café's operations. Audit logging captures a wide array of data, including the user’s unique identifier (such as their employee name, ID number, or login username), a precise description of the exact action performed, the full timestamp (including date, hour, minute, and second), the specific device or station where the action occurred, the outcome of the action, and any additional notes or reasons the user may have entered.

For example, if a cashier decides to void an order of two cappuccinos, one blueberry muffin, and a chocolate croissant because the customer changed their mind at the last minute, the audit log will permanently record the cashier’s name, the exact time of the void, the original total amount of the sale, the reason selected from a dropdown menu (such as “Customer Changed Mind”), and even the items that were removed from the transaction. This creates a complete, organized, and tamper-resistant digital trail that business owners and managers can easily search, filter, and review at any time. Audit logging is essential for maintaining strong accountability among staff, detecting unusual or suspicious patterns (like too many voids during a single shift), supporting thorough fraud investigations, assisting with government tax audits, resolving customer complaints or disputes fairly, training new employees by showing real examples of correct procedures, and helping small café owners gain deeper insights into their daily operations, peak hours, and staff performance.

“CyberLedger” is the powerful, built-in transaction security and audit logging subsystem specially developed as the secure core and protective backbone of StockSecure POS. It functions like an unbreakable, intelligent digital vault that carefully captures, encrypts, stores, processes, and safeguards every payment transaction record along with all operational activities inside a cryptographically protected local database space. Every action is secured using advanced encryption techniques, digital signatures, and checksum validations, making it extremely difficult and easily detectable for anyone to alter, delete, or fake records without leaving clear evidence of tampering.

For instance, when a customer pays for a full breakfast set consisting of avocado toast with poached eggs, fresh orange juice, a butter croissant, and a flat white coffee using a contactless debit card during a busy morning, CyberLedger instantly and securely records the full transaction details, the cashier involved, the exact time down to the second, the payment method, and any applied taxes or discounts and automatically links it to the corresponding real-time inventory deductions. This secure storage not only helps café owners maintain highly trustworthy records for daily accounting, monthly financial reporting, legal compliance, insurance claims in case of disputes, or internal performance reviews, but also builds long-term confidence that their business data remains protected and reliable even when operating entirely within a small local network environment without constant cloud dependency. The sophisticated architecture of CyberLedger ensures that every piece of data is not only stored securely but is also easily retrievable for audits or analyses, enhancing operational efficiency.

“Internal Fraud” refers to any dishonest, unauthorized, unethical, or illegal actions carried out by employees or other insiders within the café business, typically motivated by personal gain or to cover up mistakes. This can include physically stealing inventory by quietly taking items off the shelf without recording them as sold, manipulating or deleting sales records to pocket cash from the drawer, secretly granting large unauthorized discounts or completely free items to friends and family members, performing fake voids or refunds after a customer has already paid, and then keeping the money, or improperly accessing and viewing sensitive customer information such as phone numbers, email addresses, or previous order history.

In a real-life café scenario, an employee working the evening shift might remove several fresh pastries and two bottles of premium cold brew from the display case without logging any sale and either consume them personally or sell them informally to acquaintances outside. StockSecure POS actively combats internal fraud by combining powerful real-time monitoring tools, highly detailed CyberLedger audit trails that cannot be easily erased, strict role-based permissions, and automatic alerts for suspicious activities such as one staff member performing an unusually high number of voids or discounts during their shift allowing owners to quickly spot red flags, investigate fairly using concrete evidence, and take appropriate corrective actions like additional training, warnings, or role adjustments before small losses turn into significant financial damage. The proactive measures incorporated into StockSecure POS ensures that café owners can maintain a secure and trustworthy environment for both their staff and customers.

“Inventory Monitoring” is the ongoing, careful, and automated process of continuously tracking stock levels, ingredient consumption rates, product availability, and usage patterns throughout the entire café operation. It provides managers and owners with an accurate, always-current picture of exactly how much of each item, ranging from whole coffee beans, fresh milk, and syrup flavors to disposable cups, straws, pastries, sandwiches, and cleaning supplies, remains available at any given moment. The system automatically records usage as each sale is completed and can generate timely low-stock alerts via on-screen notifications or simple reports.

For example, if the café experiences a strong afternoon rush and sells 65 iced caramel lattes in just two hours, the inventory monitoring feature will instantly reduce the recorded stock of milk, espresso beans, caramel syrup, ice cubes, and plastic cups accordingly. It may then automatically send a low-stock alert to the manager’s dashboard when the milk supply drops below 8 liters or when only 15 packs of coffee beans remain, prompting a timely reorder. This smart monitoring helps small café businesses prevent embarrassing stockouts during peak customer hours, significantly reduce food and ingredient waste from over-ordering or spoilage, make smarter and more data-driven purchasing decisions based on actual sales trends, maintain consistently high service quality, and ultimately improve profitability by keeping popular menu items reliably available. The insights gained from inventory monitoring can also inform menu adjustments and promotional strategies, enabling café owners to optimize their offerings and better serve their customers.

“An offline web application” is a modern, reliable browser-based software solution that operates efficiently and fully without requiring a constant or active internet connection, making it highly practical for real-world business environments. It runs smoothly using a dedicated local server computer and a secure internal database entirely within the café’s own private network, so all essential functions continue without disruption even during internet outages. In practical situations, this means that during a heavy rainstorm that cuts off Wi-Fi service or a temporary internet provider maintenance period, baristas and cashiers can still quickly enter detailed customer orders, process cash payments and offline-supported card transactions, update inventory records in real time, print professional receipts, and manage basic daily operations without losing any sales or frustrating waiting customers.

Once internet access is restored, the system can safely and securely synchronize any necessary data with external services if required for reporting or backups. This robust offline capability is particularly valuable for small cafés located in suburban areas, rural towns, or buildings with unstable connectivity, ensuring uninterrupted business continuity, protecting daily revenue, and preventing stressful situations where staff would otherwise have to resort to manual paper-based methods. The offline functionality of StockSecure POS reinforces the café’s resilience against external disruptions, allowing staff to focus on providing excellent service regardless of technological challenges.

“A Point-of-Sale System (POS)” is the central, user-friendly computer interface or touchscreen workstation that serves as the main hub where all customer sales transactions are handled quickly, accurately, and professionally in a café setting. It acts as the modern, digital replacement for the old-fashioned cash register and integrates seamlessly with other critical modules such as inventory tracking, audit logging, and security features. In StockSecure POS, a barista can easily and intuitively select multiple menu items, for example, a “Large Iced Mocha,” a “Ham and Cheese Croissant,” and a “Strawberry Smoothie” customize them with special requests like extra shots or no sugar, calculate the total with applicable taxes, accept various payment methods, including cash, credit cards, debit cards, or mobile wallets, and complete the entire sale within seconds.

The system also automatically connects to inventory and audit functions, creating a smooth, error-reducing workflow that significantly speeds up service during hectic breakfast or lunch rushes, minimizes common human mistakes like incorrect pricing, and provides a pleasant, efficient experience for both staff and customers. The intuitive interface of StockSecure POS is designed to minimize training time for new employees, allowing them to become proficient quickly and contribute positively to the café's operations. Additionally, the integration of various payment methods ensures that customers have a seamless checkout experience, enhancing overall satisfaction.

“Real-time inventory management” means that stock quantities, ingredient levels, and product availability are updated instantly and automatically the very moment any transaction is successfully completed, ensuring the system always reflects the most current and accurate information with virtually no delay. A practical everyday example in a busy café would be a group of customers purchasing a large order that includes three Americanos, two chocolate chip muffins, and four bottles of sparkling water immediately after payment is confirmed, the system deducts the precise amounts of coffee beans, hot water, milk (if added), muffin stock, and bottled water inventory from the database.

This real-time accuracy empowers managers to check current stock levels on their dashboard, mobile device, or computer at any time of day, quickly place reorders for fast-selling items before they run out, identify slow-moving products that might benefit from special promotions or menu adjustments, reduce overstocking costs, prevent waste from expired ingredients, and make confident, data-backed business decisions that enhance overall café profitability and customer satisfaction. The ability to monitor inventory in real time also allows café owners to respond proactively to trends and customer preferences, ensuring that they can adapt their offerings to meet demand effectively.

“Receipt printing” is the fast, automated process of generating and physically printing a clear, neat, and professional transaction document for every customer immediately after a sale is finalized. The printed receipt usually contains rich details such as the café’s full name, address, phone number, and logo, the exact date and precise time of the purchase, a complete itemized list of all products bought with their individual prices, any discounts, promotions, or special offers applied, the subtotal, tax breakdown, final total amount, the payment method used, a unique transaction reference number, and often a warm thank-you message, loyalty program points earned, or an invitation to return soon.

For instance, after a regular customer buys a slice of fresh carrot cake, a medium cappuccino with an extra shot, and a takeaway sandwich for later, they receive a nicely printed receipt that not only serves as their proof of purchase for any future returns or exchanges but also helps the café maintain high professional standards and build customer trust. All receipts are simultaneously stored digitally and securely within the CyberLedger system for easy future reference, accounting reconciliation, or detailed sales analysis. The professional presentation of receipts can also enhance the café's branding and customer experience, reinforcing a sense of quality and reliability.

“A Role-Based Access Control System” is an intelligent and flexible security framework that carefully restricts system permissions, features, and actions according to each employee’s specific job role and responsibilities, ensuring that individuals can only access and perform the tasks directly necessary for their position while being blocked from everything else. This greatly minimizes risks from both accidental errors and intentional misuse. For example, a part-time cashier or barista may be permitted to log into the system, take and modify customer orders, and process standard payments, but they are strictly prevented from approving large discounts over $10, viewing full daily or monthly financial reports, or making any changes to audit logs.

A shift supervisor might receive additional rights to issue moderate discounts, perform certain inventory adjustments, and view basic sales summaries, whereas only the café owner or senior manager has complete full access to all sensitive reports, system configuration settings, user management tools, and detailed audit log reviews. By carefully implementing these role-based rules, StockSecure POS significantly reduces opportunities for internal fraud, supports better teamwork and accountability, helps with staff training by clearly defining boundaries, and creates a more secure, well-organized, and professional working environment overall. This structured approach to access control not only enhances security but also fosters a culture of trust and responsibility among staff.

“StockSecure POS” is a complete, modern, user-friendly, and specially developed point-of-sale solution created from the ground up to help small cafes, coffee shops, and similar foodservice businesses run their daily operations more efficiently, securely, and profitably. It intelligently brings together a wide range of powerful yet easy-to-use tools, including fast and intuitive order processing, precise real-time inventory tracking, advanced data protection through the CyberLedger subsystem, comprehensive audit logging for full transparency, smart role-based access controls, dependable offline functionality that works without the internet, and high-quality receipt printing, all working together seamlessly in one integrated platform.

In a typical busy morning rush at a small neighborhood café, multiple staff members can rapidly serve dozens of customers, automatically track every single ingredient and item used across hundreds of transactions, prevent potential internal theft through continuous monitoring and audit trails, maintain highly accurate stock counts at all times, generate professional customer receipts instantly, and handle everything smoothly even during internet outages. Ultimately, StockSecure POS helps café owners and managers save valuable time on administrative tasks, reduce financial losses from errors or fraud, make smarter data-driven decisions about menus and purchasing, improve overall operational smoothness, and focus more energy on delivering excellent quality coffee, friendly service, and creating a warm, welcoming atmosphere that keeps customers coming back. By leveraging the robust features of StockSecure POS, café owners can not only enhance their operational efficiency but also cultivate a loyal customer base, ensuring long-term success in a competitive market.

CHAPTER II:

Foreign Studies

The growing body of international research on small and medium enterprise (SME) sustainability underscores the critical role of effective inventory management and customer retention in ensuring long-term business viability, particularly within the food and beverage sector. Among these foreign studies, the qualitative case investigation by Sarman et al. (2026) offers valuable insights into the operational challenges faced by an SME café in Malaysia and the practical strategies that can strengthen its internal resources. Their findings, grounded in the Resource-Based View, provide a relevant foundation for examining how technology-driven solutions such as StockSecure POS and CyberLedger can further address inventory control, fraud prevention, and transactional security in similar SME contexts.

Sarman et al. (2026) found that effective inventory management and customer retention strategies contribute significantly to the sustainability of SME cafés. Sarman et al. (2026) conducted a qualitative case study examining how inventory management and customer retention strategies can enhance the sustainability of small and medium enterprise (SME) cafés. Drawing on secondary data from a SULAM (Service Learning Malaysia – University for Society) business advisory project involving a local food and beverage SME café in Johor, Malaysia (referred to as Café A to preserve confidentiality), the researchers applied the Resource-Based View (RBV) as their theoretical lens. The RBV posits that a firm's internal resources and capabilities—particularly those that are valuable, rare, inimitable, and organizationally embedded—form the foundation of sustained competitive advantage and long-term viability. The study identifies two primary operational weaknesses that threaten Café A's sustainability: inadequate inventory management practices arising from limited storage capacity and reliance on frequent small-quantity procurement, and the absence of structured customer retention and feedback mechanisms. These deficiencies elevate operating costs, transportation expenses, stockout risks, and exposure to price volatility while simultaneously limiting the café's ability to monitor satisfaction, understand preferences, and foster repeat patronage.

The inventory-related findings are particularly instructive. Because Café A lacked sufficient freezer capacity, it was compelled to purchase ingredients in small volumes at higher unit prices and to make frequent trips to suppliers. This pattern generated elevated logistics costs, increased the probability of stock shortages during peak periods, and left the business vulnerable to sudden price fluctuations in perishable goods. The researchers recommend practical, low-cost interventions—acquiring additional freezer space, shifting toward bulk purchasing where feasible, and implementing simple stock-card systems—to improve visibility, reduce waste, and stabilize supply. On the customer side, the absence of loyalty programs, systematic feedback collection, and digital communication tools constrained the café's capacity to convert one-time visitors into regular patrons. Suggested remedies include loyalty cards, WhatsApp Business for order and relationship management, and Google Forms for capturing customer insights. Collectively, these measures illustrate that SME café sustainability can be strengthened by reinforcing basic internal resources rather than by relying solely on external market conditions or capital-intensive technologies.

This Malaysian case study holds direct relevance for the development and justification of StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention, and CyberLedger: Secure Transaction and Audit Monitoring System. Although Sarman et al. (2026) focus on physical storage constraints and rudimentary record-keeping, their diagnosis of inventory visibility deficits and the resulting operational inefficiencies maps closely onto the problems that StockSecure POS is designed to solve. In many SME food-service settings, inventory records remain paper-based or are maintained in rudimentary spreadsheets that lack real-time updating, multi-user accountability, and audit trails. StockSecure POS addresses these gaps by embedding inventory monitoring directly within a point-of-sale environment. Every sale, stock adjustment, wastage entry, and receiving transaction is recorded electronically, producing an immediate and accurate picture of on-hand quantities. The integration of role-based access control further elevates the system beyond mere digital stock cards. By assigning granular permissions—managers may authorize purchase orders and view full inventory histories, cashiers may only process sales and view limited stock levels, and stock clerks may perform receiving and adjustment functions under supervisory oversight—RBAC reduces opportunities for internal fraud such as unauthorized stock removals, fictitious wastage claims, or collusion with suppliers. In this respect, StockSecure POS operationalizes the RBV principle articulated by Sarman et al.: technology, when configured as a controlled internal capability, becomes a valuable and difficult-to-imitate resource that protects the firm's physical inventory assets.

The connection extends to the broader sustainability agenda. Sarman et al (2026). demonstrate that frequent small purchases and stockouts erode margins and customer experience. StockSecure POS mitigates these pressures through continuous inventory visibility, automated low-stock alerts, and historical consumption analytics that support more rational ordering decisions. When combined with the bulk-purchasing and additional storage recommendations of the Malaysian study, the system enables SMEs to move from reactive, high-cost procurement cycles toward planned, cost-efficient replenishment. Moreover, the fraud-prevention dimension of RBAC supplies a layer of internal control that the original case study does not explicitly address yet is essential for sustainability. Inventory shrinkage caused by employee misconduct can silently undermine the cost savings achieved through better storage and bulk buying. By making every inventory movement attributable to a specific role and user, StockSecure POS strengthens the resource base that RBV theory identifies as critical to sustained performance.

CyberLedger complements and extends these benefits by focusing on the transactional and audit layers of café operations. While StockSecure POS secures the physical and digital inventory flow, CyberLedger ensures that the financial counterpart of every inventory movement—sales, refunds, supplier payments, and cash handling—is recorded in a tamper-evident, continuously auditable ledger. Secure transaction monitoring detects anomalies such as unusual void patterns, after-hours adjustments, or discrepancies between recorded sales and inventory depletion, thereby reinforcing the anti-fraud architecture initiated by RBAC. Audit trails generated by CyberLedger provide managers and external advisors with chronological evidence of operational integrity, which is particularly valuable for SMEs that lack dedicated accounting staff. In the context of the customer-retention strategies advocated by Sarman et al. (2026), reliable transaction records also support loyalty-program administration: points or discounts can be accurately credited only when legitimate purchases are verified, and historical purchase data can inform personalized promotions delivered via channels such as WhatsApp Business. Thus, CyberLedger transforms the transactional data generated at the point of sale into a strategic resource that simultaneously safeguards assets and deepens customer relationships.

The theoretical alignment with the Resource-Based View further cements the connection. Sarman et al. (2026) argue that SME cafés achieve sustainability by developing and deploying internal resources that are difficult for competitors to replicate. StockSecure POS and CyberLedger constitute precisely such resources. RBAC-enforced inventory monitoring and secure audit-capable transaction systems are not generic off-the-shelf applications; they are purpose-configured capabilities that embed accountability, real-time visibility, and forensic readiness into daily operations. Once implemented and institutionalized, these systems become path-dependent organizational routines that raise the cost of imitation for rival cafés. In addition, the data they generate—consumption patterns, shrinkage rates, peak-hour transaction volumes, and customer purchase histories—constitute knowledge assets that can guide continuous improvement in procurement, menu design, and retention tactics. In this manner, the technological solutions proposed in the present research directly amplify the practical interventions recommended in the Malaysian case while addressing a critical omission: the vulnerability of inventory and cash to internal opportunism.

The SULAM experience documented by Sarman et al. also underscores the value of bridging academic inquiry with real SME constraints. Just as university students and faculty collaborated with Café A to diagnose problems and propose feasible remedies, the design of StockSecure POS and CyberLedger has been informed by the operational realities of resource-constrained food-service enterprises. Both systems deliberately emphasize low-complexity interfaces, role-appropriate access, and minimal hardware requirements so that they remain accessible to businesses that cannot afford enterprise-grade enterprise-resource-planning suites. By demonstrating that sophisticated control mechanisms can be delivered through focused, affordable digital tools, the research extends the spirit of the SULAM project: academic knowledge is translated into actionable capabilities that strengthen SME resilience.

In summary, the work of Sarman et al. (2026) establishes that inventory mismanagement and weak customer-retention infrastructure are central threats to SME café sustainability and that these threats can be mitigated through the deliberate cultivation of internal resources. StockSecure POS responds to the inventory visibility and control deficits by integrating real-time monitoring with role-based access control, thereby reducing both operational inefficiency and internal fraud. CyberLedger extends protection to the financial and audit domain, ensuring that every transaction is secure, attributable, and analyzable. Together, the two systems operationalize and expand the RBV logic advanced in the Malaysian study, converting basic operational weaknesses into sources of controlled, data-rich competitive advantage. The foreign literature therefore supplies both empirical justification and theoretical grounding for the claim that purpose-built POS and ledger technologies constitute essential internal capabilities for the long-term sustainability of SME cafés.

A second major contribution to the foreign literature underpinning this study comes from Mohapatra et al. (2026), whose work on a web-based café management system provides a clear empirical and conceptual foundation for understanding how digital integration can transform the operational efficiency of small and medium-sized café enterprises. Their research demonstrates that consolidating order processing, billing, inventory management, and customer data handling into a single platform substantially reduces manual workload, minimizes transcription and calculation errors, and accelerates service delivery. By enabling customers to view digital menus, place orders, track status in real time, and complete payments, while simultaneously allowing administrators to manage menu items, adjust prices, monitor sales records, and maintain customer information, the system illustrates the practical value of end-to-end digitalization in the café sector. Developed with contemporary web technologies—HTML, CSS, Bootstrap, Flask, and SQLite—the application delivers a responsive, secure, and user-friendly interface that supports both front-of-house and back-of-house functions within one cohesive environment. This integrated approach enhances organizational accuracy and operational coordination, offering café owners a viable pathway to replace fragmented paper-based or loosely connected digital processes with a unified management solution.

The contribution of Mohapatra et al. (2026) is highly relevant to the present research on StockSecure POS and CyberLedger precisely because it establishes the baseline benefits of process integration that any modern point-of-sale (POS) and inventory system must deliver. Their findings affirm that computerized systems improve the reliability of sales and operational records, reduce dependency on manual intervention, and create a more coherent flow of information from order capture through to financial reporting. For small and medium-sized enterprise (SME) cafés operating under resource constraints, these efficiency gains are not merely desirable; they are often essential for survival in competitive markets where speed of service, inventory accuracy, and cost control directly influence profitability and customer satisfaction. By documenting how a single digital platform can streamline daily operations, Mohapatra et al. supply both empirical justification and a technological precedent for the systems developed in this study.

Nevertheless, the Café Management System presented by Mohapatra et al. (2026) exhibits important limitations that leave critical dimensions of internal control and fraud prevention unaddressed. While the system successfully automates transactional workflows and provides secure data storage together with basic sales tracking, it does not incorporate explicit role-based access restrictions. Consequently, different categories of staff—cashiers, stock clerks, supervisors, and managers—may possess overlapping or excessively broad permissions that allow unauthorized viewing or modification of inventory records, price lists, and sensitive operational data. In the absence of granular permission structures, opportunities for internal misconduct, including unauthorized stock adjustments, price manipulations, and unrecorded inventory diversions, remain inadequately constrained. Likewise, although the system records sales activity, it does not emphasize continuous, tamper-evident monitoring or comprehensive audit trails that would enable managers to reconstruct the full sequence of events surrounding each transaction, detect anomalies in real time, or verify that inventory movements consistently align with recorded sales.

These gaps are particularly consequential for SME cafés, where limited managerial oversight and high staff turnover can amplify the risk of internal fraud and data integrity failures.

StockSecure POS directly builds upon the integrated operational foundation established by Mohapatra et al. (2026) by embedding Role-Based Access Control (RBAC) into the core functions of point-of-sale processing and inventory monitoring. RBAC assigns permissions strictly according to defined user roles—cashiers, stock clerks, and managers—so that each individual receives only the privileges necessary to perform assigned tasks. Cashiers may process sales and view current stock levels relevant to order fulfillment, yet they are prevented from altering inventory quantities, modifying unit costs, or accessing historical adjustment logs. Stock clerks may update stock levels upon delivery or perform physical counts, but they cannot override pricing or generate financial reports. Managers retain comprehensive oversight, including the ability to approve adjustments, review audit logs, and reconfigure role permissions. By enforcing the principle of least privilege, StockSecure POS converts the efficiency gains identified by Mohapatra et al. into a controlled environment in which unauthorized access to inventory records, price adjustments, and sensitive operational data is systematically limited. This controlled access strengthens inventory accuracy, reduces opportunities for internal misconduct, and introduces accountability at every stage of the stock lifecycle—features that extend the original café management concept into the domain of internal control and fraud prevention.

CyberLedger complements and further advances the transactional and monitoring capabilities highlighted in the foreign study. Where Mohapatra et al. emphasize secure data storage and sales tracking, CyberLedger introduces continuous, tamper-evident recording of every billing activity, order adjustment, and payment. Each event is time-stamped, attributed to a specific authenticated user, and stored in a manner that renders subsequent alteration detectable. Comprehensive audit trails enable café managers to reconstruct the sequence of actions surrounding any transaction, compare inventory movements against corresponding sales data, and identify discrepancies that may indicate fraud, error, or process failure. This level of visibility transforms passive record-keeping into active oversight, allowing anomalies to be detected promptly and investigated with reliable evidence. In combination with the RBAC framework of StockSecure POS, CyberLedger ensures that the integrated digital environment not only streamlines operations but also protects critical internal resources against both intentional misconduct and unintentional integrity risks.

Together, StockSecure POS and CyberLedger therefore transform the integrated café management concept presented by Mohapatra et al. (2026) into a more robust, security-oriented solution. The original system demonstrated that consolidating order processing, billing, and inventory functions yields measurable improvements in efficiency, accuracy, and coordination. The present research retains that integrative logic while systematically addressing the security and control deficiencies that remain after basic automation is achieved. By embedding RBAC within POS and inventory modules and by layering continuous, attributable audit monitoring over all transactional activity, the combined solution delivers both the operational streamlining documented by Mohapatra et al. and the internal safeguards necessary to sustain those gains over time. For SME cafés, this dual emphasis is decisive: efficiency improvements that are not protected against internal fraud or data compromise are fragile and ultimately self-limiting. A system that simultaneously accelerates service, maintains accurate stock records, restricts unauthorized access, and provides verifiable audit evidence offer a more durable foundation for performance, accountability, and long-term viability.

In summary, the work of Mohapatra et al. (2026) establishes the relevance and practical utility of digital café management systems as a foundation for handling sales and operational records. It confirms that integrated platforms reduce manual effort, minimize errors, and improve coordination—outcomes that remain central to the design of StockSecure POS and CyberLedger. At the same time, the foreign study's limited attention to role-based access restrictions, internal fraud prevention, and continuous audit monitoring creates a clear research and development opportunity. StockSecure POS responds by integrating RBAC into POS and inventory functions, thereby limiting user permissions and strengthening accountability. CyberLedger advances the monitoring dimension by providing secure, tamper-evident transaction records and comprehensive audit trails. The resulting architecture retains the efficiency benefits of integrated café management while embedding the internal controls required to protect inventory integrity and financial reliability. In doing so, it converts a valuable operational foundation into a security-conscious solution specifically tailored to the risk profile and resource constraints of SME café businesses.

In general, Sarman et al. (2026) established, through a qualitative case investigation grounded in the Resource-Based View (RBV), that the long-term sustainability of SME cafés depends critically on the strength of internal resources rather than solely on external market conditions. Their examination of Café A in Johor, Malaysia, revealed two principal operational weaknesses: inadequate inventory management caused by limited storage capacity and a reliance on frequent small-quantity procurement and the complete absence of structured customer-retention and feedback mechanisms. These deficiencies produced elevated logistics and transportation costs, higher unit prices for ingredients, increased stockout risks during peak periods, exposure to price volatility in perishable goods, and a limited capacity to convert occasional customers into loyal patrons. The researchers demonstrated that relatively low-cost interventions—expanding freezer capacity, shifting toward bulk purchasing where feasible, introducing simple stock-card systems, implementing loyalty cards, utilizing WhatsApp Business for relationship management, and employing Google Forms for customer insights—can materially strengthen the firm's resource base. In RBV terms, these improvements render internal capabilities more valuable, rare, and organizationally embedded, thereby supporting sustained competitive advantage. The study thus provides empirical evidence that inventory visibility deficits and weak customer-relationship infrastructure constitute central threats to SME café viability and that deliberate cultivation of controllable internal resources offers a practical pathway to sustainability.

Also, Mohapatra et al. (2026) contributed complementary findings focused on the operational and technological dimensions of café management. Their development and evaluation of a web-based Café Management System demonstrated that consolidating order processing, digital menu presentation, real-time order tracking, billing, inventory management, price updates, sales monitoring, and customer data handling into a single platform substantially reduces manual workload, minimizes transcription and calculation errors, and accelerates service delivery. Implemented with modern web technologies (HTML, CSS, Bootstrap, Flask, and SQLite), the system delivered a responsive, secure, and user-friendly interface that improved organizational accuracy and operational coordination. The study confirmed that end-to-end digitalization replaces fragmented paper-based or loosely connected processes with a coherent information flow from order capture through to financial reporting. For resource-constrained SME cafés, these efficiency gains were shown to be essential for survival in competitive markets where speed of service, inventory accuracy, and cost control directly affect profitability and customer satisfaction. At the same time, the research implicitly highlighted remaining vulnerabilities: the absence of explicit role-based access restrictions and the lack of continuous, tamper-evident audit monitoring left the system open to unauthorized modifications of inventory records, price manipulations, and undetected transactional anomalies.

When the findings of the two studies are combined, a coherent and reinforced justification for StockSecure POS and CyberLedger emerges. Sarman et al. (2026) identify the strategic problem—inventory mismanagement and weak internal controls that erode margins, elevate costs, and threaten sustainability—while framing the solution within the resource-based view: technology and processes must be configured as controlled, difficult-to-imitate internal capabilities. Mohapatra et al. (2026) supply the technological solution architecture—an integrated digital platform that unifies order processing, billing, inventory, and sales tracking—and empirically demonstrate the efficiency benefits of such integration. Together, the studies establish both the necessity and the feasibility of moving beyond basic digitalization toward systems that embed accountability and security.

StockSecure POS operationalizes this combined insight by embedding Role-Based Access Control (RBAC) within a POS-driven inventory monitoring environment. It directly addresses Sarman et al.'s diagnosis of inventory visibility deficits by producing real-time, electronically recorded stock levels, automated low-stock alerts, and historical consumption analytics that support rational bulk purchasing and reduce stockout risks. Simultaneously, it overcomes the control gaps left by Mohapatra et al.'s system: cashiers, stock clerks, and managers receive only the permissions required for their roles, thereby limiting unauthorized stock adjustments, fictitious wastage claims, and price manipulations. In RBV terms, RBAC-enforced inventory monitoring becomes a valuable and path-dependent organizational capability that protects physical inventory assets and raises the cost of imitation for competitors.

CyberLedger extends the synthesis into the financial and audit domain. Building on Mohapatra et al.'s emphasis on secure data storage and sales tracking, it introduces continuous, tamper-evident recording of every billing activity, order adjustment, refund, and payment, with full attribution to authenticated users. This capability detects anomalies such as unusual void patterns or discrepancies between recorded sales and inventory depletion—risks that Sarman et al. did not explicitly examine but that can silently undermine the cost savings achieved through better storage and bulk buying. The resulting audit trails further support the customer-retention strategies advocated by Sarman et al.: loyalty points and personalized promotions can be accurately administered only when transactions are verified, and historical purchase data can inform targeted communication via channels such as WhatsApp Business. Thus, CyberLedger converts transactional data into a strategic knowledge asset that simultaneously safeguards cash and inventory while deepening customer relationships.

In combination, the findings of Sarman et al. (2026) and Mohapatra et al. (2026) demonstrate that SME café sustainability requires both the reinforcement of basic internal resources (inventory visibility, cost-efficient procurement, and customer retention) and the technological integration of operational processes. StockSecure POS and CyberLedger respond by transforming these insights into purpose-built systems that deliver real-time inventory control, role-based accountability, and continuous audit monitoring. The resulting architecture retains the efficiency gains of integrated digital management while embedding the internal safeguards necessary to protect those gains against fraud, error, and data compromise. For resource-constrained SME cafés, this dual contribution—strategic resource strengthening grounded in RBV and secure technological integration—constitutes a durable foundation for operational resilience, financial integrity, and long-term viability.

Local Studies

Local Review of Related Literature

Local studies on Point-of-Sale systems in Philippine micro-enterprises remain limited, yet they provide essential empirical grounding for technology-driven solutions in the café sector. Tronqued's (2026) investigation of start-up cafés in Tabaco City, Albay, is particularly valuable because it examines actual POS adoption patterns, operational benefits, and persistent gaps in small-scale food-service businesses. These findings directly inform the design priorities of StockSecure POS and CyberLedger by confirming both the practical usefulness of POS tools and the security and control deficiencies that still require attention.

The study is highly relevant to the present research because it investigates the implementation of Point-of-Sale (POS) systems in start-up cafés and demonstrates how these systems improve operational efficiency through order processing, billing, receipt generation, automated reporting, and real-time inventory tracking. The findings show that POS systems help reduce waiting time, improve transaction accuracy, and support inventory monitoring, highlighting the importance of digital technologies in managing café operations. These findings support the operational objectives of the proposed StockSecure POS system.

Moreover, the study found that many cafés use POS systems primarily for basic transaction functions while advanced features such as inventory management, sales analytics, and customer relationship management remain underutilized. It also identified challenges related to technical limitations, operational inefficiencies, financial constraints, and staff errors caused by insufficient training. These findings indicate that although POS systems provide significant benefits, there are still opportunities to improve their functionality and management.

The research further revealed that most of the participating cafés were relatively young enterprises operating for only one to three years, with small workforces of fewer than ten employees and capitalization levels generally ranging from ₱501,000 to ₱1,000,000. Average daily patronage remained modest, with the majority serving fewer than 100 customers. Despite these resource constraints, the café owners and managers recognized the value of POS systems in streamlining daily operations. However, the limited utilization of advanced modules, combined with recurring technical issues such as poor internet connectivity, system downtime, and compatibility problems, prevented the systems from delivering their full potential. Operational inefficiencies, including weak integration with online platforms, together with subscription and maintenance costs, added further barriers. Human resource challenges, particularly staff errors arising from inadequate training, also emerged as a significant concern.

These documented gaps strengthen the justification for the present study. The proposed StockSecure POS addresses some of these limitations by incorporating Role-Based Access Control (RBAC) to regulate employee access based on assigned responsibilities, thereby reducing the risk of unauthorized transactions and internal fraud. In addition, the CyberLedger component provides secure transaction logging and audit monitoring, allowing administrators to track user activities and strengthen accountability. By focusing on security, controlled access, and continuous audit trails, the system directly responds to the inventory-monitoring weaknesses and staff-related errors identified by Tronqued (2026). While the cited study emphasizes the operational value of POS systems in local cafés, the present research extends these capabilities by embedding stronger security and fraud-prevention mechanisms, making the solution more reliable and suitable for modern café operations in similar resource-constrained settings.

The study conducted by **Boado et al. (2025)** found that coffee shop owners use point-of-sale (POS) systems to improve business operations and efficiency. Coffee shop owners' decision and use of point-of-sale (POS) system 024, which examined the perceptions and behavioral intentions of coffee shop owners in Baliwag City regarding the adoption of point-of-sale systems, provides a critically important foundation for understanding both the market context and the specific design requirements that shaped the development of StockSecure POS. This research, employing a quantitative descriptive-correlational design, surveyed a sample of coffee shop owners to assess their awareness of POS technology, their attitudes toward its adoption, and the factors that influenced their willingness or reluctance to implement such systems in their operations. The findings of this study are particularly valuable because they are grounded in the specific cultural, economic, and operational context of small food service businesses in the Philippines, making them directly applicable to the environments where StockSecure POS will be deployed.

The most fundamental finding of the Boado et al. (2025) study is that coffee shop owners in the Philippines have a broadly positive attitude toward POS systems and recognize their potential to improve business operations. Specifically, owners identified improvements in transaction speed, accuracy of financial records, inventory tracking, and reporting capabilities as key benefits of POS adoption. This recognition of the value proposition of POS technology is an important enabling factor for the introduction of StockSecure POS into this market, as it suggests that potential users already understand and appreciate the general category of solution being offered, even if they are not yet familiar with the specific capabilities of StockSecure. This baseline level of technological awareness reduces the educational burden on the StockSecure team and allows the system's marketing and training efforts to focus on the distinctive features and superior capabilities of the platform rather than on basic technology literacy.

However, the study also reveals a significant and persistent gap between owners' positive attitudes toward POS adoption and their actual implementation of these systems. Despite expressing strong intentions to adopt POS technology, many of the surveyed owners had not yet done so, and the study identified several specific barriers that accounted for this gap. Chief among these were concerns about the complexity of POS systems and the perceived difficulty of integrating them into existing workflows without disrupting daily operations. Owners expressed anxiety about the learning curve associated with new technology, particularly the time and effort required to train existing staff and the risk that the system would cause operational disruptions during the initial implementation period. These concerns reflect a rational and understandable caution on the part of small business operators who cannot afford the kind of operational downtime or service degradation that might accompany the adoption of a poorly designed or overly complex system.

The Boado et al. (2025) study's finding that adoption decisions were most strongly correlated with the technological context, specifically with perceptions of employee preparedness and infrastructural support, has direct and significant implications for the design philosophy of StockSecure POS. This finding suggests that the success of a POS system in the café market is determined not primarily by the sophistication of its features or the elegance of its technology, but by how well it fits into the existing capabilities and infrastructure of the businesses it serves. A system that requires extensive technical infrastructure upgrades, that demands a high level of prior technological knowledge from its users, or that imposes a steep and discouraging learning curve will face significant adoption barriers regardless of its technical merits. Conversely, a system that is designed from the ground up to be accessible, intuitive, and compatible with the typical operational environment of a small café will be far more likely to achieve widespread adoption.

These insights shaped the development of StockSecure POS in several important ways. The system was designed to operate on standard web browsers without requiring the installation of specialized software, reducing the infrastructure requirements for adoption to essentially nothing beyond a stable internet connection and a compatible device. The user interface was designed with non-technical users in mind, emphasizing clarity, simplicity, and guidance at every step of the workflow. The RBAC framework was deliberately aligned with the natural role structure of a typical café so that the access permissions assigned to each role correspond intuitively to what that role's occupant actually needs to do their job. This alignment minimizes the cognitive load on staff and reduces the amount of training required to bring new employees up to speed on the system, directly addressing the employee preparedness concerns that Boado et al. (2025) identified as a primary adoption barrier.

The study also highlights the importance of compatibility, defined as the degree to which a new technology fits with existing values, experiences, and needs, as a driver of adoption decisions. This finding provides strong theoretical support for the design principle that StockSecure POS should map its functionality as closely as possible to the actual operational workflows of the cafés it serves, rather than requiring those cafés to restructure their operations to accommodate the system. By designing the system around the real-world division of labor in a typical café, with roles and permissions that correspond to the positions of barista, cashier, and manager that already exist in most small coffee shops, StockSecure POS maximizes its compatibility with existing operational structures and thereby reduces the friction associated with adoption.

The Boado et al. (2025) study serves as empirical validation that the challenges StockSecure POS and CyberLedger are designed to address are not hypothetical constructs but real and persistent obstacles that prevent small café operators from benefiting from modern POS technology. By grounding the design and development of these systems in the specific findings of locally conducted research, the project team has ensured that the solutions offered by StockSecure POS are precisely calibrated to the needs, concerns, and capabilities of their target users. This evidence-based approach to system design is one of the defining strengths of the StockSecure POS project and represents a meaningful contribution to the practice of technology development for small business environments in the Philippines.

To fully appreciate the significance of the Boado et al. (2025) study and its implications for StockSecure POS, it is necessary to situate that research within the broader context of the operational challenges faced by small food service businesses in the Philippines. The Philippine food service industry, particularly the café and coffee shop segment, has experienced remarkable growth over the past decade, driven by rising incomes, urbanization, a growing middle class, and the rapid proliferation of coffee culture among younger consumers. However, this growth has also intensified competition and placed significant operational pressure on small café operators, who must manage increasingly complex supply chains, larger and more diverse menus, higher customer volumes, and a growing regulatory environment, all while operating with limited financial resources and small management teams.

Inventory management is one of the most significant operational challenges for small cafés, and it is also one of the areas where technology adoption can have the greatest impact. In a typical small café, inventory includes not only the coffee beans, milk, syrups, and other ingredients used in beverages but also the packaging materials, cleaning supplies, and food items that make up the broader operational inventory. Managing this inventory manually is time-consuming, error-prone, and inefficient. Café owners who rely on manual inventory systems often struggle to maintain accurate stock counts, identify shrinkage and waste, anticipate supply needs, and respond quickly to changes in demand. These challenges are compounded by the fact that many small café owners also manage other aspects of the business, including customer service, marketing, and financial administration, leaving little time for detailed inventory oversight.

The risk of internal fraud is another significant concern for small café operators. In a small business where the owner cannot personally supervise every transaction and every inventory movement, the temptation for dishonest employees to engage in theft, manipulation of records, or unauthorized discounts is a real and persistent threat. Research on employee theft in the food service industry suggests that shrinkage due to internal fraud can account for a substantial portion of a café's losses, often significantly exceeding losses from external theft or operational inefficiency. Yet many small café owners lack the tools and systems necessary to detect and prevent internal fraud effectively. Without detailed transaction logs, inventory reconciliation reports, and clear access controls, it can be extremely difficult for a café owner to identify patterns of fraudulent behavior or to build a documented case when disciplinary action is required.

The development of StockSecure POS, informed by the local research context provided by Boado et al. (2025) and by the broader literature on small business operations in the Philippines, represents a direct and practical response to these challenges. By providing a system that is simultaneously affordable, user-friendly, and capable of delivering enterprise-grade security and inventory management capabilities, StockSecure POS aims to close the technology gap between large food service chains and small independent cafés. This democratization of access to effective business management tools has the potential not only to improve the operational performance of individual cafés but also to strengthen the resilience and competitiveness of the broader small business sector in the Philippines.

In summary, both studies show that while café owners in the Philippines recognize the operational value of POS systems—particularly in improving transaction speed, accuracy, inventory tracking, and reporting—significant gaps remain in actual adoption and full utilization. Tronqued (2026) highlights the underuse of advanced features and persistent challenges such as technical limitations, staff errors from inadequate training, and weak inventory controls among start-up cafés, while Boado et al. (2025) reveal a clear intention-adoption gap driven by concerns over system complexity, learning curves, employee preparedness, and infrastructure constraints. Together, these findings strengthen the present research by confirming that local café operators need not only functional POS tools but also secure, role-aligned, and user-friendly systems. StockSecure POS, through its integration of Role-Based Access Control and the CyberLedger audit component, directly responds to these documented gaps by enhancing inventory monitoring, preventing internal fraud, reducing unauthorized access, and minimizing training barriers—thereby offering a practical and context-appropriate solution that builds on the operational insights of both studies.

Related Literature

The study by Ratna et al. (2025), which provides a comprehensive examination of role-based access control implementation in multi-user information systems characterized by varying role hierarchies, offers a rich and detailed theoretical framework that underpins several of the most important design decisions made in the development of StockSecure POS. **Ratna et al. (2025)** emphasized that implementing Role-Based Access Control (RBAC) enhances the security of university-based digital systems by restricting access based on user roles. The research focuses on environments where multiple users with different levels of authority and responsibility must interact with a shared information system, accessing different subsets of the system's functionality based on their organizational role. These environments are characterized by the need to balance accessibility, which requires that each user can quickly and easily access the functions they need, against security, which requires that no user can access functions beyond their authorized scope. The challenge of achieving this balance without imposing undue complexity or administrative burden is one of the central problems of information systems design, and Ratna et al.'s (2025) research makes a substantial contribution to its resolution.

One of the key insights from the Ratna et al. (2025) study is that the effectiveness of an RBAC implementation depends critically on the quality and granularity of the role definitions it employs. If roles are defined too broadly, permissions will inevitably be over-assigned to some users, creating security vulnerabilities and opportunities for misuse. If roles are defined too narrowly, the administrative burden of managing a large number of specialized roles can become unmanageable, and the system can become rigid and difficult to adapt to changing organizational needs. The study argues that the optimal approach involves defining roles at a level of specificity that corresponds naturally to the actual divisions of labor within the organization, creating roles that are meaningful to managers and users alike and that can be clearly and consistently applied without generating ambiguity or exception cases.

This principle of role definition has been applied directly in the development of StockSecure POS, where the role structure was designed through careful consultation with actual café operators and staff to ensure that it reflects the real-world operational structure of a typical Filipino café. The roles defined in the system, including café owner, manager, cashier, and barista, correspond to positions that already exist in most small cafés, and the permissions associated with each role were determined by a systematic analysis of the tasks and data access requirements of each position. By grounding the role structure in actual operational reality rather than in theoretical organizational charts, the development team has ensured that the RBAC implementation of StockSecure POS is both effective and intuitive, minimizing the gap between the system's security model and the users' mental model of their own work.

Ratna et al.'s (2025) research also explores the concept of role hierarchy, in which higher-level roles inherit the permissions of lower-level roles while also possessing additional privileges that reflect their greater organizational authority. This hierarchical structure simplifies the administration of RBAC systems by ensuring that permission sets are cumulative rather than independent, reducing the amount of redundant configuration required when defining roles at different organizational levels. In StockSecure POS, this principle is reflected in the relationship between the cashier and manager roles: a manager can do everything a cashier can do, plus additional management-level functions such as modifying prices, approving inventory adjustments, and accessing sales reports. The café owner role extends this hierarchy further, adding access to administrative functions, financial summaries, and system configuration options. This hierarchical structure ensures that the system's role definitions are both comprehensive and logically consistent, supporting both effective security management and clear organizational communication.

An additional dimension of the Ratna et al. (2025) study that deserves attention is its treatment of the relationship between RBAC and organizational accountability. The study argues that RBAC enhances accountability by creating a clear and auditable record of which role is responsible for each action performed within the system. When a discrepancy is identified in inventory records or a suspicious transaction is flagged by the audit system, the RBAC framework allows administrators to immediately identify which role's permissions were used to perform the action in question and to trace that action back to the specific user who was assigned that role at the relevant time. This accountability function is particularly important in environments where internal fraud is a concern, as it transforms the abstract possibility of being caught into a concrete and demonstrated certainty: anyone who misuses their system access will leave a clear, role-attributed audit trail that can be reviewed at any time.

The integration of this accountability function into StockSecure POS is accomplished through the CyberLedger audit monitoring system, which records every user action within the system, tagged with the role, the user identity, the timestamp, and the nature of the action performed. This comprehensive audit trail, made possible by the combination of RBAC and CyberLedger, provides café owners with a powerful investigative tool that can be used both proactively, to monitor for unusual patterns of activity, and reactively, to investigate specific incidents of suspected fraud or unauthorized access. The Ratna et al. (2025) finding that RBAC-based accountability mechanisms significantly enhance the security posture of information systems is thus directly validated by the design architecture of StockSecure POS.

The concept of a secure transaction and audit monitoring system, as embodied in the CyberLedger component of StockSecure POS, is grounded in a long tradition of research on audit trails and transaction monitoring in information systems security. Audit trails have been recognized as a fundamental security control since the earliest days of computerized information systems, serving three distinct but complementary functions: deterrence, detection, and investigation. As a deterrent, the knowledge that every action within a system is being recorded and can be reviewed by authorized personnel reduces the incentive for dishonest behavior by raising the perceived risk of detection. As a detection mechanism, regular review of audit logs can identify patterns of behavior that deviate from normal operational norms, flagging potential fraud, error, or unauthorized access for further investigation. As an investigative tool, detailed audit records provide the evidentiary foundation for disciplinary action, legal proceedings, or regulatory compliance documentation.

The design of CyberLedger reflects these three functions in a system that is simultaneously comprehensive, secure, and accessible to non-technical users. Comprehensive, in the sense that it records every transaction and every operational action performed within StockSecure POS, without exception or omission. Secure, in the sense that audit records are encrypted and protected against modification, ensuring that the audit trail itself cannot be tampered with by users who are aware that their actions are being monitored. Accessible, in the sense that café owners and authorized managers can review audit logs through an intuitive interface that presents information in clear, meaningful formats, rather than requiring them to parse raw technical data. This combination of comprehensiveness, security, and accessibility makes CyberLedger a genuinely useful operational tool rather than a passive compliance mechanism that exists primarily to satisfy regulatory requirements.

The importance of immutability in audit log design deserves particular emphasis in this context. An audit trail that can be modified by users with sufficiently high levels of system access provides only limited security value, because a sophisticated fraudster who understands the system's audit logging capabilities might attempt to cover their tracks by altering or deleting audit records after the fact. CyberLedger addresses this vulnerability by implementing a write-once architecture for audit records, in which entries can be written but never modified or deleted by any user, regardless of their role permissions. This architectural choice is informed by the broader literature on tamper-proof logging systems, which consistently identifies write-once logging as the gold standard for audit trail integrity. By ensuring that the audit record is itself protected against manipulation, CyberLedger provides a level of evidentiary reliability that can support both internal investigations and formal legal or regulatory proceedings.

The integration of CyberLedger with the RBAC framework of StockSecure POS creates a security architecture that is significantly more robust than either component would be in isolation. RBAC restricts what actions users can perform, while CyberLedger records every action that is actually performed, including those that were authorized by RBAC. This means that even actions performed by highly privileged users, such as café owners or senior managers, are fully documented in the audit trail. This is a critical safeguard against a category of fraud that RBAC alone cannot prevent: the misuse of legitimate high-level access by trusted users. A manager who has the authority to process refunds might abuse that authority to issue fraudulent refunds to themselves or their associates. A café owner who has access to the full financial reporting module might manipulate transaction records to reduce taxable income. CyberLedger's comprehensive logging ensures that these actions, if they occur, will be captured in the audit record and will be detectable through systematic review of the logs.

The practical implementation of CyberLedger within StockSecure POS also incorporates several features that enhance its usability as a real-time monitoring tool, rather than merely a retrospective investigative resource. The system generates automated alerts when it detects patterns of activity that deviate significantly from established norms, such as an unusually high number of refunds processed by a particular cashier, a significant discrepancy between recorded inventory consumption and actual stock levels, or repeated failed login attempts that might indicate an unauthorized access attempt. These real-time alerts allow café owners and managers to respond to potential issues promptly before they escalate into larger problems, rather than discovering fraud or error only during periodic manual reviews of the audit logs. This proactive monitoring capability transforms CyberLedger from a passive record-keeping system into an active security tool that contributes continuously to the operational integrity of the café.

The broader literature on technology adoption in micro, small, and medium enterprises provides essential context for understanding both the challenges and the opportunities associated with the deployment of StockSecure POS in the café market. Research in this area has consistently found that MSMEs face a distinctive set of barriers to technology adoption that differ significantly from those encountered by large enterprises. These barriers include limited financial resources, which constrain the budget available for technology investment and ongoing maintenance; limited technical expertise, which makes the selection, implementation, and troubleshooting of complex systems challenging; limited time, as MSME owners and managers are typically stretched across multiple responsibilities and cannot dedicate significant effort to technology management; and limited organizational slack, meaning that the disruption associated with implementing a new system can have an immediate and material impact on business performance in ways that would be more easily absorbed by larger organizations.

These barriers are particularly relevant in the Philippine context, where many small café operators are first-generation entrepreneurs who built their businesses from scratch and may have limited formal training in business management or information technology. For these operators, the decision to invest in a POS system is not a routine IT procurement decision but a significant strategic commitment that carries real financial and operational risk. Understanding and addressing the specific concerns of this audience is essential for any technology product that aspires to achieve widespread adoption in this market. The research on MSME technology adoption, combined with the local findings of Boado et al. (2025), provides a detailed and nuanced picture of the fears, expectations, and needs that StockSecure POS must address to be commercially and operationally successful.

Several theoretical frameworks from the technology adoption literature are particularly relevant to this analysis. The Technology Acceptance Model, originally developed by Davis in 1989 and subsequently extended and refined by numerous researchers, posits that users' decisions to adopt a new technology are primarily determined by two factors: their perception of the technology's usefulness, meaning the degree to which they believe the technology will improve their job performance, and their perception of the technology's ease of use, meaning the degree to which they believe using the technology will be free of effort. Both dimensions are critically important for StockSecure POS: the system must demonstrate clear and tangible improvements in café management efficiency and security, and it must achieve these improvements without demanding a level of technical sophistication that is beyond the reach of typical café staff.

The Diffusion of Innovations framework, developed by Rogers and widely applied in technology adoption research, offers additional insights into the factors that influence the speed and extent of technology adoption across a market. Rogers identifies five characteristics of innovations that influence their rate of adoption: relative advantage, the degree to which the innovation is perceived as better than what it replaces; compatibility, the degree to which it is consistent with existing values and practices; complexity, the degree to which it is difficult to understand and use; trialability, the degree to which it can be experimented with before a full commitment is made; and observability, the degree to which the results of the innovation are visible to others. StockSecure POS scores favorably on most of these dimensions: it offers a clear relative advantage over manual inventory management and paper-based transaction records; it is designed to be highly compatible with existing café operational structures; it has been engineered for simplicity and ease of use; it can be trialed on a limited basis before a café commits to full deployment; and its benefits, including reduced inventory discrepancies, improved transaction accuracy, and enhanced fraud detection, are observable and measurable in ways that can be communicated to potential adopters.

The literature on MSME technology adoption also highlights the importance of after-sales support and training in determining the long-term success of technology implementations. Even systems that are well-designed and easy to use can fail to deliver their promised benefits if users do not receive adequate training and ongoing technical support. This is particularly true in resource-constrained environments where staff turnover may be relatively high, creating a persistent need for onboarding new users to the system. The development of StockSecure POS has incorporated this insight by designing a comprehensive training and support framework that includes in-system guidance, documentation accessible through the user interface, and responsive technical support channels. This commitment to ongoing user support reflects the research consensus that technology adoption is not a one-time event but an ongoing process of learning, adaptation, and refinement.

Synthesis of the Reviewed Literature

All the foreign and local studies examined in this review share a set of clear and repeated findings about the daily realities of small and medium enterprise cafés. Whether the research is conducted in Malaysia, India, or different cities in the Philippines, the same operational problems surface again and again. Inventory management remains weak across every setting. Limited storage space forces owners to buy ingredients in small quantities at higher unit prices, frequent supplier trips raise transport costs, stockouts occur during busy periods, and sudden price changes in perishable goods cut into already thin margins. Customer retention is equally fragile. Without loyalty programs, systematic feedback tools, or simple digital channels for staying in touch with patrons, one-time visitors rarely become regular customers. At the same time, every study agrees that digital tools can help. When order taking, billing, inventory tracking, and sales reporting are brought together inside one platform, manual work decreases, calculation errors drop, service becomes faster, and owners finally gain a clearer picture of what is happening each day. Local café owners already recognize these advantages and generally hold positive attitudes toward point-of-sale systems, listing faster transactions, more accurate records, better inventory visibility, and clearer reporting as the main benefits they expect. These shared observations form the common ground that links research conducted in very different places and under different conditions.

These common findings appear with striking consistency when the details of each study are examined side by side. Sarman et al. (2026) describe a Malaysian café that lacked sufficient freezer capacity and therefore had to purchase ingredients in small volumes. The result was higher unit prices, repeated trips to suppliers, elevated logistics costs, greater risk of running out of stock during peak hours, and constant exposure to sudden price swings in perishable goods. The same pattern surfaces in local Philippine studies. Tronqued (2026) finds that start-up cafés in Tabaco City often struggle with incomplete inventory tracking even after adopting basic POS systems, while Boado et al. (2025) note that coffee-shop owners in Baliwag City recognize inventory monitoring as one of the most valuable features of POS technology yet frequently fail to use it to its full potential. Customer retention receives equally consistent attention. Sarman et al. point to the absence of loyalty cards, WhatsApp Business messaging, and Google Forms as missed opportunities that keep occasional visitors from becoming regulars. Local studies echo the same gap by showing that advanced customer-relationship modules inside existing POS systems remain under-used. Across every study the message is the same: stronger internal systems improve efficiency and can support long-term sustainability, but only if they match the real constraints of small cafés that operate with limited staff, limited capital, and limited technical expertise.

The studies also agree that staff-related issues and technical barriers frequently prevent the full benefits of digital tools from being realized. Training gaps lead to errors at the counter and in the stockroom. Internet problems and system downtime interrupt operations at the worst possible moments. Many owners hesitate to adopt or fully use available technology because they fear complexity, long learning curves, or disruption during the changeover period. Despite these obstacles, the shared conclusion remains positive. When a system is useful, relatively simple, and compatible with existing work routines, small café operators are willing to try it and can see measurable improvements in daily performance. Mohapatra et al. (2026) demonstrate that consolidating order processing, digital menus, real-time order tracking, billing, inventory updates, price changes, sales monitoring, and customer data into one web-based platform sharply reduces manual workload and speeds service. Tronqued reports that POS systems in young Philippine cafés improve transaction speed, receipt accuracy, and basic inventory visibility. Boado et al. find broadly positive attitudes toward POS technology among coffee-shop owners. The broader technology-adoption literature reinforces the same point: when a system is perceived as useful and relatively easy to use, small-business operators are more willing to experiment with it. This common ground of problems and partial solutions forms the foundation on which the present research builds.

Taken together, the foreign and local studies paint a coherent picture of the operational landscape of SME cafés. Inventory visibility is poor, customer relationships are weakly managed, and basic digital tools already deliver clear efficiency gains, yet those gains are often incomplete because of training gaps, technical limitations, and under-used advanced features. The consistency of these findings across different countries and research methods strengthens their reliability. It shows that the challenges faced by a café in Johor, Malaysia, are not fundamentally different from those faced by start-ups in Tabaco City or coffee shops in Baliwag City. The same internal weaknesses threaten sustainability everywhere, and the same category of digital solutions offers the most practical path forward. This shared understanding is what makes the next step of comparison both necessary and useful.

When the studies are placed side by side, important differences in focus, method, and depth become visible. Sarman et al. concentrate on physical and procedural weaknesses. They examine limited storage capacity, frequent small-quantity purchases, and the complete absence of structured customer-retention and feedback mechanisms. Their recommendations stay practical and low-cost: acquire additional freezer space, shift toward bulk purchasing where feasible, introduce simple stock-card systems, implement loyalty cards, use WhatsApp Business for relationship management, and employ Google Forms for customer insights. Their analysis is firmly grounded in the Resource-Based View, which argues that internal resources that are valuable, rare, and organizationally embedded form the foundation of sustained competitive advantage. Yet they do not examine digital access controls or continuous audit records. Their focus remains on physical storage and basic record-keeping rather than on the security architecture of digital systems.

Mohapatra et al. take a distinctly technological approach. They develop and evaluate a web-based café management system built with HTML, CSS, Bootstrap, Flask, and SQLite. The system consolidates order processing, digital menu presentation, real-time order tracking, billing, inventory management, price updates, sales monitoring, and customer data handling into a single platform. Their findings show measurable reductions in manual workload, transcription errors, and service time, together with improved organizational accuracy and coordination. However, their system still allows different categories of staff—cashiers, stock clerks, supervisors, and managers—to possess overlapping or overly broad permissions. Without granular role restrictions, unauthorized viewing or modification of inventory records, price lists, and sensitive operational data remains possible. The system also lacks continuous, tamper-evident monitoring and comprehensive audit trails that would enable managers to reconstruct the full sequence of events surrounding any transaction or to detect anomalies in real time. In short, Mohapatra et al. prove the value of integration but leave the control and security dimensions largely unaddressed.

Local Philippine studies highlight different but related concerns that reflect the specific conditions of small food-service businesses in the country. Tronqued documents actual POS adoption patterns among start-up cafés in Tabaco City that are mostly one to three years old, employ fewer than ten people, and operate with capitalization levels generally ranging from ₱501,000 to ₱1,000,000. Average daily patronage remains modest, with the majority serving fewer than 100 customers. The systems deliver clear gains in order processing, billing, receipt generation, automated reporting, and real-time inventory tracking. Yet advanced modules for inventory management, sales analytics, and customer relationship management stay under-utilized. Technical problems such as poor internet connectivity, system downtime, and compatibility issues, together with staff errors caused by insufficient training, prevent the tools from delivering their full potential. Operational inefficiencies, including weak integration with online platforms, and the ongoing costs of subscription and maintenance add further barriers.

Boado et al. focus on the gap between intention and actual adoption. Using a quantitative descriptive-correlational design, they survey coffee-shop owners in Baliwag City and find that owners hold broadly positive attitudes toward POS systems and recognize benefits in transaction speed, accuracy of financial records, inventory tracking, and reporting. Despite these positive views, many have not yet implemented the technology. The strongest barriers are concerns about complexity, the perceived difficulty of integrating new systems into existing workflows, anxiety about the learning curve, the time and effort required to train staff, and the risk of operational disruption during the initial period. The study shows that adoption decisions are most strongly correlated with the technological context, specifically with perceptions of employee preparedness and infrastructural support. A system that requires extensive technical upgrades, demands high prior technological knowledge, or imposes a steep learning curve will face significant resistance regardless of its technical merits. Compatibility with existing values, experiences, and operational structures emerges as a critical driver of adoption.

Ratna et al. (2025) supply a theoretical and practical framework that the other studies largely omit. They examine role-based access control in multi-user information systems and emphasize that the effectiveness of RBAC depends on the quality and granularity of role definitions. Roles defined too broadly create security vulnerabilities; roles defined too narrowly create administrative burden and rigidity. The optimal approach is to define roles that correspond naturally to actual divisions of labor so that they are meaningful to managers and users alike. They also explore role hierarchy, in which higher-level roles inherit the permissions of lower-level roles while adding additional privileges. Finally, they highlight the accountability function of RBAC: every action can be traced to a specific role and user, creating a clear audit record that supports both deterrence and investigation. Yet none of the café systems reviewed in the foreign or local literature fully apply these principles in an affordable, easy-to-learn package designed for resource-constrained food-service environments.

These differences create a coherent picture of what has been achieved and what remains unfinished. Foreign studies establish both the strategic importance of internal resources and the efficiency gains of integrated digital platforms. Local studies confirm that Philippine café owners already see the value of POS technology but face practical barriers of complexity, training, incomplete feature use, and limited infrastructure. Theoretical work on RBAC explains how controlled access and accountability can close security gaps, yet that knowledge has not been translated into systems that small Philippine cafés can actually adopt and sustain. The comparison therefore reveals not contradiction but incompleteness: each study illuminates part of the problem and part of the solution, but none delivers a complete, security-conscious, and context-ready package.

The research gap that emerges from this comparison is therefore clear, specific, and consequential. Existing systems improve speed and accuracy but do not systematically restrict what each staff member can do inside the system, nor do they provide continuous, unchangeable records that allow owners to detect problems while they are still small. Without role-based limits, a cashier may be able to alter stock counts or prices, a stock clerk may access financial reports, and unauthorized adjustments, fictitious wastage claims, or collusion with suppliers can occur without leaving a clear trail. Without permanent audit records, discrepancies between recorded sales and actual inventory movement, unusual void patterns, after-hours adjustments, or repeated failed login attempts remain hard to investigate and even harder to prove. These gaps leave the efficiency gains fragile. Money and ingredients can still disappear through both honest mistakes and internal misconduct. Small café owners who cannot personally supervise every transaction and every stock movement are left without the tools they need to protect the very resources that the Resource-Based View identifies as critical to long-term survival.

The proposed StockSecure POS system directly addresses the first part of this gap by embedding role-based access control into every point-of-sale and inventory function. Permissions are assigned according to the real jobs already found in small Philippine cafés. A cashier can process sales and view only the limited stock levels needed for order fulfillment but cannot change inventory quantities, modify unit costs, or open historical adjustment logs. A stock clerk can receive deliveries and perform physical counts under supervisory oversight but cannot override pricing or generate financial reports. Managers retain comprehensive oversight, including the ability to approve adjustments, review complete audit logs, and reconfigure role permissions when necessary. The café owner role extends this hierarchy further, adding access to administrative functions, financial summaries, and system configuration options. This design follows the principle of least privilege and the hierarchical logic outlined by Ratna et al., ensuring that each person sees only what their work requires. By grounding the role structure in actual operational reality rather than in abstract organizational charts, the system remains intuitive and minimizes the cognitive load on staff. As a result, the efficiency improvements already proven by Mohapatra et al. and the local studies become more durable because unauthorized actions are harder to perform and easier to detect.

CyberLedger addresses the second part of the gap by creating a continuous, tamper-evident record of every transaction, refund, order adjustment, and payment. Each entry is permanently linked to the user’s identity, role, and exact time stamp, and no one—regardless of rank—can later alter or delete it. The system implements a write-once architecture so that the audit trail itself cannot be manipulated by users who know they are being monitored. When recorded sales do not match inventory depletion, when an unusual number of voids or after-hours adjustments appear, or when patterns suggest possible fraud or error, the owner can open a clear, permanent history and see exactly what happened, who did it, and when. Automated alerts can flag deviations from normal patterns in real time, allowing managers to respond before small problems become large losses. This reliable, forensic-grade evidence is what earlier systems lacked and what small cafés need most when they cannot afford dedicated accounting or security staff. The combination of StockSecure POS and

CyberLedger also supports the customer-retention practices recommended by Sarman et al. Loyalty points and personalized promotions can be credited only against verified purchases, and historical purchase data can guide targeted messages through familiar channels such as WhatsApp Business. In this way transactional data become a strategic knowledge asset that simultaneously safeguards cash and inventory while deepening customer relationships.

By keeping the system browser-based, simple to learn, and aligned with existing café roles, StockSecure POS and CyberLedger further remove the adoption barriers identified by Boado et al. and Tronqued. No specialized software installation is required beyond ordinary devices and a stable internet connection. The user interface emphasizes clarity and guidance at every step so that staff with limited technical background can perform their tasks without extensive training. In-system help and accessible documentation reduce the ongoing support burden. The design deliberately maps functionality onto the real-world division of labor in a typical Philippine café, maximizing compatibility with existing operational structures and thereby reducing friction during adoption. In everyday language, the previous studies show that cafés have already started the journey toward better tools and that the first steps bring real improvements in speed and accuracy, but they have not yet reached a complete, secure destination. Efficiency without protection is temporary. The moment staff can change records without leaving a trace or can access functions beyond their responsibility, the gains begin to leak away through shrinkage, error, or internal opportunism. The present research finishes that journey by keeping every useful part of existing POS platforms while adding the missing security and accountability layers that earlier systems omitted.

The result is a practical, affordable pair of tools that protect both the efficiency gains already demonstrated in the literature and the inventory and cash that small café owners work so hard to safeguard. StockSecure POS turns real-time inventory monitoring into a controlled organizational capability. CyberLedger turns every transactional event into a permanent, attributable record. Together they operationalize the resource-based view insight that technology, when configured as a controlled internal resource, becomes valuable, difficult to imitate, and capable of supporting sustained performance. They convert the operational weaknesses identified by Sarman et al. into sources of visibility and control. They retain the integrative benefits proven by Mohapatra et al. while closing the security gaps those researchers left open. They answer the local adoption concerns documented by Tronqued and Boado by remaining simple, role-aligned, and low in infrastructure demand. And they apply the accountability principles articulated by Ratna et al. in a form that resource-constrained Philippine cafés can actually use day after day. In doing so the present study takes the shared findings, the visible differences, and the open gaps of the entire body of reviewed literature and converts them into a concrete, context-ready solution that directly addresses the operational realities of SME cafés.

In general, StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention, together with CyberLedger: Secure Transaction and Audit Monitoring System, serves as direct and practical proof that the problems repeatedly identified in the foreign and local studies can be solved in a way that fits the real conditions of small and medium enterprise cafés.

The earlier studies show that cafés lose money and struggle to survive because inventory is hard to see and control, staff can make changes without clear limits, and there is no reliable record that lets owners check what really happened. StockSecure POS proves that these inventory and fraud problems can be fixed by placing role-based access control inside the everyday point-of-sale and stock screens. Each person is given only the permissions that match their actual job. A cashier can sell and see limited stock levels but cannot change quantities or prices. A stock clerk can receive goods and update counts but cannot touch financial reports. Managers and owners keep full oversight. This simple structure turns the efficiency gains already shown by previous systems into gains that last, because unauthorized actions become difficult to perform and easy to notice.

CyberLedger proves that the missing audit and security layer can also be supplied without making the system complicated or expensive. Every sale, refund, adjustment, and payment is permanently recorded with the user's name, role, and exact time, and no one can later erase or change that record. When stock does not match sales or unusual patterns appear, the owner can open a clear history and see exactly what occurred. This permanent trail provides the evidence that earlier systems lacked and that small café owners need when they cannot watch every transaction themselves.

Together the two systems prove that it is possible to keep the useful parts of existing digital tools while adding the controlled access and continuous monitoring that the literature shows are still missing. They prove that a solution can be built to match the real roles already used in Philippine cafés, stay simple enough for staff with limited training, work on ordinary devices, and remain affordable for businesses with modest capital. In this way, StockSecure POS and CyberLedger stand as concrete evidence that the gaps identified across all the reviewed studies are not permanent obstacles but solvable problems and that a purpose-built, security-conscious POS and ledger system can give resource-constrained SME cafés both the efficiency they need and the protection they have been lacking.

CHAPTER III:

Research Design

The study chose a **developmental research design**, which is particularly well-suited for the specific objectives pursued in the development and evaluation of the StockSecure POS system integrated with CyberLedger technology. Developmental research design, as a methodological framework, is distinguished from purely descriptive or correlational approaches by its explicit focus on the creation, improvement, and systematic evaluation of products, systems, or processes intended to address identified real-world problems.
Unlike experimental designs that seek to measure the effect of a manipulated variable on an outcome under controlled conditions, developmental research is oriented toward the iterative construction of a working artifact, in this case, a functional point-of-sale system, and the rigorous assessment of that artifact's performance across multiple dimensions in settings that closely approximate the environments in which it will ultimately be used. This dual emphasis on creation and evaluation makes developmental research design the most appropriate framework for a project whose primary contribution is not a theoretical model or an empirical dataset but a practical technological solution designed to meet the operational and security needs of small café businesses.

The rationale for selecting a developmental research design in this particular study is grounded in the problem being addressed and the type of knowledge that the research aims to generate. The challenges faced by café operators in the Philippines—unauthorized inventory access, unmonitored transactions, inadequate audit trails, and the persistent risk of internal fraud—are fundamentally operational problems that call for operational solutions. While descriptive research could document the prevalence and severity of these challenges, and correlational research could identify the factors that predict their occurrence, neither of these approaches is capable of delivering the practical tool that café owners actually need.

Developmental research design, by contrast, treats the development of that tool as the primary research activity and treats the systematic evaluation of its effectiveness as the mechanism by which the research generates new knowledge. This means that the knowledge produced by this study is not primarily propositional it does not take the form of statements about relationships between variables but practical and procedural: it takes the form of a working system, a development methodology, and a set of design principles that can guide the creation of similar systems in other contexts.

A defining characteristic of the developmental research design employed in this study is its commitment to grounding the development process in the real operational context of the cafés for which the system is being built. This contextual grounding is achieved through multiple mechanisms: the systematic collection of data about café operations through interviews, observation, and document analysis; the active involvement of café owners and staff as stakeholders throughout the development process; and the eventual deployment and evaluation of the system in actual café environments rather than in simulated laboratory conditions. This commitment to contextual authenticity is essential for ensuring that the final product is not merely technically functional in an abstract sense but genuinely useful and usable in the specific operational conditions that characterize Philippine café businesses. The café environment presents distinctive challenges that differ significantly from those of other retail or food service contexts: the combination of high transaction volumes during peak service periods, a relatively small and often informally managed staff, the use of both cash and electronic payment methods, and the physical constraints of a compact workspace all create an operational environment that the system must be designed to accommodate.

The developmental research design also provides an appropriate framework for addressing the security dimensions of this project, which are among its most technically challenging aspects. The implementation of role-based access control within a web-based POS system and the integration of a secure audit logging subsystem in the form of CyberLedger both require careful design decisions that have significant implications for both security and usability. Developmental research design accommodates these requirements by treating the evaluation of security effectiveness as a core component of the research process, rather than an afterthought. Through systematic testing procedures that include unit testing, system testing, and user acceptance testing, the development team is able to verify that the security mechanisms of StockSecure POS perform as intended and that they do so in a way that does not impose undue burdens on the system's users. This iterative security evaluation is an integral part of the developmental research process, ensuring that the system that emerges from the research process is both technically secure and practically deployable.

Furthermore, the developmental research design adopted in this study is explicitly oriented toward generalizability and replication. While the immediate goal of the research is to create a system that meets the needs of the specific café operators who serve as the study's primary stakeholders, the broader aim is to develop a model of secure POS system design that can be adapted and applied in other café and small food service environments. By carefully documenting the development process, the design decisions made at each stage, the feedback received from stakeholders, and the outcomes of the various testing and evaluation procedures, the study generates a body of knowledge that goes beyond the specific system developed to encompass the principles and practices that underpinned its creation. This documentation and reflection is a fundamental responsibility of developmental research: the researcher is not merely an engineer who builds a system but a scholar who critically examines the process of building and learning from it and who communicates those lessons in a form that others can apply.

It is also worth noting that the developmental research design chosen for this study is consistent with a growing trend in information systems and software engineering research toward practice-based and design science approaches that treat the creation of useful artifacts as a legitimate and valuable form of scholarly contribution. Design science research, which has gained significant traction in the information systems field since the landmark work of Hevner et al. (2004), argues that the development and evaluation of innovative IT artifacts constitutes genuine scientific knowledge creation, provided that the process is conducted rigorously, that the artifacts are evaluated against explicit criteria, and that the knowledge generated is communicated to both technical and management audiences. The current study adheres to these principles by establishing clear evaluation criteria, conducting systematic testing and assessment, and situating the development work within the broader literature on RBAC, POS systems, and small business technology adoption.

System Development Methodology

The system development methodology selected for the StockSecure POS and CyberLedger project is the **Rapid Application Development model**, a proven and widely adopted software engineering approach that is particularly well-suited to the constraints and objectives of this research. Rapid Application Development, which emerged as a formal methodology in the early 1990s through the work of James Martin and others, was developed as a response to the limitations of the traditional Waterfall development model, which required developers to fully specify all system requirements before any coding began and left little room for incorporating user feedback during the development process. The Waterfall model, while well-suited to large-scale projects with stable and well-understood requirements, was found to produce systems that often failed to meet users' actual needs because those needs changed during the extended development period or because the requirements gathered at the outset did not accurately capture the nuances of real operational workflows. RAD addressed these limitations by replacing the sequential, phase-gated structure of Waterfall development with an iterative cycle of prototyping, feedback, and refinement that keeps users and stakeholders continuously engaged throughout the development process.

The decision to adopt RAD for the development of StockSecure POS was made on the basis of several factors that make this methodology especially appropriate for this particular project. First and foremost, the operational requirements of a café POS system are sufficiently complex and context-specific that it would be impractical to attempt to fully specify them in advance without extensive real-world testing and stakeholder feedback. The details of how orders flow from the point of customer entry to the barista queue, how inventory levels are tracked and updated in real time as ingredients are consumed, how receipts are formatted and generated, and how the audit log captures and stores transaction data are all details that can only be fully understood and validated through working with actual café operators and observing the system in action. RAD's emphasis on iterative prototyping provides the ideal mechanism for uncovering and resolving these implementation details progressively, reducing the risk that the final system will contain fundamental design flaws that were invisible at the requirements stage.

Second, the RAD approach is particularly appropriate for a research project that involves the integration of multiple technically complex subsystems, specifically the RBAC framework and the CyberLedger audit logging system, whose interactions with each other and with the broader POS functionality need to be carefully validated at each stage of development. By developing the system incrementally and testing each component and integration point through successive prototyping cycles, the development team can identify and resolve technical issues before they compound into more serious architectural problems.

This incremental validation is especially important for the security-critical components of the system: an RBAC implementation error that allows unauthorized access to sensitive functions, or a CyberLedger configuration problem that causes audit records to be incomplete or corruptible, could have serious consequences if discovered only after the system has been deployed in a live café environment. RAD's iterative structure provides multiple opportunities to catch and correct such issues before they reach production.

Third, the RAD methodology's emphasis on stakeholder engagement aligns well with the research project's commitment to developing a system that genuinely meets the needs of its intended users. Café owners and staff are not passive subjects of this research but active collaborators in the development process, whose practical knowledge of café operations, whose expectations about system usability, and whose feedback on prototype functionality all play a crucial role in shaping the final product. RAD's structured feedback loops, in which working prototypes are presented to stakeholders for evaluation and their responses are systematically incorporated into the next development iteration, provide a formal mechanism for this collaboration that ensures stakeholder input is not merely gathered but actually implemented in the system. This participatory approach to development is not only good engineering practice but also a fundamental requirement of developmental research design, which demands that the artifact being developed be grounded in and validated by the real-world context it is intended to serve.

Phase One: Requirements Analysis

The first phase of the RAD methodology as applied to the StockSecure POS project is requirements analysis, which encompasses the comprehensive gathering and specification of all functional, technical, and security requirements that the system must satisfy. This phase is foundational to the entire development process: the quality and completeness of the requirements gathered at this stage directly determines the quality and completeness of the system that will ultimately be developed. Requirements analysis in the context of this project involves three distinct but interrelated activities: the identification and documentation of the operational requirements of café businesses, which define what the system must enable its users to do; the specification of the role hierarchy for the RBAC framework, which defines who can do what within the system; and the articulation of the audit specifications for the CyberLedger system, which define what data must be captured, how it must be protected, and how it must be retrievable.

The operational requirements of the café business were gathered through a combination of stakeholder interviews, operational observation, and document analysis, each of which is described in detail in the data gathering section of this chapter. The goal of this requirements gathering process was to develop a comprehensive understanding of the end-to-end workflow of café operations, from the moment a customer places an order to the moment the transaction is recorded in the financial system and the inventory is updated to reflect the ingredients consumed in fulfilling the order.

This understanding needed to be sufficiently detailed and accurate to serve as the basis for system design decisions, which meant that the requirements analysts needed to understand not only the high-level process flow but also the specific data elements involved at each step, the timing and sequencing constraints on system responses, and the exception cases and error conditions that the system would need to handle gracefully.

The specification of the RBAC role hierarchy was a particularly important and challenging aspect of the requirements analysis phase, because the effectiveness of the entire access control framework depends on the accuracy and completeness with which the roles, permissions, and hierarchical relationships are defined at the outset. The development team worked closely with café owners and senior staff to document the actual division of labor in a typical café operation, identifying the distinct functions performed by each type of employee, the data and system functionality required to perform each function, and the relationships of authority and supervision that exist between different staff roles. This analysis revealed a role structure comprising four primary roles: the café owner, who has full administrative access to all system functions and data; the manager, who can perform all cashier functions and additionally has access to inventory management, pricing adjustments, and operational reports; the cashier, who processes customer transactions, handles payment recording, and generates receipts; and the barista, who views order preparation queues and updates order status as items are prepared. The precise permission sets associated with each role were documented in detail during the requirements analysis phase, forming the specification against which the RBAC implementation would be tested and validated.

The audit specifications for CyberLedger were developed through a process of consultation with café owners about the types of audit data they would find most useful, combined with an analysis of the regulatory and operational requirements for financial record-keeping in the Philippine café sector. The audit requirements that emerged from this process included the comprehensive capture of all financial transactions, including GCash electronic payment reference numbers; the recording of all inventory adjustments and the identities of the staff members who made them; the logging of all login and logout events, including failed authentication attempts; the immutable storage of all audit records with cryptographic protection against tampering; and the provision of filtering and retrieval functions that allow administrators to access audit records for any specified time period, supporting both routine monitoring and forensic investigation of suspected fraud incidents.

Phase Two: System Design

The system design phase of the RAD process translates the requirements gathered in the previous phase into a detailed architectural specification that guides the actual development of the StockSecure POS and CyberLedger systems. System design encompasses multiple levels of abstraction, from the high-level architectural decisions about how the system's major components relate to one another down to the detailed design of individual data structures, algorithms, and user interface elements. Each of these design levels must be addressed carefully and coherently, ensuring that the overall architecture is sound, that the individual components are well-designed, and that the interfaces between components are clearly specified and consistently implemented.

At the highest level of abstraction, the architecture of StockSecure POS adopts a client-server model operating within a local network environment. This architectural choice reflects a deliberate decision to balance the performance, reliability, and security requirements of the system against the infrastructure constraints typical of small café businesses in the Philippines. A cloud-based architecture, while offering advantages in terms of accessibility and centralized management, was rejected on the grounds that it would introduce dependency on internet connectivity, which cannot be guaranteed in all of the environments where the system will be deployed, and that it would raise data security concerns associated with storing sensitive financial data on third-party servers. The local server architecture, by contrast, ensures that all data remains within the physical premises of the café, that the system can continue to operate during internet outages, and that the café owner retains complete control over the storage and management of their business data.

Within this client-server architecture, the server component of StockSecure POS is responsible for all data processing and business logic, including the evaluation of RBAC permissions, the execution of inventory updates, the management of the transaction database, and the operation of the CyberLedger audit logging subsystem. By centralizing all business logic on the server, the architecture ensures that security controls are applied consistently regardless of which client device a user is accessing the system from, and that the integrity of the business data is maintained even if individual client devices are lost, stolen, or compromised. The client components, which run in standard web browsers on desktop computers or tablets used by café staff, are responsible only for presenting the user interface and transmitting user actions to the server for processing. This separation of concerns between client and server is a fundamental principle of secure web application design and is essential for ensuring that the RBAC controls implemented on the server cannot be bypassed by a technically sophisticated user who might attempt to manipulate client-side code.

The database design for StockSecure POS was developed with particular attention to the integrity constraints and relationships required to support both the operational functions of the POS system and the audit logging functions of CyberLedger. The database schema was designed to ensure that every transaction, inventory adjustment, and user activity generates a corresponding record that is linked to the relevant user, role, and timestamp, creating the complete and traceable audit trail that is central to the system's fraud prevention capabilities. The use of SQLite3 as the database engine reflects a pragmatic choice suited to the deployment context: SQLite3 provides a reliable, efficient, and well-supported relational database that operates as a single file on the server, requiring no separate database server process and no complex database administration. This simplicity is well-suited to the resource-constrained environment of a small café server and reduces the administrative burden associated with maintaining the system over time.

The user interface design was developed with a strong emphasis on role-specific simplicity and operational efficiency. Each role in the RBAC system is presented with a distinct interface that shows only the functions and data relevant to that role, hiding the complexity of the full system behind a clean and intuitive façade that presents staff members with exactly what they need to do their jobs and nothing more. The cashier interface is dominated by the order entry and payment processing workflow, providing a clear and efficient path through the most frequently performed transactions. The barista interface is focused on the order queue, providing real-time visibility into pending orders and a simple mechanism for updating order status as items are prepared. The manager interface adds access to inventory management tools and operational reports, presented in a dashboard format that allows quick review of key performance indicators. The administrator interface, accessible only to the café owner, provides the full suite of system management functions, including user account management, role assignment, pricing configuration, and access to CyberLedger's audit log review module.

Phase Three: Rapid Prototyping

The rapid prototyping phase of the RAD methodology is where the abstract specifications produced in the design phase are transformed into tangible, testable artifacts that can be evaluated by stakeholders and used to gather feedback for subsequent development iterations. Rapid prototyping is distinguished from full-scale system development by its focus on speed and adaptability: prototypes are developed quickly with the explicit expectation that they will be modified and improved in response to feedback, and the scope of each prototype is deliberately limited to focus attention on the most critical or uncertain aspects of the system design. This targeted approach to prototyping allows the development team to resolve the most important design questions early in the process, before significant investment has been made in components that may need to be reworked.

The prototyping process for StockSecure POS was structured as a series of short iterative cycles, each of which targeted a specific subset of the system's functionality. The first prototyping cycle focused on the core transaction processing workflow, developing a basic order entry and receipt generation interface that could be evaluated by cashiers and café owners for its alignment with the actual order-taking and payment processes used in the café. This initial prototype deliberately excluded the RBAC framework and the CyberLedger logging system, allowing stakeholders to focus their attention on the fundamental transaction workflow without being distracted by security-related interface elements. The feedback gathered from this first prototype session provided valuable insights into the sequencing of order entry steps, the information required on receipts, and the formatting of the order queue interface viewed by baristas.

The second prototyping cycle incorporated the RBAC authentication and authorization framework, introducing the login interface, role-based permission controls, and role-specific views that are central to the system's security architecture. This prototype allowed stakeholders to experience first-hand how the different role-based interfaces felt from the perspective of each staff position, providing a basis for evaluating whether the permission structures and interface designs were appropriately aligned with actual job responsibilities. Feedback from this session was particularly valuable in refining the boundary between the cashier and manager roles, as stakeholders identified several functions that had initially been assigned to the manager role but were more naturally performed by cashiers in the actual café workflow, and vice versa. These adjustments were incorporated into the role specification before the next development iteration, ensuring that the final RBAC configuration would accurately reflect real operational practice.

Subsequent prototyping cycles progressively added the inventory management module, the CyberLedger audit logging interface, the reporting and analytics dashboard, and the system administration tools, with each cycle incorporating the feedback gathered in the previous session and expanding the scope of the prototype to encompass the next area of functionality. By the final prototyping cycle, all major system components had been demonstrated to stakeholders and had been refined in response to their feedback, providing a high degree of confidence that the system under development was well-aligned with their needs and expectations. The iterative nature of this process also had an important ancillary benefit: it gave café staff members multiple opportunities to interact with and become familiar with the system before its formal deployment, contributing to the employee preparedness that the Boado et al. study identified as a critical factor in successful POS adoption.

Phase Four: Full System Development

The full development phase builds directly on the validated prototypes produced in the preceding phase, transforming them into a complete, production-quality system that incorporates all of the features, security controls, and performance optimizations required for deployment in a live café environment. This phase is characterized by a more systematic and rigorous approach to development than the prototyping phase, reflecting the shift from exploratory design work to the construction of a finished product. While the iterative spirit of RAD is maintained throughout this phase, with ongoing stakeholder feedback informing development decisions, the primary focus shifts from discovering what the system should do to ensuring that it does those things reliably, efficiently, and securely.

The development of the RBAC implementation in the full development phase required particular care and attention to the security implications of every design decision. The RBAC framework is implemented at the server level using Python and the Flask web framework, ensuring that access control checks are applied at the point where all requests are processed before any business logic is executed or any data is returned to the client. Every route in the Flask application that provides access to a protected function or data resource is decorated with a custom authorization check that verifies that the currently authenticated user possesses a role with the requisite permissions for that resource. This approach ensures that it is impossible for a user to access a protected function simply by navigating directly to its URL, as would be possible in a client-side access control implementation. The authorization checks are implemented using a combination of Flask's session management capabilities and a custom role-permission mapping stored in the SQLite3 database, allowing the permission structure to be reviewed and modified through the administrative interface without requiring changes to the application code.

The development of the CyberLedger audit logging subsystem presented its own set of technical challenges, particularly in ensuring that the audit log is comprehensive, accurate, and tamper-resistant. The comprehensiveness requirement was addressed by instrumenting every route in the Flask application with logging calls that capture the identity of the requesting user, the role under which they are operating, the nature of the action being performed, the specific data records affected by the action, and the timestamp at which the action occurred.

This instrumentation was designed to be automatic and invisible to end users: all logging operations occur in the background without any user interaction or awareness, ensuring that the completeness of the audit record does not depend on users' willingness to perform explicit logging actions. The accuracy requirement was addressed by ensuring that audit records are written to the database in the same transaction as the actions they record, using SQLite3's transaction management capabilities to ensure that it is impossible for an action to be performed without a corresponding audit record being created.

The tamper-resistance requirement for the CyberLedger audit records was addressed through a combination of database-level and application-level controls. At the database level, audit records are stored in a dedicated table with no delete permissions granted to any application user, ensuring that records cannot be removed through normal application operations. At the application level, each audit record includes a cryptographic hash of its content, computed using a secure hash function, which allows any modification to the record to be detected by comparing the stored hash with a freshly computed hash of the record's current content. Additionally, each audit record includes a reference to the hash of the previous record in the sequence, creating a chain structure that makes it computationally infeasible to modify any record in the log without also recomputing the hashes of all subsequent records. This blockchain-inspired integrity verification mechanism provides a level of tamper-evidence that goes significantly beyond simple database access controls and ensures that the audit trail is reliable and trustworthy as an evidentiary record.

The development of the inventory management module required careful attention to the concurrency and consistency requirements that arise in a busy café environment where multiple transactions may be occurring simultaneously on different client devices. The SQLite3 database's native support for transactions and write locking was used to ensure that inventory updates are applied atomically, preventing the race conditions and data corruption that could occur if two clients attempted to update the same inventory record simultaneously. The real-time inventory update mechanism was implemented using a combination of server-side transaction logic and client-side polling, ensuring that each client's view of the inventory reflects the most current state of the database and that discrepancies between displayed and actual inventory levels are minimized. The inventory alert system, which notifies administrators when stock levels fall below specified thresholds, was implemented as a background process that continuously monitors inventory levels and generates alerts as needed, ensuring that café managers are promptly informed of potential stock shortages without having to actively monitor the inventory dashboard.

Phase Five: Testing

The testing phase of the RAD methodology encompasses the systematic verification and validation of all components and integrations of the StockSecure POS and CyberLedger systems, ensuring that the system performs correctly under normal operating conditions and behaves appropriately in the face of unusual or adversarial inputs. Testing in the context of this project is structured across three levels unit testing, system testing, and user acceptance testing, each of which addresses a different dimension of system quality and serves a distinct purpose in the overall quality assurance process.

The design and execution of this multi-level testing strategy reflect the recognition that no single testing approach can provide comprehensive assurance of system quality and that different types of defects are most effectively detected at different levels of testing granularity.

Unit testing, the most granular level of the testing strategy, focuses on verifying the correct behavior of individual components and functions in isolation from the rest of the system. In the context of StockSecure POS, the components subjected to unit testing include the RBAC authentication and authorization functions, the inventory update logic, the transaction recording procedures, and the CyberLedger logging and hash verification functions. Each of these components is tested by supplying it with a carefully designed set of test inputs and verifying that the resulting outputs and state changes match the expected behavior specified in the requirements documentation. Unit tests for the RBAC functions, for example, include tests that verify that authenticated users are granted access to the functions permitted by their role, that they are denied access to functions outside their role permissions, and that unauthenticated requests are correctly rejected and redirected to the login interface. Unit tests for the CyberLedger functions verify that audit records are correctly created for each loggable event, that the cryptographic hash values are correctly computed and stored, and that the chain integrity verification mechanism correctly detects any modifications to audit records.

System testing, the next level of the testing strategy, evaluates the behavior of the complete integrated system, verifying that all components work together correctly to support the end-to-end workflows that the system is designed to enable. System tests for StockSecure POS cover the complete transaction processing workflow, from order entry through payment recording to receipt generation and inventory update, ensuring that each step in the workflow is correctly performed and that the relevant audit records are created at each stage. System tests also cover the RBAC framework as it operates across the complete system, verifying that the role-based restrictions enforced by individual components compose correctly to produce the intended overall access control behavior. For example, a system test might verify that a user authenticated in the cashier role can successfully complete a transaction but cannot access the inventory adjustment function and that an attempt to access an unauthorized function generates an appropriate error response and a corresponding audit record.

Performance testing constitutes an important component of the system testing phase, ensuring that the StockSecure POS system can handle the transaction volumes and concurrent user loads typical of a busy café without degrading to unacceptable response times. The Apache JMeter load testing tool is used to simulate multiple concurrent users performing typical system operations, measuring the response times for key transactions under various load conditions and identifying any performance bottlenecks that require optimization. The performance testing protocol is designed to reflect the peak operational conditions of a café during lunch or afternoon rush periods, when multiple cashiers may be simultaneously processing transactions while baristas are updating order statuses and managers are reviewing inventory levels.

The results of performance testing are used to set performance benchmarks against which the deployed system can be evaluated and to identify any architectural or implementation changes required to meet the performance requirements specified in the system design.

User acceptance testing, the final level of the testing strategy, moves the evaluation of the system from the technical domain into the operational domain, engaging actual café staff and owners in a structured assessment of whether the system meets their business needs and operational requirements. User acceptance testing is conducted in a controlled but realistic setting, using realistic test data and operational scenarios that closely approximate the conditions of actual café service. Participants in the UAT process include representatives of each user role in the RBAC system: café owners, managers, cashiers, and baristas, each of whom evaluates the system from the perspective of their specific role and responsibilities. The UAT process is structured around a series of predefined test scenarios that cover all major system functions, with participants completing each scenario and providing structured feedback on the system's usability, functionality, and alignment with their actual operational workflows.

Phase Six: Deployment

The deployment phase of the RAD methodology involves the installation and configuration of the StockSecure POS system in the actual café environment where it will be used, the training of café staff on its operation, and the management of the transition from the café's existing operational systems to the new POS platform. Deployment is one of the most operationally sensitive phases of the development process, as it involves making changes to the live operational environment of a functioning business and carries the risk of disrupting normal service if not managed carefully. The deployment strategy developed for StockSecure POS was designed to minimize operational disruption by phasing the transition, beginning with a parallel operation period during which both the existing processes and the new system are used simultaneously, allowing staff to build confidence in the new system before full cutover.

The technical aspects of deployment involve the installation of the Flask application server on a dedicated local server device, the initialization of the SQLite3 database with the required schema and initial data, the configuration of the local network to ensure that all client devices can reliably access the server, and the setup of backup procedures to protect against data loss in the event of hardware failure. The local server deployment model was chosen in preference to cloud hosting for reasons already discussed in the system design section, but it does require careful attention to server reliability and data backup, as the café owner is responsible for the physical security and operational maintenance of the server hardware. The deployment documentation provided to café owners includes detailed guidance on server maintenance, backup procedures, and troubleshooting common technical issues, ensuring that they can manage the system effectively without requiring ongoing technical support for routine operations.

Staff training is a critical component of the deployment phase, and the approach to training adopted for StockSecure POS reflects the research findings that employee preparedness is one of the most significant factors determining the success of technology adoption in small businesses. The training program developed for StockSecure POS is structured around the role-based architecture of the system, with separate training modules for each user role that focus exclusively on the functions and workflows relevant to that role. Cashier training covers the order entry and payment processing workflow, receipt generation, and the basic principles of the RBAC system as they apply to the cashier role. Barista training covers the order queue interface and the order status update process. Manager training covers inventory management, operational reporting, and the additional transaction and user oversight functions available to the manager role. Administrator training, provided to café owners, covers the complete range of system functions, including user account management, CyberLedger audit log review, and system configuration.

The training delivery approach was designed to be practical and hands-on, reflecting the reality that café staff members learn most effectively by doing rather than by reading documentation or watching demonstrations. Each training session involves guided practice with the actual system, using realistic scenarios derived from the café's actual operational workflows. Participants are given the opportunity to make mistakes and experience the system's error handling and access control mechanisms in a safe training environment, building their understanding of the system's behavior and their confidence in their ability to use it correctly. The training program also includes a reference guide for each role, provided in both printed and digital formats, that summarizes the key workflows and provides step-by-step instructions for the most common tasks. This reference material is designed to be used by staff members who encounter unfamiliar situations after training, providing a quick and accessible reminder of the correct procedure without requiring them to seek assistance from a trainer or technical support.

Phase Seven: Evaluation and Refinement

The evaluation and refinement phase of the RAD methodology is the final stage of the development cycle for StockSecure POS and CyberLedger, but it is also the beginning of an ongoing process of continuous improvement that is expected to continue throughout the operational life of the system. Evaluation involves the systematic collection and analysis of data about the system's performance and impact in the real-world café environment, using both quantitative metrics and qualitative stakeholder feedback to assess the extent to which the system is achieving its intended objectives. Refinement involves translating the insights gained from evaluation into specific improvements to the system's design, functionality, and deployment, implementing these improvements through additional development cycles and redeploying the updated system for continued evaluation.

The evaluation framework for StockSecure POS is structured around the five criteria established in the system requirements: functionality, usability, reliability, efficiency, and security. Each criterion is assessed through a combination of automated testing, structured stakeholder feedback, and operational data analysis, providing a multi-dimensional picture of the system's performance that is more comprehensive and reliable than any single evaluation method could provide alone. The evaluation process is conducted over an extended period following deployment, allowing the system's performance to be assessed under the full range of operational conditions that the café experiences over a typical business cycle, rather than only during the initial post-deployment period when operations may not yet have fully normalized.

The feedback gathered during the evaluation phase is systematically analyzed to identify patterns of user difficulty, functional gaps, and performance limitations that warrant attention in the next refinement cycle. This analysis is conducted collaboratively with café owners and staff, ensuring that the prioritization of refinement activities reflects the actual operational priorities of the system's users rather than the technical preferences of the development team. The iterative nature of the RAD methodology means that the evaluation and refinement cycle can be repeated multiple times as the system matures and as the operational context evolves, allowing the system to be progressively improved over time in response to changing user needs, emerging security threats, and new technological opportunities.

System Architecture

The system architecture of StockSecure POS integrated with CyberLedger represents a carefully considered set of design decisions that balance the competing demands of security, performance, usability, and operational practicality in the context of a small café environment. The architecture adopts a client-server model in which a central application server handles all data processing and business logic, while multiple client devices running standard web browsers provide the user interface through which café staff interact with the system. This architectural pattern is well-established in enterprise web application development and has been specifically adapted for the resource-constrained and security-sensitive environment of a small café business, with particular attention to the requirements imposed by the RBAC framework and the CyberLedger audit logging subsystem.

The client layer of the architecture encompasses all of the devices used by café staff to access the StockSecure POS system: desktop computers at the cashier stations, tablets used by managers for on-the-floor inventory oversight, and any additional devices used by administrators for system management and audit review. All of these client devices access the system through standard web browsers, including Google Chrome, Microsoft Edge, and Mozilla Firefox, without requiring the installation of any specialized client software.

This browser-based access model significantly reduces the complexity and cost of deploying and maintaining the system across multiple client devices, as updates to the application are applied centrally on the server and are immediately available to all clients without requiring individual device updates. The responsive web interface design ensures that the system provides an appropriate and usable experience across the range of device types and screen sizes that may be found in a typical café environment, from large desktop monitors at cashier stations to smaller tablet screens used by managers.

The application server is the core of the StockSecure POS architecture, responsible for processing all client requests, enforcing the RBAC access control policies, executing business logic, managing the database, and operating the CyberLedger audit logging subsystem. The server runs the Flask web application framework on Python, providing a lightweight but highly capable platform for implementing the RESTful API endpoints through which the client interfaces communicate with the server-side business logic. Flask was chosen for this role because of its simplicity, its extensive ecosystem of security-focused libraries and extensions, and its suitability for deployment on modest hardware, which is consistent with the typical server infrastructure available to small café businesses. The server's responsibilities are organized into discrete application modules, each handling a specific domain of functionality: the authentication module manages user sessions and enforces login requirements; the RBAC module evaluates access control decisions for every protected resource; the transaction module processes order entries, payment recordings, and receipt generation; the inventory module manages stock levels and generates alerts; the reporting module produces operational summaries and analytics; and the CyberLedger module handles all aspects of audit logging, including record creation, hash computation, and audit retrieval.

The CyberLedger subsystem operates as a secure audit backbone that is deeply integrated into every layer of the StockSecure POS application architecture. Rather than being an optional add-on or a separate system that receives data from the main POS application, CyberLedger is woven into the core processing logic of the application, ensuring that audit records are generated as a natural by-product of every system operation rather than as a separate, optional step. This deep integration is essential for the comprehensiveness and reliability of the audit trail: if the logging function were implemented as an optional or disconnected process, it would be vulnerable to failures that might cause audit records to be lost without any visible indication to administrators or to deliberate circumvention by users who understood the architecture well enough to exploit its gaps. By integrating CyberLedger at the application level, the architecture ensures that the generation of audit records is inseparable from the processing of the operations they record, providing a level of reliability and comprehensiveness that is essential for a forensically useful audit trail.

The database layer of the architecture uses SQLite3 to provide persistent storage for all of the application's data, including transaction records, inventory levels, user accounts and role assignments, system configuration parameters, and the CyberLedger audit log. The database schema is organized into distinct logical domains that correspond to the major functional areas of the application, with careful attention to the integrity constraints and referential relationships required to ensure data consistency.

The audit log tables are particularly carefully designed: they include fields for the user identity, role, action type, affected records, timestamp, and cryptographic hash of each entry, and they are protected by database-level constraints that prevent the deletion or modification of audit records through the normal data manipulation operations available to the application.

The network layer of the architecture operates entirely within the local network of the café, without any connection to the public internet being required for normal operation. The local server is connected to the café's local area network, to which all client devices are also connected, either through wired Ethernet connections or through the café's local Wi-Fi network. The choice to operate exclusively within the local network rather than over the internet provides significant security benefits, as it eliminates the exposure of the application to the range of network-based attacks that target internet-connected applications, including SQL injection attempts, cross-site scripting, credential stuffing, and distributed denial of service attacks. It also ensures that the system continues to operate normally during internet outages, which are a practical concern in many Philippine café locations and could otherwise cause significant operational disruption if the system were cloud-hosted.

Tools and Technologies Used

Programming Languages

The selection of programming languages for the StockSecure POS and CyberLedger systems was carefully driven by the competing demands of robust security, strong performance, high developer productivity, and long-term maintainability. After thorough evaluation, **Python** emerged as the clear choice for the primary backend language. It powers all server-side application logic, including the sophisticated Role-Based Access Control (RBAC) framework, complex business process workflows (P1 through P6), the tamper-evident CyberLedger audit logging system, database interactions with SQLite3, hardware integration for receipt printing, PDF generation with ReportLab, and encryption routines using cryptography and Fernet. This decision has proven instrumental in delivering a reliable, secure, and practical POS solution tailored to the real-world needs of Dulci Cafe.

Python's greatest strength lies in its exceptionally clean and readable syntax, which significantly reduces the cognitive load when implementing security-critical components. In a system where even small mistakes could compromise financial data or audit integrity, Python's straightforward, English-like code makes complex logic easier to write, review, and debug. Developers can focus on solving business and security problems rather than wrestling with verbose or cryptic syntax. For instance, the RBAC permission checks and the HMAC-SHA256 chaining logic in CyberLedger are expressed clearly and concisely, lowering the risk of subtle implementation flaws that might go unnoticed in more complex languages. This readability also accelerates code reviews and makes it easier for new team members or future maintainers to understand the system quickly — a vital advantage for a small business like Dulci Cafe that may not have a large dedicated development team.

Beyond syntax, Python's rich standard library provides powerful, battle-tested tools out of the box. Modules for cryptographic operations, JSON handling, regular expressions, threading for background tasks (such as the auto-backup daemon and low-stock monitoring), and secure random number generation allow the team to build most core functionality with minimal reliance on external packages. This reduces the attack surface and eliminates concerns about unmaintained third-party dependencies. When additional capabilities are needed — such as bcrypt password hashing via Werkzeug, Fernet encryption, or ESC/POS printer commands — the ecosystem offers well-maintained, community-vetted libraries that integrate seamlessly. Python's performance is more than sufficient for a café environment, easily handling concurrent transactions, real-time inventory updates, and report generation during peak hours without requiring premature optimization.

The language's massive, active community further strengthens its suitability. Security vulnerabilities are identified and patched rapidly through coordinated efforts, while extensive documentation, tutorials, and Stack Overflow discussions speed up problem-solving. This ecosystem maturity translates into faster development cycles and greater confidence in long-term stability. For Dulci Cafe, Python also supports smooth offline-first operation, easy deployment on modest hardware, and straightforward integration with tools like python-dotenv for secure configuration management.

In practice, choosing Python has enabled the StockSecure POS team to deliver a system that feels both enterprise-grade in its security and compliance features and refreshingly approachable for everyday café operations. The language's balance of simplicity and power has accelerated implementation of critical features like real-time stock alerts, professional PDF reports, and immutable audit trails while keeping the codebase clean and maintainable. This thoughtful technology choice ultimately helps owners focus more on serving customers and growing their business, knowing their POS system rests on a solid, trustworthy, and future-proof foundation.

At the heart of the StockSecure POS user experience for Dulci Cafe lies the thoughtful combination of **HTML5** and **CSS3**, which together provide the structural backbone and visual polish that makes the system feel modern, intuitive, and welcoming to staff who use it hour after hour. HTML5 supplies clean, semantic markup that gives every page a logical, meaningful structure. Instead of generic divs, the system uses proper elements like <header>, <nav>, <main>, <section>, <article>, and <button> to define role-specific interfaces. This semantic approach not only improves accessibility for screen readers and future maintainability but also helps search engines and development tools understand the content while working seamlessly with Jinja2's server-side rendering to deliver fully formed, role-appropriate pages right from the first load.

A cashier logging in sees a focused transaction interface with clearly marked product grids, a prominent cart area, and large, touch-friendly payment buttons. A manager opening their dashboard encounters well-organized sections for sales summaries, low-stock alerts, and staff activity—all structured in a way that feels natural and easy to scan even during a busy shift. Because the HTML is generated server-side based on RBAC decisions, each user receives exactly the content they need, with no hidden or restricted elements cluttering the page.

CSS3 then brings these structured pages to life with consistent, professional styling that reflects Dulci Cafe's warm and clean brand identity. Modern CSS features such as Flexbox and CSS Grid enable flexible, elegant layouts that adapt gracefully to different content volumes. Carefully chosen typography, color schemes (with high-contrast modes for visibility under varying café lighting), subtle hover effects, and clear visual hierarchy guide users' attention exactly where it needs to go—whether highlighting a low-stock item in red or emphasizing the grand total during checkout.

One of the most practical advantages is CSS3's powerful responsive design capabilities. Using media queries, flexible units (rem, em, percentages), and mobile-first principles, the interface automatically adjusts to the full range of devices used in the café: large desktop monitors at the main counter, mid-sized tablets carried by roaming managers or baristas, and even smaller screens for quick checks. Buttons enlarge on touch devices, product cards reflow into comfortable grids, and side menus collapse into accessible hamburger navigation—eliminating the need for frustrating zooming, scrolling, or pinching. This responsiveness ensures that staff can complete transactions quickly during morning rushes without fighting the interface, reducing errors and customer wait times.

Together, HTML5 and CSS3 create more than just a pretty frontend — they deliver a reliable, performant, and human-centered interface that respects the realities of café work. Fast loading, consistent behavior across devices, and thoughtful visual feedback all contribute to higher staff satisfaction, fewer training headaches, and smoother daily operations. By combining these foundational web technologies with the security and logic of the Python backend, StockSecure POS offers an experience that feels both professionally robust and pleasantly easy to use, helping Dulci Cafe deliver great service while maintaining strong internal controls.

In the StockSecure POS system developed for Dulci Cafe, **JavaScript (ECMAScript 2015/ES6 and later versions)** plays a vital role on the client side, transforming what could have been a static, clunky web interface into a fast, responsive, and genuinely pleasant tool that café staff actually enjoy using during hectic service hours. While the backend—built on Flask, Jinja2, and Python—handles all the heavy lifting for security, business logic, and data integrity, JavaScript steps in to create the smooth, real-time experience that makes daily operations feel effortless rather than frustrating.

When a cashier logs in and starts a new order, JavaScript immediately springs into action. It powers dynamic product search with instant filtering as the cashier types, automatically updates the running subtotal, tax, and grand total in real time as items are added or removed, and handles quick modifiers like "add sugar" or "make it iced" without requiring page reloads. Client-side form validation catches simple mistakes—such as missing customer details or invalid discount codes—before they even reach the server, providing friendly inline feedback that helps new staff learn the system quickly and reduces unnecessary server round-trips.

One of the most appreciated features is the seamless asynchronous communication with the server using the modern Fetch API and async/await patterns. When a sale is completed, JavaScript sends the order data to the backend, then instantly updates the interface to show the next customer screen while the receipt prints in the background. Managers benefit enormously from the interactive **Chart.js** dashboards: JavaScript makes the sales trend graphs zoomable, filterable by date or shift, and responsive across devices—from the large monitor in the back office to a tablet carried around the floor. Hovering over a bar instantly displays exact figures, helping owners make quick decisions about staffing or promotions without digging through reports.

Beyond functionality, JavaScript enhances accessibility and user experience in meaningful ways. Keyboard shortcuts for power users, real-time low-stock highlights that gently pulse on the cashier screen, and smooth animations for adding items to the cart all contribute to a less stressful environment during morning rushes or lunch peaks. The code is deliberately kept lightweight and focused—avoiding complex client-side state management frameworks—to maintain performance even on older café computers.

Security remains paramount: JavaScript never performs critical operations like password validation, RBAC enforcement, or ledger updates. Those stay firmly on the server where they belong. Client-side code only handles presentation, basic validation, and user-friendly interactions, with all sensitive data protected through server-side rendering and automatic escaping. This clear separation gives developers confidence that even if a technically curious staff member tries to inspect or manipulate the browser console, they cannot bypass permissions or alter financial records.

By thoughtfully applying modern JavaScript, StockSecure POS bridges the gap between powerful backend capabilities and the human reality of a busy café. It creates an interface that feels responsive, forgiving, and intuitive—qualities that translate directly into faster service, fewer errors, happier staff, and more time for what matters most: serving great coffee and building customer relationships. In the context of this study, JavaScript proves essential not just as a programming language but as the humanizing layer that makes a secure, compliant enterprise system feel approachable and efficient for real people working long hours on their feet.

Frameworks & Libraries

StockSecure POS is built on a thoughtfully curated Python technology stack that prioritizes security, simplicity, maintainability, and performance while delivering a robust, compliant point-of-sale experience tailored for Dulci Cafe and similar establishments. At its foundation lies Flask 3.0, the lightweight and highly flexible micro web framework chosen specifically because it avoids imposing rigid architectural patterns or excessive built-in features common in more opinionated frameworks. Flask provides a clean core focused on HTTP request routing, session management, and response handling, allowing the development team to implement custom role-based access control (RBAC), comprehensive audit logging through CyberLedger, and multi-location café configurations without fighting against framework constraints. This minimal core significantly reduces the application's attack surface, improves deployment efficiency on modest café hardware, and ensures seamless scalability as the business grows, all while maintaining excellent performance under concurrent cashier and manager usage during peak hours.

Complementing Flask is Jinja2, its default server-side templating engine, which dynamically generates every HTML interface ranging from the login screen and transaction terminals to managerial dashboards and owner reports according to the authenticated user's specific role and permissions. By rendering content exclusively on the server, StockSecure enforces strict access control principles: restricted elements such as advanced audit tools or sensitive financial summaries are never transmitted to the client, even in hidden form, thereby eliminating client-side bypass risks. Jinja2's automatic HTML escaping, backed by MarkupSafe, further fortifies the system against Cross-Site Scripting (XSS) attacks by converting potentially malicious user inputs or database content into safe HTML entities, while its support for template inheritance, macros, and custom filters keeps the front-end codebase clean, consistent, and easily maintainable across devices.

Underpinning Flask's functionality is Werkzeug, the comprehensive WSGI utility library that supplies critical security primitives, most notably its bcrypt-based password hashing utilities. These ensure user credentials are stored using an adaptive, computationally expensive algorithm that resists brute-force, dictionary, and rainbow-table attacks even in the event of a database breach. Werkzeug also delivers secure session cookie management, robust input parsing to mitigate malformed request attacks, and safe file-handling routines used during data exports and backups, effectively preventing path traversal vulnerabilities. For data visualization, Chart.js powers the interactive operational dashboards accessible to managers and owners, rendering responsive bar charts, line graphs, pie charts, and doughnut visualizations directly in the browser via HTML5 Canvas. These charts provide real-time insights into sales trends, inventory consumption, transaction patterns by time of day, void activity, and staff performance, enabling informed decisions on staffing, purchasing, menu optimization, and promotions without any server-side image generation overhead.

When formal documentation is required, ReportLab takes center stage for high-quality server-side PDF generation, producing professional, print-ready reports such as daily sales breakdowns with tax summaries, monthly inventory variance analyses, low-stock alerts with color-coded tables, and complete CyberLedger audit exports. Operating entirely on the backend, ReportLab supports embedded logos, vector graphics, barcodes, precise layout control, repeating headers and footers, and complex multi-page structures that align with BIR compliance standards, ensuring that sensitive data exits the system only in finalized, tamper-evident formats suitable for archiving, emailing, or regulatory submission. Image processing needs are elegantly handled by Pillow, the modern Python Imaging Library fork, which generates secure, math-based CAPTCHA challenges for login and password reset flows. These challenges feature distorted arithmetic expressions with randomized fonts, colors, noise, and warping to thwart automated bots and OCR attempts, all created in memory and streamed directly to the client without disk persistence, maintaining a minimal attack surface while preserving user privacy by avoiding external CAPTCHA services.

Sensitive data at rest receives strong protection through the cryptography library's Fernet implementation, which delivers authenticated symmetric encryption (AES-128-CBC combined with HMAC-SHA256) for fields like employee payroll details, supplier banking information, and configuration secrets. Unique ciphertexts for identical plaintexts, automatic tampering detection via authentication tags, and environment-variable-managed keys ensure defense-in-depth even if the underlying database is fully compromised. ItsDangerous complements this by providing stateless, cryptographically signed tokens for CAPTCHA validation and password reset links, embedding expiration timestamps and preventing replay or forgery attacks without requiring additional database storage. Finally, python-dotenv securely manages all configuration database credentials, encryption keys, Flask secrets, and SMTP settings by loading them from a Git-ignored .env file at startup, enabling identical code to run across development, staging, and production environments with environment-specific secrets, dramatically reducing the risk of credential leakage while simplifying deployment and maintenance.

Collectively, this integrated stack of Flask, Jinja2, Werkzeug, Chart.js, ReportLab, Pillow, cryptography, itsdangerous, python-dotenv, and MarkupSafe creates a cohesive, secure-by-design ecosystem that balances powerful functionality with minimal complexity. Each component was selected for its proven security track record, lightweight footprint, and seamless interoperability, resulting in a POS system that not only meets the day-to-day operational demands of a busy café but also exceeds compliance, auditability, and data-protection expectations in a high-risk financial environment. This architecture delivers confidence to café owners that their business data remains protected, their reports are professional and reliable, and their staff workflows stay efficient and intuitive, all while providing a solid foundation for future enhancements such as additional integrations or multi-branch expansions.

Python Standard Library Modules

The development of StockSecure POS makes extensive use of Python's comprehensive standard library, drawing on a range of built-in modules that provide the low-level functionality required by the application without introducing dependencies on external packages whose security and reliability would need to be independently assessed. The functools module provides higher-order function utilities that are used extensively in the RBAC implementation to create decorator functions that can be applied to Flask route handlers to enforce access control requirements cleanly and consistently. The datetime module provides the date and time handling capabilities required for timestamping transactions, generating audit records, and implementing the date-based filtering functions of the CyberLedger audit retrieval interface. The contextlib module provides utilities for implementing context managers that are used to ensure the correct initialization and cleanup of database connections and other resources that require explicit lifecycle management.

The sqlite3 module, which provides Python's interface to the SQLite database engine, is the foundation of all data persistence in StockSecure POS. This module's DB-API 2.0-compliant interface is used for all database operations, including schema creation and migration, transaction management, query execution, and result retrieval. The module's support for parameterized queries, in which variable values are passed as parameters rather than being interpolated directly into query strings, is used consistently throughout the application as a defense against SQL injection attacks, which are among the most common and dangerous vulnerability classes in web applications. The os module provides the operating system interfaces required for file path manipulation, environment variable access, and process management operations used in the application's startup, configuration, and maintenance functions. The secrets module provides cryptographically secure random number generation capabilities that are used for generating session tokens, CSRF protection tokens, and other security-sensitive random values that require a higher quality of randomness than the standard random module can provide.

Their module, which provides Python's regular expression processing capabilities, is used in several parts of the application for input validation and sanitization, ensuring that user-supplied data meets the expected format requirements before being processed or stored. Regular expression validation is applied to inputs such as user names, email addresses, GCash reference numbers, and product identifiers, rejecting inputs that do not match the expected pattern and thereby preventing a range of input-based attack techniques. The **JSON module** handles the serialization and deserialization of data exchanged between the server-side Python code and the client-side JavaScript, ensuring that this data transfer occurs in a standardized and safely handled format. The html module provides HTML escaping functions that are used as a secondary defense against cross-site scripting attacks, complementing the automatic escaping provided by the Jinja2 template engine. The shutil module provides high-level file operations used in the application's backup and data export functions, while the threading and time modules support the background processing tasks that handle real-time inventory monitoring and alert generation.

Data Gathering Techniques

The strategy employed in this research was designed to provide the development team with a comprehensive and accurate understanding of the operational environment, user needs, and security requirements that the StockSecure POS system must address. This strategy draws on three complementary data collection methods stakeholder interviews, operational observation, and document analysis, each of which captures a different dimension of the complex operational reality of a café business and contributes distinct and non-redundant insights to the overall picture. The use of multiple data collection methods, a practice known as triangulation in research methodology, enhances the validity and reliability of the findings by ensuring that the conclusions drawn from the data are supported by multiple independent sources rather than depending entirely on any single method, which might introduce its own systematic biases or blind spots.

Interviews with café owners and employees constituted the primary data gathering method for the requirements analysis phase of the project, providing direct access to the tacit knowledge and practical experience of the people who manage and work in the café environments where StockSecure POS will ultimately be deployed. The interview protocol was designed to elicit detailed information about three core areas of the informants' experience: the specific challenges they face in managing inventory and preventing unauthorized access to sensitive business data; the specific functions and features they would require from a POS system to address these challenges effectively; and the role structures and hierarchical relationships that define the division of labor in their café operations. The interviews were conducted in a semi-structured format, using a prepared set of open-ended questions to ensure that all critical topics were covered while allowing informants the flexibility to elaborate on issues that they considered particularly important or problematic.

The interview findings provided an invaluable foundation for the RBAC role specification, revealing the nuances of role definition and permission assignment that could not have been reliably inferred from general knowledge about café operations. For example, the interviews revealed that in many small cafés, the functions traditionally associated with the cashier role and the barista role are sometimes performed by the same individual, particularly during slower service periods, creating a need for the system to support flexible role assignment without compromising the access control architecture. The interviews also revealed that café owners have highly specific and sophisticated ideas about what constitutes suspicious behavior in their operations, the kinds of transaction patterns, inventory discrepancies, and access events that would raise a red flag in their minds, which provided directly actionable guidance for the design of the CyberLedger alert generation logic.

Operational observation provided a complementary and in some respects deeper source of data than the interviews, capturing aspects of café workflows that informants might not think to mention because they are so deeply embedded in everyday practice that they are effectively invisible. The observation sessions were conducted during both peak service periods, typically the morning coffee rush and the lunchtime peak, and quieter off-peak periods, providing a representative picture of how the café operates across the full range of conditions the system will need to support. During observation sessions, the research team documented the sequence and timing of key operational events, including order entry, ingredient retrieval, beverage preparation, payment processing, and receipt generation, as well as the informal communication patterns between staff members that coordinate these activities. This detailed process mapping was essential for designing a POS workflow that integrates naturally with the café's existing operational rhythms rather than disrupting them.

Document analysis of existing café records, including inventory management documents, sales logs, incident reports, and financial summaries, provided a third layer of data that was particularly valuable for understanding the current state of inventory management practice and the historical patterns of discrepancy and loss that the system is designed to prevent. The analysis of inventory records revealed the frequency and magnitude of stock discrepancies, the categories of products most susceptible to loss or theft, and the procedures currently used to investigate and document discrepancies. The analysis of sales logs revealed the volume and timing of transactions, the distribution of payment methods, and the patterns of product sales that would be expected in a well-functioning system, providing a baseline against which the system's anomaly detection logic could be calibrated. The analysis of incident reports, where available, provided direct evidence of the types of internal fraud and unauthorized access events that café owners have encountered, informing the design of CyberLedger's alert logic and audit retrieval interface.

Testing Procedures

The testing procedures employed in the validation of StockSecure POS and CyberLedger are designed to provide comprehensive assurance of the system's correctness, security, performance, and usability across the full range of conditions it will encounter in operational deployment. The testing strategy is structured around three complementary levels of testing unit testing, system testing, and user acceptance testing that together provide a thorough and multi-dimensional assessment of system quality. Each level of testing targets a different scope of system behavior, uses different evaluation criteria and methods, and involves different participants, ensuring that the testing program captures both the technical correctness of individual components and the operational appropriateness of the system as a whole.

Unit testing is conducted by the development team throughout the implementation phase, with tests written and executed concurrently with the development of each individual component. The unit testing program for StockSecure POS is organized around the principal functional modules of the application: the authentication module, the RBAC authorization module, the transaction processing module, the inventory management module, the receipt generation module, and the CyberLedger logging module. For each module, a comprehensive suite of test cases is developed that covers both the expected normal behavior of the module and the edge cases and error conditions that the module must handle correctly. The unit tests for the RBAC authorization module, for example, include test cases that verify the correct behavior of the system when a user attempts to access a resource for which their role does possess the required permission, when they attempt to access a resource for which their role does not possess the required permission, when no user is currently authenticated, and when the database query that retrieves the user's role assignment returns an unexpected result due to a data integrity issue.

System testing evaluates the complete integrated system, verifying that all components interact correctly to produce the intended end-to-end behaviors across the full range of supported workflows. System testing for StockSecure POS covers every major operational scenario that the system is designed to support: the complete order-to-receipt workflow for a cash transaction, the complete order-to-receipt workflow for a GCash electronic payment including reference number recording, the inventory update workflow triggered by transaction processing, the inventory alert notification workflow triggered by stock levels falling below defined thresholds, the user account creation and role assignment workflow, the audit log generation and retrieval workflow, and the various error handling and access denial scenarios that arise when users attempt unauthorized actions. Each scenario is executed by the testing team using a standardized script that specifies the input data, the sequence of actions, and the expected outcomes at each step, ensuring that the testing is reproducible and that the results can be compared reliably across different test runs.

Security testing, which constitutes a specialized component of the system testing phase, systematically evaluates the effectiveness of the access control and audit logging mechanisms under a range of adversarial conditions designed to probe for vulnerabilities. The security test plan includes tests for all of the common web application vulnerability categories identified in the OWASP Top Ten security risks framework, including SQL injection, cross-site scripting, broken authentication, insecure direct object references, security misconfiguration, and insufficient logging and monitoring. For each category, the testing team uses a combination of automated scanning tools and manual penetration testing techniques to identify any implementation weaknesses that could allow an attacker to circumvent the system's security controls. The results of security testing are used to prioritize remediation activities and to verify that identified vulnerabilities have been successfully addressed in subsequent development iterations.

User acceptance testing involves actual café staff and owners in a structured evaluation of the system's fitness for purpose from an operational perspective. The UAT protocol is developed in collaboration with café owners and is organized around a set of realistic operational scenarios that represent the tasks that each user role will need to perform on a typical business day. UAT participants are given minimal instruction on the operation of the system prior to the testing session, allowing the testing to assess the intuitiveness and discoverability of the interface under conditions that simulate the experience of a new user. Participants are asked to complete each scenario and to verbalize their thought processes and any difficulties they encounter as they do so, providing the development team with rich qualitative data about usability issues that might not be apparent from quantitative success rate metrics alone. Structured feedback forms are used to collect ratings and comments on specific aspects of the system's usability, functionality, and alignment with operational workflows, providing quantitative data that can be analyzed and compared across participants and scenarios.

Evaluation Criteria

The evaluation of StockSecure POS and CyberLedger is structured around five core criteria that collectively encompass the full range of qualities that the system must demonstrate to be considered a successful solution to the operational and security challenges it was designed to address. These criteria functionality, usability, reliability, efficiency, and security were developed through a combination of review of the relevant information systems evaluation literature and consultation with café owners and staff about the qualities they consider most important in a POS system. Each criterion is operationalized through a specific set of metrics and evaluation methods that provide objective and interpretable evidence of the system's performance on that dimension, enabling a rigorous and defensible assessment of the system's overall fitness for purpose.

Functionality, the first and most foundational evaluation criterion, assesses the completeness and correctness of the system's implementation of all the features and workflows specified in the requirements documentation. A system that is highly usable, reliable, efficient, and secure is of limited value if it does not actually perform the functions that users need it to perform, and conversely, a system that performs all required functions correctly is providing its users with genuine value even if it falls short in other dimensions. The assessment of functionality is conducted through structured functional testing that systematically exercises every major feature of the system, including order processing, payment recording, receipt generation, real-time inventory management, role-based access control, CyberLedger transaction logging, and audit record retrieval. Each feature is evaluated against a detailed specification of its required behavior, and a functionality score is computed as the proportion of specified behaviors that are correctly implemented and performed without errors across a representative set of test inputs.

Usability, the second evaluation criterion, assesses the degree to which the system's interface is intuitive, efficient, and satisfying for the users of each role. Usability is evaluated through the structured stakeholder feedback gathered during user acceptance testing, supplemented by analysis of task completion rates, error rates, and task completion times recorded during UAT sessions. The evaluation framework follows the principles of ISO 9241-11, the international standard for usability, which defines usability as the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use. Effectiveness is measured by task completion rates; efficiency is measured by task completion times; and satisfaction is measured by structured feedback ratings on standardized usability scales applied to each user role's interface. The usability evaluation is conducted separately for each role, cashier, barista, manager, and administrator, reflecting the role-specific nature of the interfaces that each user group will actually work with.

Reliability, the third evaluation criterion, assesses the consistency and correctness of the system's data management and operational performance under the demanding conditions of peak café service. Reliability is particularly important in the POS context because failures or inconsistencies in transaction recording or inventory updating during peak service periods can have immediate and serious consequences for the café's business operations, potentially resulting in unrecorded sales, incorrect inventory levels, and gaps in the CyberLedger audit trail. The assessment of reliability involves extended operational testing during simulated peak service conditions, with particular attention to the consistency of data between the transaction records, the inventory database, and the CyberLedger audit log. Test cases are designed to verify that every transaction that is processed through the POS system is correctly recorded in all three data stores and that the inventory levels and audit records remain consistent with the transaction history even under conditions of high concurrent load.

Efficiency, the fourth evaluation criterion, assesses the performance of the system in terms of response times and throughput under operational load conditions. In a busy café environment, where customers expect prompt service and cashiers need to process transactions quickly, the performance of the POS system is a direct determinant of service quality. A system that introduces unacceptable latency into transaction processing will create queues, frustrate customers, and reduce the café's ability to serve customers efficiently during peak periods. The efficiency assessment is conducted using Apache JMeter, a widely used open-source load testing tool that simulates multiple concurrent users performing typical system operations and measures the response times experienced by each user. Performance benchmarks are established for the key transaction types order entry, payment recording, receipt generation, inventory update, and audit log query, and the system is evaluated against these benchmarks under load conditions representing typical peak service scenarios. Response time targets are set in consultation with café owners, who have direct experience of the performance standards that their operations require.

Security, the fifth and in many respects most critical evaluation criterion, assesses the effectiveness of the system's access control and audit logging mechanisms in protecting sensitive business data and preventing unauthorized actions. Security evaluation is inherently more complex than the other evaluation criteria because it involves assessing not only what the system does under normal operating conditions but also how it behaves when subjected to deliberate attempts to circumvent its controls. The security evaluation of StockSecure POS encompasses three principal assessments: verification that the RBAC framework correctly enforces the specified access control policies across all user roles and all protected system functions; verification that the CyberLedger system correctly captures, encrypts, and stores all specified audit events in a tamper-resistant manner; and penetration testing that systematically probes the system for vulnerabilities that might allow unauthorized access, data manipulation, or audit log compromise. Each of these assessments is conducted by the development team in collaboration with security-aware stakeholders, using a combination of automated testing tools and manual testing techniques that collectively provide a comprehensive assessment of the system's security posture.

The evaluation results are synthesized into an overall assessment of the system's readiness for operational deployment, with the findings from each criterion presented both individually, to allow focused analysis of specific strengths and weaknesses, and in aggregate, to provide an overall picture of system quality. Areas where the evaluation reveals gaps between the system's actual performance and the required performance standard are prioritized for attention in the refinement phase, with the severity of the gap and the operational significance of the criterion determining the urgency of the remediation effort. This structured, criteria-based approach to evaluation ensures that the assessment is objective, transparent, and actionable, providing café owners and the development team alike with a clear and trustworthy basis for decisions about deployment readiness and ongoing improvement priorities.

In summary, the methodology presented in this chapter establishes a comprehensive, rigorous, and well-grounded framework for the development and evaluation of StockSecure POS and CyberLedger. By combining the developmental research design's emphasis on artifact creation and real-world evaluation with the Rapid Application Development methodology's commitment to iterative prototyping and stakeholder engagement, and by grounding both in the specific operational and security requirements of Philippine café businesses revealed through systematic data gathering, the research process is designed to produce a system that is simultaneously technically sound, operationally appropriate, and genuinely useful to the small business operators who are its intended beneficiaries. The testing and evaluation framework ensures that this usefulness is demonstrated empirically rather than merely asserted, providing a credible and defensible basis for the research's contributions to both academic knowledge and practical technology development for the MSME sector.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Every well-built system begins not with code, but with a clear and deliberate picture of how everything is supposed to work. Before a single line of program logic is written, before a single cable is connected, and before a single interface is designed, a development team must sit down and ask a very important set of foundational questions: Who will be using this system? What processes will take place inside it? How will the data be organized, stored, and protected? And how will all the different parts of the system communicate with one another? Answering these questions thoroughly and honestly at the outset is what separates a well-engineered solution from one that collapses under real-world pressure.

This chapter provides exactly those answers for the StockSecure POS system developed specifically for Dulce Cafe, a food and beverage establishment located in Manduriao, Iloilo. Rather than jumping straight into technical code or hardware installation, Chapter 3 takes the reader through three carefully designed visual models that lay out the entire system in a way that both technical developers and non-technical business owners can fully understand and appreciate. These models are not mere formalities or academic exercises. Their working blueprints are precise, deliberate, and purposeful representations of a system designed to solve real operational problems faced by a real local business.

The three diagrams presented in this chapter are the Data Flow Diagram (DFD) at Level 0, the Entity Relationship Diagram (ERD), and the Use Case Diagram. Each one tells a different part of the same story.

The **Level 0 Data Flow Diagram**, commonly known as the **Context Diagram**, offers the broadest, most accessible view of the entire StockSecure POS system. It steps back from the internal details and presents the system as a single, cohesive unit—a black box — surrounded by the external actors and entities it interacts with. This high-level perspective clearly illustrates the major people, devices, and external systems that exchange data with StockSecure, along with the primary flows of information moving in and out. For a study and real-world implementation like the one developed for Dulci Cafe, the Level 0 DFD is incredibly important because it establishes the system's scope, boundaries, and purpose in a way that everyone — from café owners and managers to developers and auditors — can quickly understand.

In the context diagram for StockSecure POS, the central process is labeled simply as **"StockSecure POS System."** Surrounding it are the key external actors: **cashiers/baristas, managers, administrators/owners, customers, thermal printers,** and external notification channels (such as email or SMS for alerts). Data flows are shown as arrows connecting these actors to the system. For example, a cashier sends customer order details and payment information into the system and receives a printed receipt and updated order confirmation in return. A manager receives real-time low-stock alerts and dashboard summaries while sending updated product information or approved discounts back into the system. The owner or administrator provides system configurations and user accounts and receives comprehensive reports and full CyberLedger audit exports.

This big-picture view is essential when building and documenting a complex system like StockSecure. It helps prevent scope creep by clearly defining what is inside the system versus what lies outside. During the study phase, the Level 0 DFD served as a foundational communication tool. Café owners could look at it and immediately see how the system supports their daily reality — faster checkouts during morning rushes, automatic stock monitoring that prevents running out of popular items, and strong audit capabilities that satisfy BIR compliance requirements — without getting lost in technical database schemas or code-level processes.

For the development team, the context diagram acted as a guiding star. It highlighted critical external integrations, such as hardware printing via python-escpos and pyusb, and ensured that security boundaries were considered from day one. By visualizing data flows early, the team could design strong input validation at P1 (Authenticate) and enforce RBAC at P2 before any detailed coding began. It also made it easier to identify potential risks — for instance, ensuring that sensitive financial data flowing to reports never leaves the system unprotected.

What makes the Level 0 DFD especially valuable is its humanizing quality. It reminds everyone involved that technology exists to serve people. Behind every data flow is a real person: a tired barista trying to serve customers quickly, a manager worried about running out of ingredients during peak hours, or an owner needing trustworthy numbers at the end of a long day. The diagram shows how StockSecure supports these humans — reducing stress, preventing costly mistakes, and giving staff and owners greater confidence and control.

Ultimately, the Level 0 DFD anchors the entire study. It connects high-level business goals with technical implementation, ensures all stakeholders share a common understanding, and provides a clear reference point throughout development, testing, and future enhancements. By starting with this broad view, StockSecure POS was built not just as powerful software but as a practical, secure, and supportive tool that truly fits the rhythm and needs of a busy café like Dulci Cafe.

While the use case diagram maps out who can do what in the system, the **entity-relationship diagram (ERD)** shifts the focus inward to reveal the carefully organized internal structure of the StockSecure POS system's data. It provides a clear visual model of the core **entities** (the key objects the system must track), the **attributes** (specific pieces of information stored about each entity), and the **relationships** between them. Unlike user-facing diagrams, the ERD is purely concerned with how information is logically organized, stored, retrieved, updated, and protected so that the system remains accurate, consistent, and performant at all times.

For a point-of-sale system like StockSecure that simultaneously manages high-volume customer transactions, real-time inventory levels, employee accountability, and tamper-evident audit trails, a well-designed ERD is absolutely critical. It forms the foundation for the SQLite3 database and ensures that every sale, stock movement, and administrative change is recorded reliably without data inconsistencies or integrity issues.

The ERD for StockSecure POS defines five primary entities that directly correspond to the system's main data stores:

- **Users** (DS3): Stores staff information with attributes including userID (primary key), username, password_hash (securely hashed with bcrypt), full name, role (cashier, manager, admin, owner), status (active, locked, inactive), last_login_timestamp, email, and phone.
- **Products** (DS2): Represents the café's menu and inventory with productID, name, description, category, unit_price, current_stock, min_stock_threshold, last_updated, and is_active.
- **Transactions** (DS1): Captures every customer sale with transaction ID, timestamp, user ID (foreign key to the cashier who processed it), total amount, tax amount, discount amount, payment method, payment reference, and status (completed, voided, or refunded).
- **Sale_Items**: A supporting entity that details each line item within a transaction, linking transactionID and productID with quantity_sold, unit_price_at_time_of_sale, and line_subtotal.
- **CyberLedger_Audit**: The heart of the compliance and security layer, containing eventID, timestamp, userID, action_type (e.g., "Process Sale," "Update Product," "Login Attempt"), affected_entity, affected_id, before_snapshot, after_snapshot, and chain_hash (HMAC-SHA256 cryptographic link to the previous record).

The relationships between these entities are clearly defined to maintain data integrity. A **one-to-many** relationship exists between Users and Transactions (one staff member can process many sales). Products maintain a **one-to-many** relationship with Sale_Items, allowing accurate stock deduction across multiple transactions. The CyberLedger_Audit entity has flexible **many-to-one** associations with Users and polymorphic links to any changed entity, creating a complete, chronological, and cryptographically verifiable history of all system activity.

This structure delivers powerful real-world benefits for Dulci Cafe. Thanks to proper normalization and foreign key constraints, the system automatically maintains accurate stock levels after every sale, prevents orphaned records, and enables fast, accurate reporting—such as “Sales by Shift," “Top-Selling Items," or “Inventory Variance." The tight integration with the CyberLedger chain ensures every action is auditable and protected against tampering, giving owners and regulators confidence in the data.

By providing this clear, logical data model early in the project, the ERD helped the development team design an efficient, secure, and scalable database from the start. It also serves as an excellent communication tool, allowing non-technical café owners to understand how their daily operations translate into organized, trustworthy information. Ultimately, the ERD transforms raw business data into a reliable backbone that supports fast service, smart inventory decisions, strong internal controls, and regulatory compliance—making StockSecure POS not just a transaction tool, but a trusted partner in running a successful café.

The **Use Case Diagram** completes the functional picture of the StockSecure POS system by visually mapping out the specific actions that different types of users can perform. It directly answers the essential question: “Who is allowed to do what? ” In a multi-role environment like Dulci Cafe, where cashiers, managers, and administrators share the same system but carry very different levels of responsibility, this diagram is especially valuable. It makes the boundaries of access visible, clear, and enforceable from the earliest stages of design through to daily operations.

The diagram centers on three primary **actors**—real people with distinct roles:

- **Cashier/Barista**: These frontline staff are the most frequent users. Their use cases focus on daily operations such as **Process Sale (P3)**, which includes selecting menu items, customizing orders, calculating totals, processing payments (cash, card, or digital), printing thermal receipts, and viewing basic order history for their current shift. They can also update their own login session but have no access to administrative or reporting features.
- **Manager**: This role expands capabilities to support oversight and decision-making. Managers can perform use cases like **monitor stock (P5)**, respond to **low stock alerts (P6)**, view interactive Chart.js dashboards showing sales trends and performance metrics, approve special discounts or refunds, generate daily sales summaries, and monitor staff activity during their shift. They act as the operational bridge between the counter and higher management.
- **Administrator/Owner**: The highest-privilege actor with full system control. This role includes all of **Manage System (P4)**—creating and editing user accounts, assigning roles, managing the complete product catalog and pricing, configuring system settings (tax rates, alert thresholds, and backup schedules), reviewing the full CyberLedger audit trail, and generating professional compliance reports via ReportLab.

The use case diagram uses simple ovals for each use case and stick-figure actors connected by lines to show permitted interactions. It explicitly links these use cases to the underlying processes (P1: Authenticate and P2: RBAC Check), ensuring that every action is first verified through secure login and role-based permissions. For example, the use case “View Full Audit Log” connects only to the administrator, while “Process Customer Order” connects to both cashier and manager.

This visual representation is critically important for several reasons. It reinforces the system’s **role-based access control (RBAC)** by design, reducing the chance of over-privileged accounts. During development, it guided the team in implementing proper server-side controls and Jinja2 template rendering so that users only see the menus and buttons relevant to their role. For café owners, it provides an easy-to-understand overview of how responsibilities are distributed, making staff training simpler and more effective.

By clearly documenting these distinctions, the use case diagram helps prevent errors, minimizes internal risks, and supports smooth daily operations at Dulci Cafe. A new cashier can quickly learn their scope without feeling overwhelmed, while managers and owners gain confidence that sensitive functions remain protected. Ultimately, it transforms abstract security concepts into a practical, human-friendly framework that supports efficient teamwork while safeguarding the café's financial and operational integrity. The diagram becomes a living reference that continues to guide system use, future updates, and compliance efforts long after implementation.

Together, these three diagrams form a complete and layered blueprint for the StockSecure POS system combined with the CyberLedger audit monitoring subsystem. Think of them the way an architect thinks about a building before a single brick is laid. Before construction begins, an architect produces floor plans, electrical layouts, structural diagrams, and plumbing schematics. Each drawing illuminates a different aspect of the same structure. No single drawing tells the whole story on its own, but together they give the construction team everything they need to bring the building to life accurately, safely, and according to specifications. This chapter serves that same purpose for the StockSecure POS system at Dulce Cafe. These diagrams are the architectural drawings that guided every development decision that follows.

Why is this kind of systematic planning especially important for Dulce Cafe? Because small and medium food and beverage businesses like Dulce Cafe operate in an environment where the margin for error is thin and the consequences of poor system design are immediate and tangible. Unlike large corporations with dedicated IT departments and multiple layers of oversight, small establishments typically rely on a handful of employees and a single point-of-sale terminal to handle all their daily transactions. This creates both operational vulnerability and significant security risk.

Small businesses in the food service industry commonly struggle with two closely related and deeply damaging problems: keeping accurate track of inventory and protecting themselves from internal theft or fraud. These are not hypothetical concerns. They are documented realities. A cashier who voids too many transactions without a corresponding reason. A manager who adjusts stock levels without prior authorization from ownership. A system that processes hundreds of transactions per day but keeps no logs, no timestamps, and no accountability trail. These are real vulnerabilities that hurt real businesses every single day, eroding profit margins, distorting financial records, and undermining the trust between employers and employees.

The StockSecure POS system was designed from the ground up to address these pain points directly and systematically. Every feature of the system   from the way transactions are recorded to the way inventory adjustments are flagged   reflects a deliberate design choice made in response to an identified operational weakness. The CyberLedger subsystem, which runs alongside the core POS functionality, extends this commitment to accountability by creating a continuous, tamper-resistant audit trail of every significant action performed within the system. No transaction is unrecorded. No adjustment goes unlogged. No suspicious pattern goes undetected.

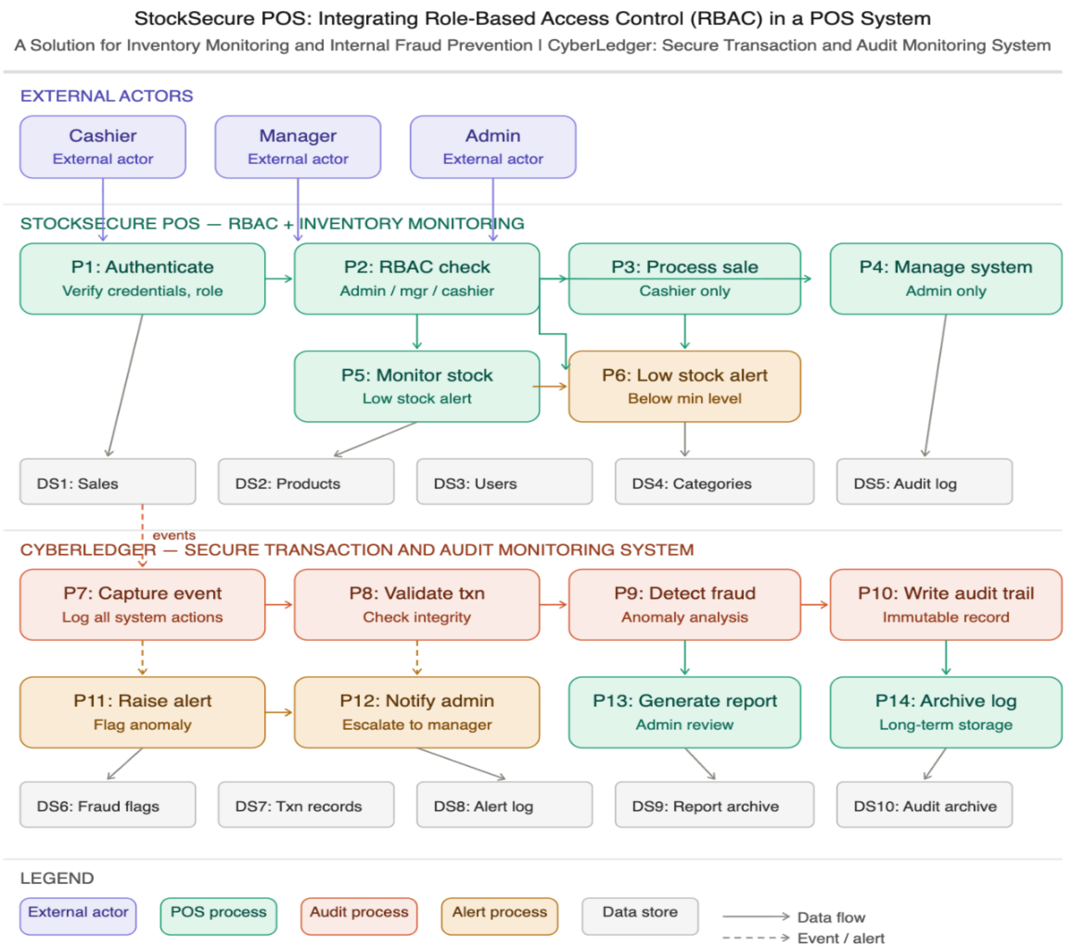
The diagrams in this chapter explain exactly how the system achieves these goals. They show which entities interact with the system and how, how the data flows between processes, how the database is structured to support both operational efficiency and forensic traceability, and how the roles and permissions framework ensures that each user can only do what they are authorized to do. By the end of this chapter, the reader will have a thorough conceptual understanding of StockSecure as a whole a foundation that makes the implementation details presented in subsequent chapters fully meaningful and contextually grounded.

Data Flow Diagram (DFD)

A data flow diagram, commonly called a DFD, is a graphical representation that shows how information moves through a system. It is one of the oldest and most widely used tools in system analysis and design. Unlike a flowchart, which tends to focus on the step-by-step logic of a process, a DFD focuses on the flow of data: where it comes from, where it goes, what transforms it, and where it ends up being stored.

The Level 0 DFD is the highest-level version of this diagram. It gives the reader a bird's-eye view of the entire system without getting buried in too many details. It identifies all the major external actors who interact with the system, all the core processes that happen inside the system, and all the data stores where information is kept. Think of a Level 0 DFD as a summary of the entire system on a single page.

For the StockSecure POS system, the Level 0 DFD is organized into two major subsystems shown side by side on the same diagram: the StockSecure POS layer, which handles everything related to sales, inventory, and role-based user access; and the CyberLedger layer, which handles all the audit monitoring, fraud detection, and alert management functions. *Figure 1. Data Flow Diagram (DFD Level 0): StockSecure POS + CyberLedger*



At the very top of the DFD are three boxes labeled "External Actors." These represent the real human beings who interact with the system on a daily basis at Dulce Cafe. They are external to the system itself, meaning the system does not control them, but it does respond to their inputs and produce outputs that they receive. Understanding who these actors are is the first step in understanding how the entire system operates.

The Cashier

The cashier is the frontline employee of Dulce Cafe. This is the person who stands at the counter, takes orders from customers, rings up purchases, processes payments, and hands customers their receipts. In the context of the StockSecure POS system, the cashier is the most frequent user of the system. However, the cashier's access is deliberately limited. The system is designed so that cashiers can only do what they need to do to complete a sale: they can scan or select items, apply discounts where allowed, accept payment, and generate receipts. They cannot view sales reports from previous days, they cannot change product prices, and they absolutely cannot access audit logs or modify user accounts.

This intentional limitation is one of the cornerstones of the StockSecure POS philosophy. By restricting what a cashier can see and do, the system dramatically reduces the risk of accidental errors and deliberate manipulation. A cashier cannot void a transaction that has already been finalized without triggering a logged alert. This protects both the business and the honest cashier from false accusations.

The Manager

The manager at Dulce Cafe occupies a middle tier of system access. Managers have more visibility into the system than cashiers, but they do not have full administrative control. In practical terms, this means a manager can view the inventory dashboard to see which products are running low, can review daily or weekly sales reports, can receive and respond to low-stock alerts, can void or review flagged transactions, and can view audit logs to check for suspicious activity.

The manager role is important from a fraud-prevention standpoint for a specific reason: managers are the first line of human oversight. When the CyberLedger subsystem detects an anomaly, such as an unusual number of discounts applied by a single cashier in one shift or a transaction that appears to reverse immediately after completion, the alert is first escalated to the manager. The manager then reviews the flagged event and decides whether to investigate further or escalate to the administrator.

The Admin

The Admin, short for Administrator, has the highest level of access in the entire StockSecure POS system. At Dulce Cafe, this is likely the owner or a trusted senior operations manager. The admin can do everything the manager can do, plus additional functions that are off-limits to everyone else: creating and managing user accounts, assigning roles to staff members, changing product information and prices, accessing full audit archives, generating comprehensive reports, and managing system-wide settings.

The concentration of critical functions in the Admin role is intentional. It creates a clear accountability structure. If a product price was changed inappropriately, the audit log will show that only an admin could have done it. If a new user account was created with unauthorized privileges, the system will have a complete record of who created it and when. This makes internal investigations much simpler and much more credible.

The middle section of the DFD shows the core operational processes of the StockSecure POS system. These are the processes that handle what happens every single day at Dulce

Cafe: employees logging in, transactions being processed, inventory being tracked, and alerts being generated when stock levels get dangerously low.

Process ID	Process Name	Description
P1	Authenticate	Verifies the identity and role of every user who attempts to log in to the system. Checks credentials against the DS3: Users data store.
P2	RBAC Check	Determines what the authenticated user is allowed to do based on their role (Admin, manager, or cashier). Enforces role-based permissions.
P3	Process Sale	Handles the complete lifecycle of a sales transaction. Available to cashiers only. Reads from DS2: Products and writes results to DS1: Sales.
P4	Manage System	Reserved for admin users only. Covers user management, product configuration, reporting, and system settings.
P5	Monitor Stock	Continuously checks current inventory levels against minimum thresholds. Triggers alerts when stock falls below acceptable levels.
P6	Low Stock Alert	Fires when P5 detects a product below its minimum level. Notifies the manager so restocking can be arranged promptly.

Table 1. StockSecure POS Processes (P1–P6)

P1: Authenticate Verifying Who You Are

Every interaction with the StockSecure POS system begins at P1: Authenticate. This is not merely a login screen. It is the gateway that decides whether a person is who they claim to be and whether they have any business being inside the system at all. When an employee sits down at a terminal and enters their username and password, the system checks those credentials against the Users data store (DS3). If the credentials match and the account is active, the user is granted access. If the credentials are wrong, the system records a failed login attempt. After a set number of consecutive failures, the account is automatically locked to prevent brute-force attacks.

This process is critically important in a cafe environment where employees may share terminal space. By requiring individual login credentials for every session, the system ensures that every transaction, every action, every change is always attributed to a specific person. There is no such thing as an anonymous action in StockSecure POS.

P2: RBAC Check—Knowing What You Are Allowed to Do

Once a user is authenticated, the system does not simply open all the doors and let them roam freely. Process P2, the RBAC (Role-Based Access Control) Check, determines what that specific user is permitted to do based on the role assigned to their account. A cashier who successfully logs in will find themselves on a simplified transaction screen. They cannot navigate to the user management panel, and the audit log viewer does not appear in their menu. A manager who logs in will see their expanded dashboard but will not find the user creation tools that only an admin can access.

Role-based access control is not just a technical feature; it is a management philosophy. It reflects the idea that information and functionality should be available only on a need-to-know, need-to-do basis. Giving everyone access to everything might seem convenient, but it dramatically increases the risk of accidental data corruption and deliberate fraud. The RBAC check at P2 enforces the boundaries that protect Dulce Cafe's business data and financial integrity.

P3: Process Sale   The Heart of Daily Operations

P3 is where the business actually happens. Every time a customer orders a cup of coffee at Dulce Cafe, P3 is the process that records it. The cashier selects the items ordered, the system calculates the subtotal, applies applicable taxes and any authorized discounts, the customer pays, and a receipt is generated. All of this flows through P3: Process Sale.

Behind the scenes, this process is doing more than just arithmetic. It is reading from the Products data store (DS2) to confirm that each item exists and to get its current price. It is writing to the sales data store (DS1) to create a permanent record of the transaction. And critically, every action taken during this process is being captured by the CyberLedger subsystem as a system event, which flows into the audit monitoring layer. This means every sale at Dulce Cafe is not just a business transaction; it is also an auditable event in the CyberLedger record.

P4: Manage System Administrative Control

P4 is the administrative nerve center of the StockSecure POS system. Only users with the Admin role can access this process. Through P4, the admin can add new staff members to the system and assign their roles, update product names, descriptions, prices, and stock levels, generate comprehensive business reports, view the full audit trail, and configure system-wide settings such as minimum stock thresholds and alert recipients.

The isolation of these critical functions within P4, accessible only to admins, creates what security professionals call a "least privilege" environment. Every user has exactly as much access as they need to do their job, and not one bit more. This is the operational heart of internal fraud prevention at Dulce Cafe.

P5 and P6: Monitor Stock and Low Stock Alert—Never Run Out Again

One of the most practical and immediately valuable features of the StockSecure POS system for Dulci Cafe is the intelligent inventory monitoring capability built into **P5: Monitor Stock** and **P6: Low Stock Alert**. These two processes work together in the background as a vigilant digital assistant, constantly watching over the café's supplies so managers can focus on customers rather than constantly counting stock.

P5: Monitor Stock runs continuously as a lightweight, threaded background service. After every sale processed through P3 and after any manual stock adjustments, it instantly cross-checks the current quantity of every product in the inventory against the minimum stock threshold set by the administrator. These thresholds are flexible — for example, the admin might set 30 units for popular espresso beans, 15 liters for fresh milk, or 20 pieces for a best-selling pastry. The moment any item drops to or below its configured minimum, P5 automatically triggers **P6: Low Stock Alert**.

P6 then delivers timely, actionable notifications through multiple channels: a prominent visual highlight on the manager's dashboard, an on-screen alert banner, and optional email or SMS notifications to designated recipients. In real-world terms, this means the manager at Dulci Cafe no longer has to manually check shelves during busy hours. Instead, the system proactively notifies them— “Low Stock: Espresso Beans (12 remaining)” or “Popular Croissants running low—only 8 left”—giving them enough time to reorder or adjust the menu before customers are disappointed.

This proactive approach prevents embarrassing and costly stockouts that frustrate customers and hurt revenue. It also reduces waste by helping managers avoid over-ordering while ensuring popular items remain available. When combined with Chart.js analytics, managers gain deeper insights, such as which items sell fastest during peak hours, allowing smarter purchasing decisions and better menu planning.

Supporting these processes are the five core **data stores** that form the structured backbone of the entire StockSecure POS system. These are not simple files on a hard drive—they are well-organized, relational SQLite3 databases designed for reliability, speed, and data integrity, even during offline operation:

- **DS1: Sales/Transactions** – Stores complete records of every customer purchase with full details for reporting and auditing.
- **DS2: Products** – Manages the complete inventory catalog, including current stock levels, pricing, categories, and minimum thresholds used by P5.
- **DS3: Users** – Holds staff accounts, roles, and login history for authentication and accountability.
- **DS4: Configuration/Settings** – Contains system-wide parameters such as tax rates, alert preferences, and backup schedules.
- **DS5: CyberLedger_Audit** – The secure, cryptographically chained audit trail that records every action for compliance and security.

All sensitive fields across these stores are protected with Fernet encryption, while every change is logged immutably in the CyberLedger chain. Together, P5, P6, and the five data stores create a reliable safety net that helps Dulci Cafe maintain smooth operations, minimize losses, and keep customers happy—turning inventory management from a constant worry into a quiet, dependable strength of the system.

Data Store	Contents and Purpose
DS1: Sales	Records every completed transaction. Each entry includes a transaction ID, the user who processed it, the total amount, payment method, and timestamp.
DS2: Products	Maintains the catalog of all items available for sale at Dulce Cafe, including current stock levels, prices, minimum stock thresholds, and category assignments.
DS3: Users	Stores all employee account information, including usernames, hashed passwords, assigned roles, and account status (active/locked).
DS4: Categories	Organizes products into logical groupings such as beverages, pastries, and merchandise, enabling clearer reporting and inventory management.
DS5: Audit Log	Captures every significant action performed within the system, including who did it, when, what changed, and from which IP address.

Table 2. StockSecure POS Data Stores (DS1–DS5)

The lower half of the DFD Level 0 diagram shows the CyberLedger subsystem. This is the security and audit intelligence layer of the StockSecure POS platform. While the POS layer handles the business operations, the CyberLedger layer watches over those operations constantly, looking for anything that seems unusual, suspicious, or out of the ordinary.

The name **"CyberLedger"** is intentionally evocative of a traditional accounting ledger, a book that records every financial transaction in permanent ink, impossible to erase or alter without leaving a trace. The CyberLedger does exactly this for the digital transactions at Dulce Cafe. Every event is captured, every anomaly is flagged, every alert is logged, and every audit trail is archived in an immutable record.

Process ID	Process Name	Description
P7	Capture Event	Logs every system action as it occurs. Receives event data from all POS layer processes in real time.
P8	Validate Txn	Checks the integrity of each captured transaction. Verifies that data has not been altered and that values are logically consistent.
P9	Detect Fraud	Runs anomaly analysis on transaction patterns to identify suspicious behavior such as excessive voids, unusual discounts, or off-hours activity.
P10	Write Audit Trail	Creates an immutable record of all validated and flagged events. Once written, this record cannot be modified by any user.
P11	Raise Alert	Flags a detected anomaly for human review. Stores the flag in DS6: Fraud Flags and initiates the notification chain.
P12	Notify Admin	Escalates a raised alert to the manager and, if necessary, the admin. Records the notification in DS8: Alert Log.
P13	Generate Report	Compiles audit data into readable reports for admin review. Stores generated reports in DS9: Report Archive.
P14	Archive Log	Moves older audit records to long-term storage in DS10: Audit Archive, maintaining system performance while preserving history.

Table 3. CyberLedger Processes (P7–P14)

Imagine a security camera that not only records footage but also has a built-in analyst who watches the recordings in real time and calls the manager the moment something looks wrong. That is essentially what the CyberLedger does for Dulce Cafe.

P7: CyberLedger Event Logging – Complete and Tamper-Evident Audit Trail

Every single action that occurs within the StockSecure POS system is automatically captured as a secure "event" by **P7: CyberLedger Event Logging**. Whether a cashier processes a sale, a manager views a sales report, an administrator updates a product price, or even a failed login attempt occurs, P7 records it instantly and immutably. This logging is completely automatic and unavoidable — users cannot disable it, bypass it, or selectively choose which activities to record. The system ensures full transparency by design.

P7 works silently in the background, capturing critical details for every event: the exact timestamp, the userID responsible, the action type, the affected entity (such as a specific transaction or product), before-and-after values where relevant, and the device or session information. Each event is then cryptographically chained into the **HMAC-SHA256 Ledger Chain**, creating a blockchain-style, tamper-evident audit trail. Any attempt to alter or delete a record would break the cryptographic chain, making manipulation immediately detectable.

This relentless, comprehensive logging is essential for Dulci Cafe. It provides owners and managers with complete visibility into daily operations, strengthens internal accountability, and serves as powerful protection against fraud or disputes. During BIR audits or internal reviews, the CyberLedger delivers clear, trustworthy evidence of all system activity. By making auditability an unbreakable part of every process, P7 reinforces the system's core promise: nothing happens in StockSecure POS without leaving a secure, permanent, and verifiable record.

P8: Event Validation – Protecting Data Integrity and Preventing Fraud

P8: Event Validation acts as the intelligent guardian that examines every event captured by P7 (CyberLedger event logging) before it is permanently committed to the audit trail. This critical validation layer ensures that each recorded action is logical, consistent, and compliant with business rules. Rather than blindly accepting every event, P8 performs real-time integrity checks that catch both honest mistakes and potential malicious attempts.

For example, if a cashier accidentally or intentionally applies a 100% discount to an order, P8 would flag the transaction as suspicious because it violates configured business rules (such as maximum allowable discount percentages). Similarly, any sale showing a negative total, a product being sold with a negative quantity, or a stock level dropping below zero without proper adjustment would immediately trigger an alert. The system cross-checks relationships between entities—ensuring that a sale can only deduct stock that actually exists and that only authorized users can perform high-risk actions.

This validation layer is essential for maintaining the trustworthiness of the entire StockSecure POS system. It protects Dulci Cafe from costly data errors that could lead to incorrect inventory counts, inaccurate financial reports, or compliance issues. More importantly, it serves as a strong deterrent against internal fraud by making deliberate manipulations extremely difficult to conceal. Flagged events are logged with high priority in the CyberLedger and can notify managers instantly, allowing quick investigation and resolution.

By combining automatic validation with comprehensive logging, P7 and P8 together create a robust defense that ensures every transaction and system action is not only recorded but also verified for accuracy and legitimacy. This gives café owners peace of mind that their data remains clean, reliable, and audit-ready at all times.

P9: Fraud Detection – Advanced Pattern Analysis for Proactive Protection

P9: Fraud Detection represents the most sophisticated and intelligent layer of the CyberLedger subsystem in StockSecure POS. While P8 focuses on validating individual events, P9 takes a broader, long-term view by analyzing patterns across many transactions and user actions over time. This process runs continuously in the background, using rule-based and statistical analysis to identify potentially suspicious behavior that would be nearly impossible to spot when examining single transactions in isolation.

For example, P9 flags cases such as an unusually high number of voided or refunded transactions by one specific cashier during their shift, unusually frequent or large discounts being applied on certain shifts or by particular staff members, or sequences where a sale is processed and then immediately reversed. It can also detect patterns like repeated “no-sale” openings of the cash drawer, suspicious timing of transactions (such as large voids right before end-of-shift), or sudden changes in inventory adjustments that don’t align with normal sales activity.

By examining these aggregated patterns, P9 helps Dulci Cafe move from reactive to proactive fraud prevention. When suspicious activity is detected, the system generates targeted alerts to managers or owners, complete with relevant context and supporting data from the CyberLedger trail. This allows quick investigation without disrupting normal operations. The process respects privacy and role boundaries—only authorized users can view detailed fraud reports—while maintaining the system’s strong commitment to accountability.

Integrated with the immutable HMAC-SHA256 audit chain, P9 strengthens internal controls, deters dishonest behavior, and provides café owners with greater confidence in their team and financial data. In a high-volume environment like Dulci Cafe, where many small transactions occur daily, this intelligent oversight layer is invaluable for protecting profitability and maintaining trust.

P10–P14: Ensuring Immutable Records, Smart Alerts, and Long-Term Audit Management

When **P9: Fraud Detection** identifies a suspicious pattern, the CyberLedger system activates a coordinated response through processes P10 to P14, turning raw detection into actionable protection and long-term compliance.

P10: Writing an audit trail immediately creates an immutable, permanent record of every validated event and flagged anomaly. Using the HMAC-SHA256 cryptographic chain, it writes the full details—timestamp, user, action, before/after values, and chain hash—into the CyberLedger. Once written, this record cannot be modified or deleted by any user, including administrators. This guarantees that even if someone attempts to cover their tracks later, the original evidence remains tamper-proof and forever verifiable.

P11: Raise Alert then flags the anomaly for human review. It stores the alert details in **DS6: Fraud Flags** with severity level, supporting evidence, and links to the relevant CyberLedger events. This step initiates the notification workflow, ensuring nothing suspicious slips through unnoticed.

P12: Notify Admin handles escalation intelligently. The on-duty manager receives the first notification through a prominent dashboard banner, email, or SMS, allowing quick investigation during operations. If the alert is high-severity or remains unacknowledged, it automatically escalates to the administrator or owner. All notifications are logged in **DS8: Alert Log** for complete traceability.

P13: Generate Report empowers administrators with clear visibility. It compiles relevant CyberLedger data, flagged events, and supporting details into professional, readable reports using ReportLab. These PDF reports—complete with summaries, charts, and timestamps—are stored in **DS9: Report Archive** for easy access during audits or management reviews.

Finally, **P14: Archive Log** maintains system performance by automatically moving older audit records to **DS10: Audit Archive** while preserving full cryptographic integrity. This long-term storage ensures Dulci Cafe can retain years of history for compliance without slowing down daily operations.

Together, P10 through P14 create a robust, automated safety net that protects the café's financial integrity, supports regulatory requirements, and gives owners confidence that their system is always watching, recording, and ready to respond.

CyberLedger is the fraud detection and audit monitoring subsystem that operates in parallel with StockSecure POS. It continuously captures system events, validates transaction integrity, detects anomalies using pattern-based analysis, and produces immutable audit trails and management reports. CyberLedger acts as the security backbone of the entire platform, providing the evidence trail required for compliance, investigation, and administrative review.

Data Store	Contents and Purpose
DS6: Fraud Flags	Stores records of all anomalies detected by P9. Each entry identifies the type of anomaly, the transaction involved, and the user implicated.
DS7: Txn Records	Maintains the validated transaction records written by P8. These serve as the authoritative record of what happened in the POS system.
DS8: Alert Log	Records every alert that was raised and every notification that was sent, including whether the alert was acknowledged and by whom.
DS9: Report Archive	Stores generated audit reports for admin review and historical reference.
DS10: Audit Archive	Long-term storage for older audit records that have been transferred out of the active system by P14. Preserved indefinitely.

Table 4. CyberLedger Data Stores (DS6–DS10)

P1 is the entry point for all system users. Before any action can be taken within StockSecure POS, every user must be authenticated. This process verifies the identity of the user by checking their credentials (username and password) against the Users data store (DS3).

- **Input** user-provided credentials from cashier, manager, or Admin
- **Output** verified identity and role assignment passed to P2
- **Data Store** DS3: Users (credential and role lookup)
- **Security Implication: Failed** authentication blocks all downstream processes, preventing unauthorized system access

The DFD references ten data stores (DS1-DS10) distributed across the two subsystems. These stores represent the persistent data repositories that the system reads from and writes to during operation.

StockSecure POS Data Stores (DS1-DS5)

Data Store	Name	Contents	Read By	Written By
DS1	Sales	Transaction records, amounts, timestamps, cashier ID	P3, P13	P3
DS2	Products	Product catalog, SKUs, stock quantities, thresholds	P2, P5	P3, P5
DS3	Users	User accounts, hashed passwords, role assignments	P1, P2	P4
DS4	Categories	Product categories, min stock levels, alert states	P5	P5, P6
DS5	Audit Log	System-level activity log, admin actions	P4	P1, P4

Table 5. StockSecure POS Data Stores (DS1-DS5)

Data Store	Name	Contents	Read By	Written By
DS6	Fraud Flags	Detected anomalies, severity scores, status	P9, P13	P11
DS7	Txn Records	Validated transaction details, integrity hashes	P8, P9, P13	P8
DS8	Alert Log	Notification records, escalation timestamps	P12, P13	P12
DS9	Report Archive	Generated compliance and fraud reports	P13	P13
DS10	Audit Archive	Immutable long-term audit records	P14	P10, P14

Table 6. CyberLedger Data Stores (DS6-DS10)

The DFD distinguishes between two types of data flows: solid-line data flows representing standard data movement between processes and data stores, and dashed-line event/alert flows representing conditional, triggered communications that occur only when specific conditions are met.

The primary operational flow follows a linear progression through the authentication and transaction pipeline:

1. A user (cashier/manager/admin) initiates a system interaction
2. P1 authenticates credentials against DS3: Users

3. P2 evaluates role-based permissions and grants appropriate access
4. P3 processes a sale transaction, writing to DS1 and updating DS2
5. P5 monitors stock levels in DS2 and DS4 following each transaction
6. P7 captures the complete event trail from all upstream processes
7. P8 validates transaction integrity against DS7: Txn Records
8. P9 performs fraud anomaly analysis using DS6 and DS7
9. P10 writes the immutable audit record to DS10: Audit Archive
10. P14 archives consolidated records for long-term compliance storage

Alert flows are conditionally activated when anomalies or threshold breaches are detected:

- **P5 to P6** activated when product stock falls below minimum threshold
- **P8 to P11** activated when transaction integrity validation fails
- **P9 to P11 activated** when a behavioral anomaly is detected
- **P11 to P12** activated when a fraud flag is formally raised
- **P12 to Admin/Manager:** Delivers real-time human escalation notification

The combined DFD architecture reflects several key security design principles that underpin both StockSecure POS and CyberLedger.

Security Principle	Implementation in the DFD	Benefit
Least Privilege	RBAC at P2 restricts each actor to minimum required access	Reduces attack surface and insider threat risk
Separation of Duties	Cashier processes sales; Admin manages the system; and CyberLedger audits independently.	Prevents single-actor fraud or error
Defense in Depth	Authentication (P1), Authorization (P2), Monitoring (P7-P9), Alerting (P11-P12)	Multiple independent security layers
Non-Repudiation	Immutable audit trail at P10 and DS10: Audit Archive	Tamper-proof evidence for investigation
Automated Detection	CyberLedger's P8 and P9 operate without human intervention	Rapid, consistent fraud detection at scale
Audit Accountability	Every system action captured by P7 and logged to DS10	Full traceability of all user actions

Table 7. CyberLedger & StockSecure POS benefits

The Level 0 DFD presented in Chapter 3 illustrates a carefully architected, dual-subsystem platform that addresses the complete lifecycle of a secure POS transaction from initial user authentication through fraud detection, audit logging, and compliance reporting.

StockSecure POS ensures that all retail operations are conducted by verified, authorized actors operating strictly within their designated role boundaries. The RBAC model enforced through P1 and P2 creates a robust access control foundation that minimizes the risk of unauthorized system interaction, data manipulation, and operational errors.

CyberLedger transforms the raw event stream produced by StockSecure POS into a structured, validated, and analyzed audit record. Its automated pipeline detects anomalies that would be invisible to manual review, escalates threats to human administrators in real time, and maintains an immutable historical record that supports forensic investigation and regulatory compliance.

Together, the two subsystems create a comprehensive security ecosystem where operational efficiency and internal fraud prevention operate in parallel rather than in conflict. The clean architectural separation of POS processes (P1-P6) handling operations and CyberLedger processes (P7-P14) handling security ensures that neither subsystem creates a single point of failure or a single point of trust.

Entity Relationship Diagram (ERD)

While the data flow diagram showed us how information moves through the StockSecure POS system, the entity relationship diagram, or ERD, shows us how information is organized and stored. If the DFD is the system's map of roads and traffic flow, the ERD is the blueprint of the filing system in the back office where all the documents are kept.

An ERD is a visual representation of the database structure. It shows all the tables (called "entities") that exist in the database, all the columns (called "attributes") within each table, and most importantly, all the relationships between the tables. These relationships tell us how one piece of data connects to another, which is what allows the system to answer questions like "Show me all the items in this particular sale" or "Which user processed the most transactions last week?"

For Dulce Cafe's StockSecure POS system, the ERD defines five main entities: USERS, PRODUCTS, CATEGORIES, SALES, SALE_ITEMS, and AUDIT_LOG. Each of these corresponds directly to a real-world concept at the cafe, and together they form a complete, logically organized database that supports all the features described in the DFD.

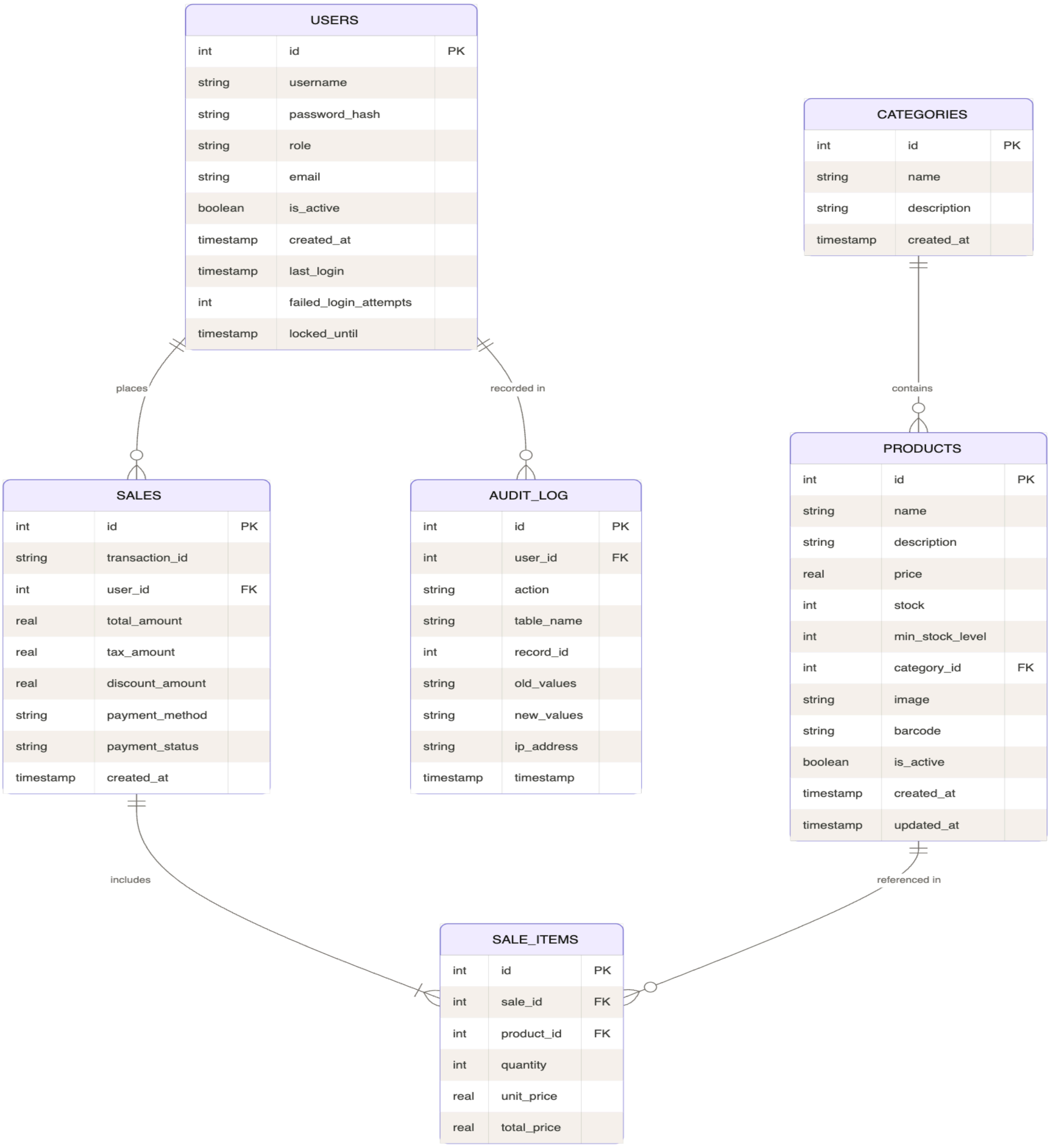


Figure 2. Entity Relationship Diagram (ERD): StockSecure POS Database Structure

The USERS Entity

The **USERS** table (DS3) is perhaps the most security-critical table in the entire StockSecure POS database. It acts as the single source of truth for every person who has access to the system — cashiers, baristas, managers, and administrators at Dulci Cafe. Since every action performed in the system is permanently linked to a specific user through the CyberLedger audit trail, the integrity and protection of this table directly determine the overall security, accountability, and trustworthiness of the entire application.

The following key attributes are maintained for each user:

- **id** (int, Primary Key): A unique, auto-generated number that permanently identifies each user account.
- **username** (string): The unique login name used by the employee to access the system.
- **password_hash** (string): The bcrypt-hashed version of the user's password. Plain-text passwords are never stored — only the irreversible cryptographic hash.
- **role** (string): The user's assigned role (Admin, Manager, or Cashier), which drives the Role-Based Access Control (RBAC) system and determines exactly what the user is allowed to see and do.
- **email** (string): The user's contact email address, used for password resets and system notifications.
- **is_active** (boolean): Indicates whether the account is currently active. Inactive accounts cannot log in.
- **created_at** (timestamp): Records when the account was created.
- **last_login** (timestamp): Tracks the most recent successful login, helping administrators monitor account usage.
- **failed_login_attempts** (int): Counts consecutive failed login attempts.
- **locked_until** (timestamp): Specifies when an account locked due to too many failed attempts will automatically unlock.

The combination of failed_login_attempts and locked_until provides simple yet effective protection against brute-force attacks. After a configured number of failed attempts, the account locks for a period, making automated password guessing impractical. All changes to the USERS table are securely logged in the CyberLedger, ensuring full auditability and strong internal controls.

The PRODUCTS Entity

The **PRODUCTS** table serves as the central catalog of everything Dulci Cafe sells. Each row represents a single menu item or product, containing all the information the StockSecure POS system needs to handle sales, inventory tracking, display, and reporting.

This table is essential for daily operations. It powers the cashier's product selection screen, drives real-time stock updates during every sale, triggers low-stock alerts, and provides the foundation for accurate sales analytics and inventory reports.

The key attributes maintained for each product are:

- **id** (int, Primary Key): Unique identifier for each product.
- **name** (string): The product name as displayed on the POS screen and printed on customer receipts.
- **description** (string): A short, helpful description of the item.
- **price** (real): The current selling price. Only Admin or Owner roles can modify this field for security and pricing control.
- **stock** (int): The current available quantity in inventory. This value is automatically decreased whenever the product is sold through **P3: Process Sale**.
- **min_stock_level** (int): The threshold quantity that triggers low-stock alerts via **P5: Monitor Stock** and **P6: Low Stock Alert**.
- **category_id** (int, Foreign Key): Links the product to its parent category in the **CATEGORIES** table.
- **image** (string): Path or reference to the product image displayed on the POS interface.
- **barcode** (string): The product's barcode value, enabling fast scanning at the cashier terminal.

- **is_active** (boolean): Indicates whether the product is currently available for sale (inactive products are hidden from the POS screen).
- **created_at** and **updated_at** (timestamps): Track when the product was added and last modified.

By maintaining rich, well-organized product data with strong relationships to categories and sales records, the PRODUCTS table ensures smooth frontline operations, accurate inventory management, and reliable business intelligence for Dulci Cafe. All changes to this table are securely logged in the CyberLedger for full auditability.

The CATEGORIES Entity

The **CATEGORIES** table is a simple yet highly valuable supporting structure in the StockSecure POS database. It organizes all menu items into logical, easy-to-understand groups such as “Hot Beverages,” “Cold Beverages,” “Pastries,” “Sandwiches,” “Desserts,” and “Merchandise.”

This categorization delivers immediate practical benefits: cashiers can quickly locate items on the POS interface through well-organized tabs or filters, speeding up order taking during busy hours. For managers and owners, it transforms raw sales data into meaningful insights — such as “Hot Beverages accounted for 45% of today’s revenue” — making reports more actionable for inventory planning and menu optimization.

Each category record includes:

- **category_id** (primary key)
- **name**
- **description**
- **created_at** timestamp

The **category_id** foreign key in the **PRODUCTS** table ensures every product belongs to exactly one category, maintaining clean data relationships throughout the system.

The SALES Entity

The **CATEGORIES** table is a simple but essential supporting structure in the StockSecure POS database. It groups related products into logical and intuitive categories such as “Hot Beverages,” “Cold Beverages,” “Pastries,” “Sandwiches,” “Desserts,” and “Merchandise.”

This well-organized categorization delivers clear, practical benefits to daily operations at Dulci Cafe. On the POS interface, cashiers can quickly navigate through neatly grouped tabs or filtered product grids, significantly speeding up order taking during peak hours.

Instead of scrolling through a long, unorganized list, they can tap “Hot Beverages” and instantly see all coffee and tea options. For managers and administrators, these categories turn sales data into meaningful business intelligence—enabling clear reports such as “Hot beverages contributed 38% of total revenue this week” or “Pastries are our top-performing category.”

As shown in the ERD, each category record includes:

- **id** (primary key)
- **name**
- **description**

- **created_at** timestamp

The **category_id** foreign key in the **PRODUCTS** table ensures every product is linked to exactly one category, maintaining clean data relationships and supporting efficient filtering, reporting, and inventory management across the system.

By bringing structure and clarity to the product catalog, the CATEGORIES table improves both frontline efficiency and managerial decision-making, making StockSecure POS more user-friendly and insightful for everyone at Dulci Cafe.

The SALE_ITEMS Entity

The **SALE_ITEMS** table is a vital component in the StockSecure POS database that makes detailed receipt generation, transaction history review, and in-depth sales analysis possible. While the **SALES** table (shown in the diagram) records the overall transaction summary — such as the total amount, tax, discount, payment method, and cashier — the **SALE_ITEMS** table breaks each sale down into its individual line items.

This granular structure allows managers and administrators to retrieve any past transaction and see exactly what was ordered: the specific products, quantities purchased, unit prices at the time of sale, and line subtotals. As clearly illustrated in the ERD, SALE_ITEMS maintains strong relationships through two foreign keys—sale_id linking back to the **SALES** table and product_id referencing the **PRODUCTS** table.

Key attributes visible in the diagram include:

- **id** (primary key)
- **sale_id** (foreign key)
- **product_id** (foreign key)
- **quantity**
- **unit_price**
- **total_price**

Thanks to this design, the system can generate accurate, itemized thermal receipts, support customer inquiries, analyze best-selling products, and provide complete evidence for audits. Without the SALE_ITEMS table, the system would lose the ability to reconstruct the full details behind every sale, significantly reducing transparency and reporting value for Dulci Cafe.

The AUDIT_LOG Entity

The AUDIT_LOG table is the CyberLedger's primary data structure within the database. It records every significant action that any user performs within the system. Its attributes are designed to answer the five key questions of any investigation: Who? What? When? Where? And what changed?

- id (int, Primary Key): Unique identifier for each log entry.
- user_id (int, Foreign Key): The ID of the user who performed the action. Links back to the USERS table.

- action (string): A description of what action was performed, such as "sale_completed," "product_updated," "login_failed," or "transaction_voided."
- table_name (string): Which database table was affected by the action.
- record_id (int): The specific record within that table that was affected.
- old_values (string): The values that existed before the change was made, stored as a formatted text record.
- new_values (string): The values that exist after the change was made.
- ip_address (string): The network address of the device from which the action was performed. This helps identify if an action was taken from an unusual location.
- timestamp: The exact date and time the action occurred.

The old_values and new_values fields deserve special mention. By recording what a value was before a change and what it became after the change, the AUDIT_LOG creates a complete history of every modification in the system. If a product price is changed from one amount to another, both the old price and the new price are preserved in the audit log. If a cashier voids a transaction, the original transaction details are preserved in old_values. This means that no modification, however minor, is truly "gone" from the system. It exists in the audit record.

What makes the ERD truly useful is not just the tables themselves but the lines connecting them. These lines represent the relationships between entities, and they carry important information about how records in one table connect to records in another.

From Entity	To Entity	Relationship Description
USERS	SALES	One user (cashier) can place many sales. Each sale is placed by exactly one user. (One-to-Many)
USERS	AUDIT_LOG	One user can have many audit log entries recorded for them. Each entry is recorded for exactly one user. (One-to-Many)
CATEGORIES	PRODUCTS	One category can contain many products. Each product belongs to exactly one category. (One-to-Many)
SALES	SALE_ITEMS	One sale can include many sale items (line items). Each sale item belongs to exactly one sale. (One-to-Many)
PRODUCTS	SALE_ITEMS	One product can appear in many sale item records across many transactions. Each sale item references exactly one product. (One-to-Many)

Table 7.

These relationships are enforced through foreign key constraints in the database. A foreign key is a column in one table that points to the primary key of another table. If a sale_items record contains a product_id of 42, the database guarantees that a product with ID 42 actually exists in the PRODUCTS table. This prevents data corruption and ensures that the system always maintains consistent, trustworthy records.

Use Case Diagram

The use case diagram is the most human-centered of the three diagrams in this chapter. While the DFD focused on data flows and the ERD focused on data structure, the Use Case Diagram focuses on interactions: specifically, what each type of user can do within the system, and how the system responds to each of those actions.

In software engineering, a "use case" is a specific scenario of interaction between a user and a system that achieves a particular goal. "A cashier processes a sale" is a use case. "A manager views a low-stock alert" is a use case. "The system detects a fraudulent transaction pattern and raises an alert" is a use case. The Use Case Diagram collects all of these scenarios and organizes them visually so that anyone, whether a software developer, a business owner, or a new staff member, can quickly understand what the system does and who is responsible for what.

The use case diagram for the StockSecure POS system at Dulce Cafe is organized around four actors: the Cashier, the Manager, the Admin, and the System (representing the automated CyberLedger functions that operate without direct human initiation). All the use cases are contained within a large box labeled "System Boundary: StockSecure POS + CyberLedger," which visually separates the things that happen inside the system from the external actors who interact with it.

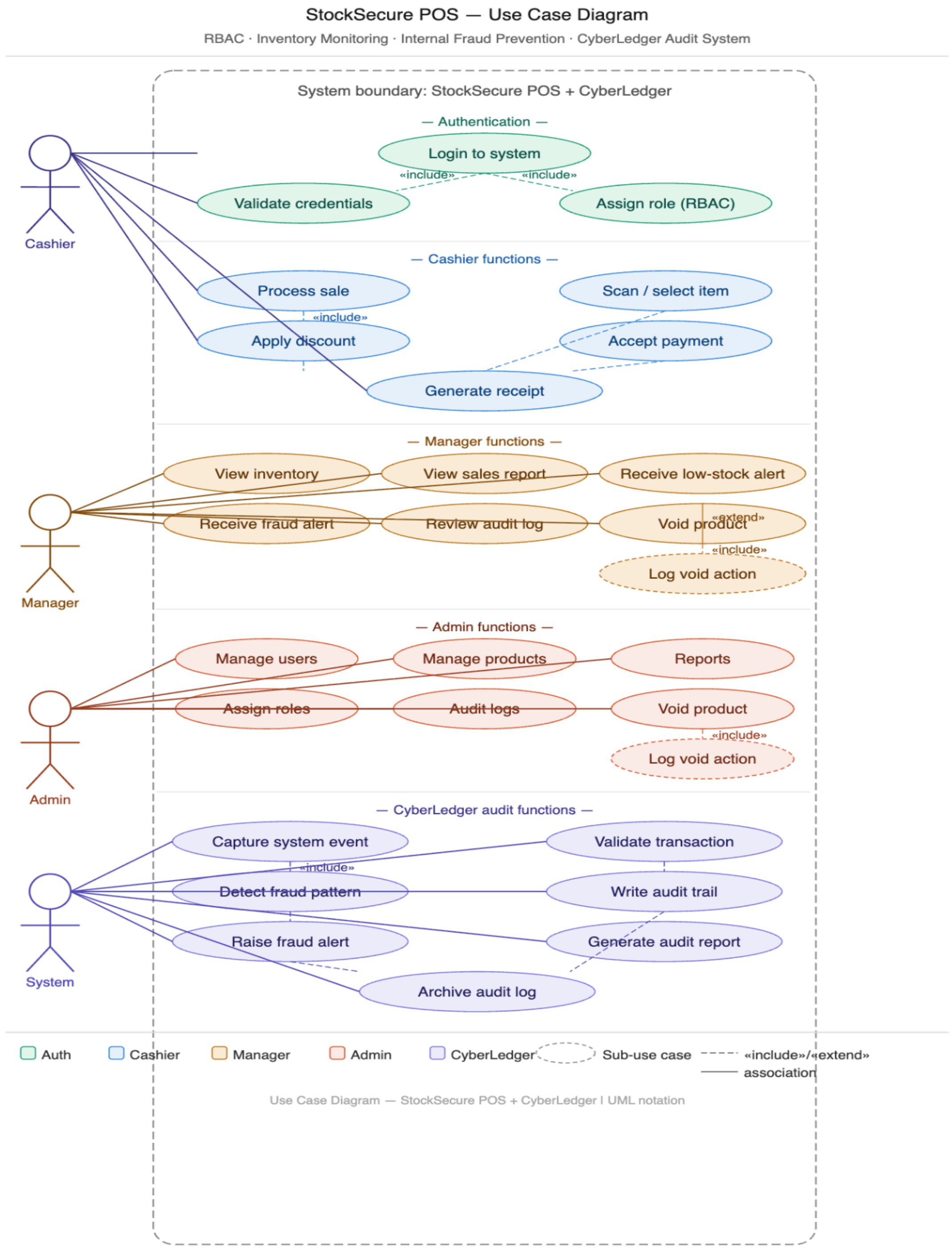


Figure 3. Use Case Diagram: StockSecure POS + CyberLedger (UML Notation)

The topmost section of the use case diagram is labeled "Authentication." This section contains the use cases that every single user must pass through before they can do anything else in the system. All three human actors, the Cashier, the Manager, and the

Admin, connect to the "Login to system" use case with association lines, indicating that all of them must log in.

The "Login to system" use case includes two sub-use cases connected with <<include>> arrows: "Validate credentials" and "Assign role (RBAC)." The <<include>> notation in UML means that whenever the main use case happens, the included use case also always happens as part of it. In other words, you cannot log in to the StockSecure POS system without your credentials being validated and without a role being assigned to your session. These are not optional steps; they are mandatory parts of the login process.

"Validate credentials" means the system checks whether the username and password combination entered by the user matches what is stored in the USERS database table. "Assign role (RBAC)" means that once the credentials are confirmed, the system reads the user's role field and configures the session with the appropriate access permissions for that role. This two-step process happens invisibly and instantly every time someone logs in, but it is the foundation of everything else the system does.

Cashier Functions   Fast, Focused, and Secure

The Cashier section of the Use Case Diagram shows all the actions that a user with the Cashier role can perform. These use cases are deliberately limited in number and scope. A cashier at Dulce Cafe does one thing: serve customers and process their purchases. The system design reflects this by giving cashiers exactly the tools they need for that task and nothing more.

Use Case	Description
Process sale	The primary function of the cashier. Initiates a new sales transaction and serves as the parent use case for all related steps.
Scan / select item	The cashier scans a product barcode or selects it from the on-screen menu. <<include>> under Process sale.
Apply discount	Applies an authorized discount to the transaction. <<include>> under Process sale. Only permitted discounts can be applied.
Accept payment	Records the payment method and confirms the transaction total. <<include>> under Process sale.
Generate receipt	Produces a printed or digital receipt for the customer. <<include>> under Process sale.

Table 6. Cashier Use Cases

Notice that "Scan / select item," "Apply discount," "Accept payment," and "Generate receipt" are all connected to "Process sale" with <<include>> arrows. This tells us that processing a sale is not a single action but a sequence of mandatory steps that must all occur for a sale to be completed correctly. A sale that accepts payment but does not generate a receipt is incomplete. A sale that adds items but never accepts payment is invalid. The <<include>> relationships enforce the proper sequence of the transaction lifecycle.

An important security note: the "Apply discount" use case, while available to cashiers, operates within strict limits set by the admin. Cashiers cannot invent their own discount rates. The system only allows discounts from a pre-approved list, and every discount application is logged in the audit trail. If a cashier is applying discounts unusually frequently or at unusually high rates, the CyberLedger's anomaly detection will flag the pattern.

Manager Functions   Oversight Without Full Control

The Manager section of the Use Case Diagram shows a broader set of actions than the Cashier section, reflecting the manager's oversight responsibilities without granting the full administrative power of the Admin role. Managers at Dulce Cafe are responsible for ensuring daily operations run smoothly, responding to operational problems, and being the first line of review when the CyberLedger raises an alert.

Use Case	Description
View inventory	Access the inventory dashboard to see current stock levels for all products, filter by category, and identify items below minimum levels.
View sales report	Review sales summaries by date range, cashier, product category, or payment method.
Receive low-stock alert	Receive and acknowledge notifications generated by P6 when product stock falls below configured minimums.
Receive fraud alert	Receive and review notifications from the CyberLedger when P11 raises a fraud anomaly flag.
Review audit log	View the audit trail to investigate flagged activities or verify that operations are proceeding correctly.
Void product (extended)	Remove a product from a transaction that has not yet been finalized. Carries a <<extend>> relationship, meaning it can optionally occur in specific scenarios.
Log void action	When a void is performed, the system automatically logs the void action in the audit trail. <<include>> under Void product.

Table 7. Manager Use Cases

The "Void product" use case uses a <<extend>> relationship rather than <<include>>. In UML, <<extend>> means that the extended use case happens only under certain conditions, not always. A manager does not void products during every transaction review, only in specific situations where a legitimate reason exists to do so. The <<extend>> notation captures this conditional nature. And whenever a void does occur, the <<include>> relationship with "Log void action" ensures it is always recorded in the audit log, making the void traceable and accountable.

Admin Functions   Full Control with Full Accountability

The Admin section of the Use Case Diagram shows the most extensive set of actions available to any human user in the StockSecure POS system. The admin at Dulce Cafe, likely the owner or a trusted senior manager, has the authority and the responsibility to configure the entire system, manage all user accounts, and access the complete audit history.

Use Case	Description
Manage users	Create, modify, deactivate, and delete user accounts. Assign and change roles.
Manage products	Add new products to the catalog, update prices, descriptions, and stock quantities, and activate or deactivate products.
Reports	Generate and download comprehensive business and audit reports covering any time period.
Assign roles	Formally assign or change the role of any user in the system. Directly controls RBAC permissions.
Audit logs	Access the complete, unfiltered audit log for the entire system, including archived records.
Void product	The admin also has void capability. <<include>> Log void action ensures all admin voids are also logged.
Log void action	Automatically triggered whenever an admin voids a product. Cannot be bypassed.

Table 8. Admin Use Cases

A critical point about the Admin role from a security perspective: having the highest level of access does not make the admin exempt from accountability. Every action the admin takes, whether creating a new user account, changing a product price, or voiding a transaction, is recorded in the AUDIT_LOG exactly the same way as any other user's actions. If an admin inappropriately modified data, the audit trail would show it. This principle, that even the most powerful user is still auditable, is what makes the CyberLedger truly effective as a fraud prevention tool.

CyberLedger Audit Functions   The Automated Guardian

The bottom section of the Use Case Diagram is labeled "CyberLedger audit functions" and is associated with the "System" actor rather than a human actor. This reflects the fact that these use cases are initiated automatically by the software itself, not by any human action. The CyberLedger operates continuously in the background of the StockSecure POS system, watching every transaction, every login, every change, and every void.

Use Case	Description
Capture system event	Automatically logs every POS system action in real time. <<include>> Detect fraud pattern.
Detect fraud pattern	Analyzes captured events using anomaly detection rules to identify suspicious behavior. Always runs when an event is captured.
Validate transaction	Checks the logical integrity of each transaction, including total calculations, discount validity, and status consistency.

Use Case	Description
Write audit trail	Creates permanent, immutable records of all events. These records cannot be edited or deleted by any user.
Raise fraud alert	When a fraud pattern is detected, the system formally raises an alert and initiates the notification process.
Generate audit report	Compiles audit data into structured reports on a scheduled or on-demand basis.
Archive audit log	Periodically moves older audit records to long-term archival storage while keeping recent records readily accessible.

Table 9. CyberLedger System Use Cases

The most important relationship in the CyberLedger section is the <<include>> arrow connecting "Capture system event" to "Detect fraud pattern." This means that every single time any event is captured, the fraud detection analysis runs automatically. There is no batch processing delay, no manual trigger required, and no way to perform an action in the POS system that bypasses the fraud detection layer. This real-time monitoring capability is what makes the CyberLedger genuinely effective at early fraud detection rather than only discovering problems days or weeks after they occur.

How the Three Diagrams Work Together

At this point, the reader has been introduced to all three supporting diagrams for Chapter 3: the DFD, the ERD, and the Use Case Diagram. Each one has been explained individually, but the full power of these diagrams only becomes clear when we understand how they work as a unified set of system blueprints.

Think of a scenario at Dulce Cafe to illustrate this. It is 11:30 in the morning. A customer orders an iced latte and a chocolate croissant. The cashier who has logged in with their individual credentials, authenticated through P1, and assigned the cashier role through P2 proceeds to process the sale through the "Process sale" use case described in the Use Case Diagram. The cashier selects the iced latte and the chocolate croissant using the "Scan / select item" function, accepts payment via GCash, and generates a digital receipt.

As this transaction completes, multiple things happen simultaneously in the background. The SALES table in the ERD receives a new row recording this transaction. Two SALE_ITEMS rows are created linking this sale to the iced latte and croissant products. The stock field in the PRODUCTS table for both items decreases by one unit each. If the croissant's stock was already at its minimum level, P5: Monitor Stock immediately triggers P6: Low Stock Alert, which sends a notification to the manager's dashboard.

Meanwhile, the CyberLedger has captured this entire transaction as a system event through P7. P8 validates the transaction amounts. P9 runs a pattern analysis and, finding nothing unusual about a normal iced latte and croissant sale, takes no further action. P10 writes a permanent audit trail entry noting that this cashier processed this transaction at this exact time for this exact amount. The AUDIT_LOG table in the ERD receives a new row.

This entire sequence, from the moment the cashier tapped the first item to the moment the customer received their receipt, took perhaps 45 seconds. But within those 45 seconds, the DFD processes P1 through P10 were activated, five tables in the ERD were updated, four use cases in the Use Case Diagram were executed, and two subsystems, the StockSecure POS layer and the CyberLedger layer, worked seamlessly together to complete the transaction and record it permanently. This is the integrated system that the three diagrams collectively describe.

What Makes StockSecure POS Right for Dulce Cafe?

A legitimate question to ask at this point is, why does a small cafe in Manduriao, Iloilo, need a system with this level of complexity and security? The answer lies in understanding that the challenges of managing inventory and preventing internal fraud are not exclusive to large corporations. In fact, they hit small businesses harder because a small business has less financial resilience to absorb losses.

Consider the impact of inventory shrinkage at a small cafe. If a single bottle of premium cold brew coffee goes unaccounted for every day because it is being consumed by staff without proper recording, that is lost revenue every single day. Over a month, that adds up. Over a year, it can represent a meaningful percentage of the cafe's profit margin. The StockSecure POS system's inventory monitoring features address this directly by tracking every product unit sold through the formal transaction system.

Consider the risk of cashier fraud. Without a system that logs every transaction and flags anomalies, a dishonest cashier has many opportunities for manipulation: pocketing cash from a transaction that they void after the customer leaves, applying unauthorized discounts to their own orders or to orders placed by friends, or simply undercharging certain customers while pocketing the difference. Each of these fraud vectors is addressed directly by features described in the diagrams of this chapter.

Dulce Cafe chose to invest in the StockSecure POS system not because they suspect their employees of dishonesty but because proper internal controls protect everyone: the business, the honest employees who work hard every day, and the customers who deserve a fairly priced product. The diagrams in this chapter show exactly how those controls work.

Understanding the Diagram Notations and Legend

For readers who are not yet familiar with the standard conventions used in these types of system diagrams, this section provides a brief guide to the visual notation used throughout the three diagrams. Understanding these conventions helps in reading and interpreting similar diagrams in any future system documentation.

DFD Notation

Symbol	Meaning
Rounded rectangle (blue)	External actor. A person or entity outside the system that interacts with it by sending or receiving data.
Rounded rectangle (green)	POS process. A function or transformation that occurs within the StockSecure POS layer.
Rounded rectangle (red/orange)	Audit process. A function within the CyberLedger subsystem.
Rounded rectangle (amber)	Alert process. Specifically handles the detection and delivery of alerts.
Rectangle (white outline)	Data store. A repository where data is stored persistently.
Solid arrow	Data flow. Shows the direction of data movement between elements.
Dashed arrow	Event or alert flow. Shows the flow of event notifications and alerts.

Table 10. DFD Legend and Notation Guide

Symbol	Meaning
Rectangle with header	Entity. Represents a database table.
PK label	Primary Key. The unique identifier for each row in a table.

Symbol	Meaning
FK label	Foreign Key. A column that references the primary key of another table.
Single vertical line (|)	The "one" side of a relationship. Exactly one record.
Crow's foot symbol	The "many" side of a relationship. Multiple records.
Circle + crow's foot	Zero or more (optional). The relationship may involve zero records on this side.
Relationship label	Text along the connecting line describing the nature of the relationship.

Table 11. ERD Notation Guide

Use Case Diagram Notation

Symbol	Meaning
Stick figure	Actor. A user or external system that interacts with the use cases.
Oval/Ellipse	Use case. A specific function or capability of the system from the user's perspective.
Rectangle (dashed border)	System boundary. Separates what is inside the system from external actors.
Solid line	Association. Connects an actor to the use cases they can initiate.
<<include>>	The connected use case is always included when the main use case occurs.
<<extend>>	The connected use case occurs only under specific, optional conditions.
Dashed oval	Sub-use case. A supporting use case that is part of a larger use case.

Table 12. Use Case Diagram Notation Guide

Security Analysis Through the Diagrams

The three diagrams of Chapter 3 are not merely technical documentation. They are also a blueprint for security. Examining the diagrams through a security lens reveals the multiple layers of protection that the StockSecure POS system provides for Dulce Cafe.

The Security Layers

1. Identity Layer (P1 / USERS entity / Login use case): Every user must prove who they are before entering the system. Failed attempts are counted and accounts are locked after repeated failures. No anonymous actions are possible.
2. Authorization Layer (P2 / role field in USERS / RBAC assignment): Authenticated identity is not enough. Every action requires the appropriate role. A cashier who somehow obtained a manager's credentials would still not be able to access admin functions, because the role is set at account creation, not at login.
3. Transaction Integrity Layer (P3, P8 / SALE_ITEMS and SALES entities / Process sale use case): Every transaction is recorded with complete detail at the line-item level. There is no way to process a sale that is not fully documented. Partial or inconsistent records trigger validation failures.
4. Anomaly Detection Layer (P9 / AUDIT_LOG entity / Detect fraud pattern use case): Pattern analysis runs continuously in real time. Individual transactions that pass integrity checks may still be flagged when viewed as part of a pattern of behavior.
5. Immutable Audit Layer (P10 / AUDIT_LOG entity / Write audit trail use case): All records, once written to the audit trail, cannot be modified or deleted by any user, including the admin. This creates a reliable forensic record.
6. Alert and Escalation Layer (P11, P12 / DS6, DS8 / Raise fraud alert use case): Detected anomalies do not simply disappear into a log file that no one reads. They trigger active notifications to responsible parties who must acknowledge and respond.

Common Fraud Scenarios and How the System Responds

To make the security analysis concrete, it is helpful to walk through several common fraud scenarios that affect small food businesses and explain specifically how the StockSecure POS system and CyberLedger address each one.

Scenario 1: Cashier Void Fraud

A dishonest cashier processes a sale, takes the customer's cash, and then after the customer leaves, voids the transaction in the system and pockets the cash.

System response: The void action triggers "Log void action" in the Use Case Diagram. P7 captures the void event. P9 analyzes the pattern and notes that this cashier has voided an unusually high number of transactions within a short period. P11 raises a fraud alert. P12 notifies the manager. The manager reviews the audit log (DS5/DS8) and can see each void transaction with its timestamp and the cashier's user ID. The manager investigates and takes appropriate action.

Scenario 2: Unauthorized Discount Application

A cashier regularly applies large discounts to orders placed by friends or family, effectively giving them partial freebies.

System response: Every discount application is logged through P7. P9 compares the cashier's discount rate against statistical norms for their role and historical patterns. Unusually frequent or unusually large discounts trigger a fraud flag in DS6. The manager receives an alert and can use the "View sales report" use case to pull a report filtered by this cashier's discount activity, revealing the pattern clearly.

Scenario 3: Inventory Theft

Products are consumed or removed from inventory without being recorded as a sale.

System response: The PRODUCTS entity tracks stock levels in real time. P5: Monitor Stock compares actual recorded stock against expected levels based on sales. Unexplained discrepancies between what was sold (SALE_ITEMS records) and what is physically missing will be visible in inventory reports. The admin can run a report comparing opening stock, recorded sales, and closing stock for any product and any date range.

Scenario 4: Unauthorized Price Modification

A user attempts to reduce a product's price temporarily to benefit a friend's purchase.

System response: Only Admin users can access P4: Manage System, which includes product price changes. If an admin makes a price change, P7 captures the event. The AUDIT_LOG records the old price value and the new price value along with the admin's user ID and the exact timestamp. The "audit logs" use case allows the admin's actions to be reviewed by the business owner. If the business owner is the only admin and notices an unexplained price change, the audit log provides irrefutable evidence.

Section 3.7: Inventory Monitoring in Detail

The inventory monitoring component of the StockSecure POS system deserves dedicated discussion because it addresses one of the most immediate and practical operational needs of Dulce Cafe. Running out of a popular menu item during peak hours is not just inconvenient for customers; it represents lost revenue that cannot be recovered. The StockSecure POS system addresses this through a combination of real-time stock tracking and proactive alerting.

The Stock Monitoring Cycle

The inventory monitoring cycle in the StockSecure POS system operates continuously and automatically. Here is how it works in plain terms:

7. The admin sets a minimum stock level for each product when the product is added to the system. For example, the admin might set a minimum of 10 units for a particular type of coffee bean blend. This value is stored in the min_stock_level field of the PRODUCTS entity.
8. Every time a cashier processes a sale that includes this product, P3: Process Sale decrements the stock field in the PRODUCTS table by the quantity sold.
9. P5: Monitor Stock continuously checks the current stock field against the min_stock_level field for every product in the catalog.
10. The moment stock falls to or below the minimum level, P5 triggers P6: Low Stock Alert.
11. P6 generates an alert notification and delivers it to the manager's dashboard. The notification identifies the specific product, its current stock level, and its minimum threshold.
12. The manager sees the alert through the "Receive low-stock alert" use case and can take immediate action to arrange a restock.

This cycle eliminates the need for manual stock checks, which are time-consuming and error-prone. It also means that the manager does not need to be watching the inventory dashboard continuously; the system watches for them and alerts them only when action is needed. This is an efficient use of the manager's time and attention.

The Relationship Between Inventory and Fraud Prevention

An often-overlooked but powerful benefit of StockSecure POS's real-time inventory tracking is its significant contribution to fraud prevention. By automatically deducting every unit sold through **P3: Process Sale** from the current stock levels in the **PRODUCTS** table, the system maintains an accurate, up-to-the-minute record of what should be physically present in the café. When the actual physical count of an item is noticeably lower than the system's calculated stock, the resulting discrepancy immediately becomes visible to managers.

This kind of mismatch—whether a few missing pastries or several bottles of premium syrup—serves as an early warning signal that products may have left the store without being properly recorded as a sale. Such gaps are classic indicators of internal theft, "sweethearting" (giving free items to friends), or unauthorized giveaways.

This tight integration between inventory monitoring (**P5** and **P6**) and the broader CyberLedger security framework is a deliberate key design principle of the StockSecure POS system. The two functions powerfully reinforce each other:

- Accurate, real-time inventory data makes fraud much harder to conceal, as any unrecorded removal creates a clear, auditable trail.
- The system's fraud detection capabilities (**P9**) actively monitor for suspicious patterns, such as repeated stock discrepancies linked to specific users or shifts, prompting timely investigation rather than allowing them to be casually dismissed as "accounting errors."

For Dulci Cafe, this creates a strong deterrent effect while maintaining a professional and trusting work environment. Staff know the system is always watching, and managers gain confidence that both operational efficiency and financial integrity are continuously protected. By turning inventory accuracy into a powerful security tool, StockSecure POS helps safeguard the café's profits and reinforces a culture of accountability and transparency.

CHAPTER 4:

4.2 Overview of the Developed System

The developed system, **StockSecure POS: Integrating Role-Based Access Control (RBAC) in a Point-of-Sale System with CyberLedger Secure Transaction and Audit Monitoring**, is a web-based Point-of-Sale (POS) and inventory management system specifically developed for Dulcis Café. The system was designed to address the operational problems identified during the conduct of the study, including manual sales recording, inefficient inventory monitoring, lack of employee accountability, inadequate security controls, and the absence of a reliable audit mechanism. By integrating modern information technology with cybersecurity principles, the system provides an efficient, secure, and centralized solution for managing the café's daily business operations.

The primary objective of StockSecure POS is to automate and simplify the café's business processes while strengthening security through Role-Based Access Control (RBAC) and CyberLedger. Traditional POS systems mainly focus on processing customer transactions;

however, the developed system extends these capabilities by incorporating inventory automation, secure user authentication, comprehensive audit logging, and access restriction based on employee responsibilities. This integration helps reduce human errors, minimize opportunities for internal fraud, and improve overall operational efficiency.

The system was developed using **Visual Studio Code (VS Code)** as the primary integrated development environment (IDE). Visual Studio Code provided a flexible and efficient coding environment that enabled the development team to organize, write, debug, and maintain the system's source code. Its built-in debugging tools, extension support, version control integration, and code management features significantly improved the productivity of the developers throughout the system development process.

For the backend infrastructure, the system utilizes **Appwrite**, an open-source backend-as-a-service (BaaS) platform. Appwrite serves as the central backend responsible for managing the system database, user authentication, session management, role-based permissions, cloud storage, and secure API communication. Instead of developing a backend server from scratch, Appwrite provides built-in services that simplify development while ensuring that data remains secure and easily manageable. Through Appwrite's authentication services, every employee is required to log in using individual user credentials before accessing the system. This prevents anonymous access and establishes accountability for every recorded transaction.

One of the most important security features implemented in the system is **Role-Based Access Control (RBAC)**. Rather than allowing every employee unrestricted access to all system functions, RBAC assigns permissions according to the user's designated role within the café. Administrators have full access to all modules, including user management, inventory configuration, audit reports, and system settings. Managers are authorized to monitor sales reports, approve refunds, oversee inventory, and generate business analytics. Cashiers are only allowed to process customer orders, receive payments, and print receipts. Inventory personnel are responsible for monitoring stock levels, recording deliveries, updating inventory records, and managing supplier information, while auditors are limited to viewing audit logs and transaction histories without having permission to modify business records. This separation of responsibilities follows the principle of least privilege, ensuring that employees can only perform tasks necessary for their assigned duties while reducing the possibility of unauthorized access or fraudulent activities.

The system also incorporates **CyberLedger**, a secure transaction and audit monitoring component developed to provide complete transparency and accountability throughout business operations. Every significant action performed within the system—including user logins, logout sessions, sales transactions, inventory adjustments, refunds, void transactions, stock updates, report generation, and administrative changes—is automatically recorded with corresponding timestamps and user identification. These audit records create a permanent activity history that allows management to review employee actions, investigate discrepancies, and identify suspicious activities whenever necessary. Because every transaction is linked to a specific authenticated user account, the possibility of denying responsibility for performed actions is greatly minimized.

The **Sales and Transaction Processing Module** serves as the primary interface used by cashiers during daily operations. Through this module, customer orders are entered into the system, product prices are automatically calculated, discounts are applied when applicable, and multiple payment methods can be processed efficiently. After every successful transaction, the system automatically generates an electronic receipt while simultaneously recording the sale in the database. Unlike manual recording methods, this automated process minimizes computational errors, speeds up customer service, and ensures that all transactions are accurately documented.

A major enhancement of StockSecure POS is its **Automated Inventory Management Module**. Instead of manually updating inventory after each business day, the system automatically deducts ingredients and products whenever a sale is completed. Each menu item is associated with predefined inventory quantities based on standardized recipes, allowing the system to calculate inventory consumption in real time. This automation significantly improves inventory accuracy while reducing stock discrepancies caused by manual computation. Additionally, the system monitors inventory levels continuously and notifies authorized personnel whenever stock reaches predefined minimum thresholds, enabling timely replenishment and reducing the risk of stock shortages.

The **Supplier Management Module** allows authorized users to maintain supplier records, record deliveries, update purchase information, and monitor procurement activities. By maintaining accurate supplier information and purchase history, the café can improve procurement planning, establish stronger supplier relationships, and minimize emergency purchasing caused by unexpected inventory shortages.

Another important component is the **Reporting and Analytics Module**, which provides real-time business reports that assist management in evaluating operational performance. The system generates daily sales reports, monthly sales summaries, inventory reports, audit logs, employee activity reports, transaction histories, and product performance reports. These reports enable management to monitor revenue, identify best-selling products, analyze inventory movement, evaluate employee performance, and support strategic business decision-making using reliable and updated information instead of manually prepared reports.

The overall workflow of the developed system begins with user authentication through Appwrite, where users log in using their assigned accounts. Once authenticated, the system determines the user's assigned role and grants access only to authorized modules through the RBAC mechanism. Cashiers process customer transactions using the Sales Module, which automatically records sales and updates inventory. Inventory personnel monitor stock availability and supplier deliveries, while managers supervise operations through dashboards, reports, and audit logs. Throughout the entire workflow, CyberLedger continuously records every significant activity to ensure transparency, accountability, and data integrity.

Overall, StockSecure POS serves as an integrated business management solution that combines secure transaction processing, automated inventory management, role-based system security, supplier monitoring, reporting, and audit tracking into a single platform. By utilizing Visual Studio Code for software development and Appwrite as the backend infrastructure, the system provides a scalable, reliable, and secure environment capable of supporting the daily operational requirements of Dulcis Café. The implementation of RBAC and CyberLedger distinguishes the system from conventional POS solutions by providing enhanced protection against internal fraud, unauthorized access, and data manipulation while simultaneously improving business efficiency, inventory accuracy, and management decision-making. This integrated approach demonstrates how modern software engineering and cybersecurity practices can be effectively applied to solve real-world operational challenges faced by micro, small, and medium-sized café businesses.

System Development Methodology

The researchers will use the Rapid Application Development (RAD) methodology in developing the StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention & CyberLedger: Secure Transaction and Audit Monitoring System. RAD is a software development methodology that emphasizes rapid prototyping, continuous user feedback, and iterative development. It enables developers to quickly build and refine system components while ensuring that the final product meets the needs of its users.

This approach is suitable for the proposed system because it allows the development team to continuously improve the POS, inventory monitoring, RBAC, and audit monitoring features through frequent testing and user evaluation.

The RAD methodology consists of the following phases:

1. **Requirements Planning**

In this phase, the researchers gather and analyze the requirements of the business. Interviews, observations, and discussions are conducted with business owners and potential users to identify the system's functional and non-functional requirements. The required features include sales processing, inventory monitoring, Role-Based Access Control (RBAC), secure transaction recording, audit trail generation, and fraud prevention mechanisms.

2. **User Design**

During this phase, prototypes of the system are created with the active participation of users. The researchers design the user interface, database structure, process flow, and access permissions for different user roles such as Administrator, Manager, Cashier, and Inventory Staff. Users provide feedback on the prototypes, allowing the researchers to improve the system design before full implementation.

3. **Rapid Construction**

In this phase, the actual development of the system takes place. The researchers develop the POS modules, inventory management functions, RBAC implementation, secure transaction processing, CyberLedger audit monitoring, and reporting features. Continuous testing and refinement are performed throughout development to ensure that each module functions correctly and meets user expectations.

4. **Cutover**

The completed system is deployed for actual use. Final testing, data migration, user training, and system installation are performed during this phase. Any remaining issues are corrected before the system becomes fully operational. After deployment, the researchers monitor the system and provide maintenance and updates to improve performance, security, and functionality.

Why RAD is Suitable for the Proposed System

The Rapid Application Development (RAD) methodology is appropriate for the proposed system because it allows the researchers to develop functional prototypes quickly and obtain continuous feedback from users. Since StockSecure POS includes critical features such as Role-Based Access Control (RBAC), inventory monitoring, secure transaction processing, and CyberLedger audit monitoring, frequent testing and refinement are essential to ensure accuracy, security, and usability. RAD shortens development time while maintaining software quality, making it an effective methodology for developing a secure and reliable POS system**.**

Requirements Gathering

Requirements gathering is the process of identifying, collecting, and analyzing the needs and expectations of stakeholders to ensure that the proposed system addresses existing business problems. For the **StockSecure POS** project, this involves gathering information from business owners, cashiers, inventory managers, and auditors through interviews, observations, questionnaires, and document analysis. These activities help identify current challenges such as unauthorized system access, manual inventory errors, limited transaction monitoring, and the risk of internal fraud. Understanding these issues provides a clear foundation for designing a system that meets both operational and security requirements.

The gathered requirements serve as the basis for developing a secure and efficient POS system integrated with **Role-Based Access Control (RBAC)** and **CyberLedger**. Functional requirements include secure user authentication, role-specific permissions, automated inventory updates, transaction processing, audit trail generation, and comprehensive reporting. Non-functional requirements focus on system security, reliability, performance, usability, and scalability to ensure the solution operates effectively in a business environment. By accurately defining these requirements, the proposed system can improve inventory monitoring, strengthen accountability, prevent unauthorized activities, and reduce opportunities for internal fraud.

System Planning

The study identifies the objectives, scope, feasibility, and resources required for developing the proposed StockSecure POS and CyberLedger system. The project aims to address common issues in traditional Point-of-Sale (POS) systems, such as unauthorized access, inventory discrepancies, internal fraud, and the lack of transparent transaction monitoring.

The proposed system will integrate **Role-Based Access Control (RBAC)** to ensure that users can only access features and data according to their assigned roles (e.g., Administrator, Manager, Cashier, and Inventory Staff). It will also incorporate **CyberLedger**, a secure transaction and audit monitoring module that records all user activities and financial transactions, allowing management to trace system actions, detect suspicious behavior, and maintain accountability.

The project scope includes:

- Secure user authentication and role-based authorization.
- Sales and transaction processing.
- Real-time inventory monitoring and stock updates.
- Audit trail and transaction logging.
- User activity monitoring and report generation.
- Dashboard for inventory, sales, and security analytics.

The system is intended for small and medium-sized businesses (SMEs), particularly retail stores and cafés, that require improved inventory control, secure transaction processing, and fraud prevention.

The development will follow the **System Development Life Cycle (SDLC)**, beginning with requirements gathering and analysis, followed by system design, implementation, testing, deployment, and maintenance. This structured approach ensures that the system meets business requirements while providing security, reliability, and efficiency.

System Design

The overall architecture of the proposed system was carefully planned to ensure that all components work together efficiently, securely, and reliably. The system was designed to support the core operations of a point-of-sale (POS) environment while addressing common business challenges such as inventory inaccuracies, unauthorized system access, internal fraud, and the lack of transaction transparency. The design process focused on creating a well-structured system that is easy to maintain, scalable for future enhancements, and capable of providing secure and accurate business operations.

The proposed system consists of several interconnected modules, including **Role-Based Access Control (RBAC), Sales Processing, Inventory Monitoring, Transaction Management, User Management, Audit Trail Logging, and Report Generation**. Each module has a specific function but works together to ensure that daily business operations are performed smoothly. The **RBAC module** serves as the foundation of the system's security by assigning different levels of access to users based on their designated roles. For example, cashiers are limited to processing sales transactions, while managers can monitor inventory, review reports, and approve selected operations. System administrators have full control over user accounts, permissions, and system configurations. This approach minimizes unauthorized access and reduces the risk of internal fraud by ensuring that users can only perform tasks relevant to their responsibilities.

The **Sales Processing module** was designed to record customer purchases accurately and update inventory records automatically after every successful transaction. This real-time synchronization helps eliminate manual inventory updates, reduces human error, and ensures that stock levels remain accurate at all times. The **Inventory Monitoring module** continuously tracks product availability, stock movements, and inventory adjustments, allowing managers to identify low-stock items, monitor product performance, and make informed restocking decisions. This functionality helps prevent stock shortages, overstocking, and inventory discrepancies that could negatively affect business operations.

To strengthen accountability and transparency, the system incorporates the **CyberLedger Audit Monitoring module**, which automatically records every important activity performed within the system. User logins, sales transactions, inventory updates, product modifications, and administrative actions are stored as audit logs with corresponding timestamps and user information. These records provide a complete history of system activities, making it easier to investigate errors, resolve disputes, detect suspicious behavior, and support internal audits. By maintaining an immutable record of critical transactions, the organization can improve operational transparency and strengthen fraud prevention efforts.

The database was carefully designed to securely store all information required by the system. It contains tables for user accounts, employee roles and permissions, product details, inventory records, sales transactions, customer information (when applicable), audit logs, and generated reports. Relationships between these data entities were established to maintain data integrity, reduce redundancy, and ensure efficient retrieval of information.

Security measures such as user authentication, password encryption, access validation, and transaction verification were incorporated into the database design to protect sensitive business information from unauthorized access.

To provide a clear blueprint for development, several software design artifacts were prepared during this phase. **Unified Modeling Language (UML) diagrams** were created to illustrate how users interact with the system and how different modules communicate with one another. **Entity Relationship Diagrams (ERDs)** were developed to define the relationships between database entities and ensure an organized database structure. In addition, **wireframes** were designed to visualize the layout of the system interfaces, including the cashier dashboard, inventory management screen, administrator control panel, audit monitoring page, and reporting dashboard. These interface designs focus on usability, allowing users to navigate the system efficiently with minimal training.

Overall, the system design provides a comprehensive framework for developing **StockSecure POS: Integrating Role-Based Access Control (RBAC) in a POS System, A Solution for Inventory Monitoring and Internal Fraud Prevention & CyberLedger: Secure Transaction and Audit Monitoring System**. By combining secure access control, automated inventory management, real-time sales processing, and comprehensive audit monitoring, the proposed design enhances operational efficiency, improves data accuracy, strengthens accountability, and provides a secure environment for managing business transactions.

Software Development

The software development phase focused on transforming the approved system design into a fully functional Point-of-Sale (POS) application by implementing each module using the selected programming languages, frameworks, and development tools. Development followed an incremental and modular approach, allowing each system component to be built, tested, and integrated systematically to ensure functionality, reliability, and maintainability.

The implementation began with the **user authentication and Role-Based Access Control (RBAC)** module, which established secure login procedures and restricted system access according to predefined user roles such as administrator, manager, cashier, inventory staff, and auditor. This ensured that users could only perform tasks and access information relevant to their assigned responsibilities, reducing the risk of unauthorized actions and enhancing overall system security.

Subsequent development included the **product and inventory management module**, which enabled administrators to manage product records, monitor stock levels in real time, receive low-stock notifications, and track inventory movements. This functionality provided accurate inventory monitoring and minimized discrepancies caused by manual stock recording.

The **transaction processing module** was then implemented to facilitate secure and efficient sales transactions. This component supported product selection, payment processing, receipt generation, refunds, discounts, and transaction validation while ensuring that sensitive operations required proper authorization based on user roles. All transactions were automatically recorded to maintain data accuracy and operational transparency.

To strengthen accountability, the **CyberLedger audit monitoring module** was integrated into the system. This module automatically captured and stored detailed audit logs, including user login activities, sales transactions, inventory adjustments, product modifications, and administrative actions. These records created a comprehensive audit trail that enabled management to monitor employee activities, detect suspicious behavior, investigate discrepancies, and support internal fraud prevention.

The development phase also included the implementation of a **reporting and analytics module**, which generated real-time dashboards and visual reports for sales performance, inventory status, transaction history, user activities, and audit logs. These reports provided valuable business insights to support operational monitoring and informed decision-making.

Throughout the development process, each module underwent continuous integration and functional testing to verify that system requirements were met and that all components operated correctly both independently and as an integrated system. This iterative approach ensured the delivery of a secure, reliable, and efficient POS solution capable of improving inventory management, strengthening internal controls, preventing unauthorized system activities, and maintaining accurate transaction records.

Testing

Each module of the system was tested individually before being combined with the other components. Unit testing was conducted to ensure that features such as role-based access control, inventory management, sales processing, and audit logging functioned correctly. Afterward, integration testing was performed to verify that all modules worked together without errors. System testing was then carried out to evaluate the overall performance, security, and reliability of the application. Finally, User Acceptance Testing (UAT) was conducted with selected users to confirm that the system met the intended requirements and was easy to use in actual business operations.

Deployment

After the testing phase was successfully completed, the system was deployed in its intended environment. The database was configured, the application was installed, and user accounts were created according to their assigned roles and access permissions. Necessary system settings were also completed to ensure secure access for authorized users. Before full implementation, users were given a brief orientation on the system's features and basic functions to help them operate the application effectively.

Maintenance

Once the system was fully implemented, regular maintenance was performed to ensure its continuous performance and reliability. This included monitoring the system for any issues, fixing reported bugs, updating security features, and improving existing functions based on user feedback. Routine database backups and software updates were also conducted to protect business data and maintain the efficiency of inventory monitoring, transaction processing, and audit tracking. Through continuous maintenance, the system remains secure, reliable, and capable of supporting the organization's daily operations.

System Architecture

The StockSecure POS system is designed using a client-server architecture to ensure secure, organized, and efficient business operations. The system consists of four main components: the user interface, application server, database server, and reporting module. Users such as cashiers, inventory staff, managers, and administrators access the system based on their assigned roles through Role-Based Access Control (RBAC). This ensures that each user can only perform tasks related to their responsibilities, reducing unauthorized access and minimizing internal fraud. All sales, inventory updates, and user activities are recorded in the database, while the audit monitoring module tracks transactions and system logs to improve accountability and security. This architecture allows the system to manage inventory accurately, process transactions efficiently, and provide reliable reports for decision-making.

The architecture consists of the following components:

User Interface (UI)—Allows users to interact with the system for sales, inventory, and account management.

Application Server – Processes user requests, business logic, and system operations.

Database Server – Stores transaction records, inventory data, user accounts, and audit logs.

Role-Based Access Control (RBAC) Module – Restricts system access based on user roles and permissions.

Inventory Management Module – Tracks stock levels, product information, and inventory movements.

Transaction Processing Module – Handles sales transactions, payments, receipts, and order processing.

Audit Monitoring Module – Records user activities and transactions to support accountability and fraud detection.

Reporting Module – Generates sales, inventory, and audit reports for managers and administrators.

Authentication Module – Verifies user identities through secure login before granting system access.

Backup and Recovery Module – Protects system data by creating backups and enabling data recovery when needed.

Presentation Layer

The presentation layer is rendered server-side using Flask's Jinja2 templating engine and delivered to the browser as HTML, CSS, and JavaScript. It is responsible for everything the cashier, manager, or administrator directly sees and interacts with.

- Cashier-facing point-of-sale screen (pos.html) for product search, cart building, and checkout

- Admin dashboard, product catalog, user management, sales, reports, and audit log views (templates/admin/)
- Manager dashboard, sales, reports, and order-cancellation views (templates/manager/)
- Authentication screens: login, admin login, forgot password, reset password, OTP verification
- Printer configuration screen for thermal receipt printer setup
- Shared styling via static/css/pos.css and static/css/admin.css, and client-side behavior via static/js/timezone.js
- Custom 404 and 500 error pages

Interactivity (product search, cart updates, receipt printing, dashboard stats) is driven by JavaScript calling the application layer's JSON API endpoints (routes under /api/...), so the page updates without a full reload

Application Layer

The application layer is a single Flask application (app_complete.py) that implements all business logic, authentication, authorization, and hardware integration. It exposes both page routes (returning rendered HTML) and JSON API routes consumed by the presentation layer.

- Role-based access control via @login_required, @role_required, and @manager_required decorators for admin, manager, and cashier roles
- Authentication: password hashing (Werkzeug), CAPTCHA, TOTP-based two-factor recovery, and password-reset token flow (itsdangerous)
- Sales processing: cart-to-sale conversion, stock deduction, discount handling (Senior Citizen / PWD, per RA 9994 and RA 10754), and order cancellation with reason logging
- Tamper-evident audit trail: an append-only HMAC-SHA256 hash-chained ledger (SimpleLedger) that records every sensitive action and detects retroactive edits
- Receipt generation: ESC/POS raw-byte formatting for thermal printers and ReportLab-based PDF generation for digital/report output
- Hardware integration: USB direct printing (pyusb/libusb), serial printing (pyserial), and native Windows print spooler printing (pywin32/win32print)
- Reporting and dashboard statistics aggregated from sales and ledger data
- Email notifications (password reset, OTP) via Flask-Mail over SMTP

The application exposes roughly 47 routes, split between server-rendered pages (e.g. /pos, /admin, /manager) and JSON APIs (e.g. /api/sale/create, /api/products/search, /api/ledger/verify, /api/print-receipt).

Data Layer

Persistent data is stored in a local SQLite database (database.db), accessed through a pooled/contextual connection helper rather than an ORM. SQLite was chosen for a single-terminal, offline-capable retail deployment that does not require a separate database server.

- Core transactional tables: users, categories, products, sales, sale_items, audit_log
- Security tables: simple_ledger (append-only, HMAC hash-chained) and tamper_log, protected by SQLite triggers that reject UPDATE/DELETE on the ledger

- Sensitive material at rest is protected using Fernet symmetric encryption (cryptography) and dedicated key files under keys/ (flask_secret.key, ledger_hmac.key, secret.key)
- Automated scheduled backups (backup.bat) write timestamped copies of the database to local and external locations, retained under backups/

4.5 Hardware and Software Requirements

4.5.1 Hardware Requirements

The system is designed to run on a single Windows-based point-of-sale terminal with the following minimum hardware:

Component	Minimum Requirement	Notes
Processor	Dual-core 1.6 GHz or better	Any modern x86-64 CPU is sufficient for a single-terminal Flask app
Memory (RAM)	4 GB	8 GB recommended for smoother multitasking with the browser and printer drivers
Storage	10 GB free disk space	Accommodates the application, SQLite database, product images, and scheduled backups
Display	1366 x 768 or higher	For comfortable use of the POS grid and admin/manager dashboards
Input devices	Keyboard and mouse	A barcode scanner configured as a keyboard-wedge device is supported for product lookup
Receipt printer	ESC/POS-compatible thermal printer (e.g., YICHIP POS58, USB VID 0x0FE6 / PID 0x811E)	Connected via USB (with the Zadig/libusb-win32 driver), serial (COM port), or the native Windows print spooler
Network	Not required for core operation	Only needed for outbound email (password reset / OTP) via SMTP

4.5.2 Software Requirements

The application targets Windows deployment and depends on the following software stack:

Layer	Requirement	Version / Detail
Operating System	Windows 10/11	Uses pywin32 for native print-spooler access; start.bat/stop.bat are Windows batch scripts.
Runtime	Python	3.14-compatible (requirements.txt hardened for this version)
Web framework	Flask	3.0.0, with Werkzeug 3.0.1, Jinja2 3.1.3, itsdangerous 2.1.2, click 8.1.7, and MarkupSafe 2.1.5
Database	SQLite 3	Bundled with Python's standard library (sqlite3 module); no separate DB server required

Layer	Requirement	Version / Detail
Printing	python-escpos 3.1, pyserial 3.5, pyusb 1.2.1, pywin32 306	USB printing requires the Zadig utility to install the libusb-win32 driver.
Email	Flask-Mail 0.10.0	SMTP via Gmail (smtp.gmail.com:587, TLS)
Security	cryptography 42.0.5	Fernet symmetric encryption; also used with hashlib/hmac (standard library) for the audit ledger
PDF / reporting	reportlab 4.1.0	Generates printable PDF receipts and reports
Imaging	Pillow 10.3.0	Product image handling and receipt image composition
Configuration	python-dotenv 1.0.1	Loads environment variables (e.g., mail credentials) via load_dotenv(override=True).
Client	Modern web browser (Chrome/Edge)	Renders the Jinja2-generated HTML/CSS/JS interface

4.6 Database Design

The database is a single SQLite file (database.db) initialized by init_db(). It consists of eight tables covering identity, catalog, transactions, and security/audit data, with foreign keys enforcing referential integrity between sales, sale items, products, and users.

- users—system accounts and roles (detailed below)
- categories—product category names used to group products
- products—catalog items, pricing, stock levels, and barcodes
- sales—completed/cancelled transactions, linked to the cashier who created them
- sale_items — line items belonging to a sale, linked to a product
- audit_log — general change log (old/new values) for administrative actions
- simple_ledger—append-only, HMAC-SHA256 hash-chained security ledger (updates/deletes blocked by triggers)
- tamper_log—records of any detected inconsistency in the ledger's hash chain

Users Table

The users table stores login credentials and role assignments for the three account types the system supports: admin, manager, and cashier.

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique internal identifier for the account
username	TEXT	UNIQUE, NOT NULL	Login username
password_hash	TEXT	NOT NULL	Werkzeug-generated password hash (never stored in plain text)
role	TEXT	NOT NULL, CHECK IN ('admin','cashier','manager')	Determines which routes and dashboards the account can access
email	TEXT	UNIQUE	Used for password-reset and OTP-recovery email delivery
is_active	BOOLEAN	DEFAULT 1	Allows an account to be disabled without deleting it
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation time
last_login	TIMESTAMP	—	Timestamp of the most recent successful login
failed_login_attempts	INTEGER	DEFAULT 0	Counter used for lockout after repeated failed logins
locked_until	TIMESTAMP	—	If set, login is blocked until this time has passed

Customers Table

StockSecure POS does not maintain a separate customers table. The system is designed for quick, walk-in retail transactions, so customer identity is captured as a single optional text field on the sales record rather than as a linked entity.

Column (on sales)	Type	Constraints	Description
customer_name	TEXT	nullable (added via ALTER TABLE)	Free-text name entered at checkout; defaults to "Walk-in Customer" when left blank

Because there is no customer master table, StockSecure POS does not track repeat-customer history, loyalty points, or contact details. If customer relationship features (purchase history, loyalty, contact info) are required, a dedicated customers table with a foreign key from sales.customer_id would need to be introduced.

Products Table

The products table holds the full catalog: pricing, stock levels, category assignment, and the barcode used for point-of-sale lookup.

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique internal identifier for the product
name	TEXT	NOT NULL	Product name shown on the POS grid and receipts
description	TEXT	—	Optional longer description
price	REAL	NOT NULL, CHECK(price >= 0)	Unit selling price
stock	INTEGER	NOT NULL, DEFAULT 0, CHECK(stock >= 0)	Current quantity on hand
min_stock_level	INTEGER	DEFAULT 10	Threshold used to flag low-stock items on the dashboard
category_id	INTEGER	FOREIGN KEY → categories(id) ON DELETE SET NULL	Links the product to a category
image	TEXT	—	Filename/path of the product's image under static/images/products/
barcode	TEXT	UNIQUE, encrypted at rest	Barcode used for scanner/keyboard-wedge lookup; encrypted with Fernet
is_active	BOOLEAN	DEFAULT 1	Soft-delete flag; inactive products are hidden from the POS
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation time
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Last modification time

Transactions Table

Transactions are stored in the sales table. Each row represents one completed or cancelled checkout, linked to the cashier who processed it and, where applicable, the manager who cancelled it.

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique internal identifier for the transaction
transaction_id	TEXT	UNIQUE, NOT NULL	BIR-compliant sequential Official Receipt / transaction number
user_id	INTEGER	NOT NULL, FOREIGN KEY → users(id)	Cashier who processed the sale
total_amount	REAL	NOT NULL	Final amount charged, after tax and discounts

Column	Type	Constraints	Description
tax_amount	REAL	DEFAULT 0	Tax portion of the total
discount_amount	REAL	DEFAULT 0	Discount applied (e.g. Senior Citizen / PWD, RA 9994 & RA 10754)
payment_method	TEXT	CHECK IN ('cash','card','digital')	How the customer paid
payment_status	TEXT	DEFAULT 'completed'	Status of the payment
is_cancelled	BOOLEAN	DEFAULT 0	Marks a voided/cancelled transaction
cancelled_at / cancelled_by / cancellation_reason	TIMESTAMP / INTEGER / TEXT	cancelled_by → FOREIGN KEY users(id)	Cancellation audit fields
customer_name	TEXT	nullable	Optional walk-in customer name (see Customers Table)
digital_reference / digital_amount_paid	TEXT	nullable, encrypted at rest	Reference number and amount tendered for digital payments
created_at	TIMESTAMP	DEFAULT CURRENT_TIME	Transaction timestamp

Transaction Details Table

Line-item detail for each transaction is stored in the sale_items table, with one row per product sold within a sale.

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique internal identifier for the line item
sale_id	INTEGER	NOT NULL, FOREIGN KEY → sales(id) ON DELETE CASCADE	The transaction this line belongs to
product_id	INTEGER	NOT NULL, FOREIGN KEY → products(id)	The product sold
quantity	INTEGER	NOT NULL, CHECK(quantity > 0)	Units of the product sold
unit_price	REAL	NOT NULL	Price per unit at time of sale
total_price	REAL	NOT NULL	quantity × unit_price for this line

Analytics Table

StockSecure POS does not persist a dedicated analytics table. Dashboard and reporting figures are computed on demand with aggregate SQL queries over the sales, sale_items, products, and categories tables, then rendered to the admin and manager dashboards/reports pages. The security ledger's own statistics (/api/ledger/stats) are computed the same way, from simple_ledger.

Metric	Source	Computed As
Today's sales / transactions	sales	SUM(total_amount) and COUNT(*) where date(created_at) = today
Low-stock count	products	COUNT(*) where stock <= min_stock_level and is_active = 1
Active products	products	COUNT(*) where is_active = 1
Daily sales (30-day trend)	sales	COUNT(*) and SUM(total_amount) grouped by date(created_at), excluding cancelled sales
Top products (30-day)	sale_items, products, sales	SUM(quantity) and SUM(total_price) grouped by product, ordered by units sold, top 10
Category sales (30-day)	sale_items, products, categories, sales	COUNT(DISTINCT sale) and SUM(total_price) grouped by category

Because these figures are derived at query time rather than stored, they are always consistent with the underlying sales data, but they are not retained as historical snapshots. If trend data beyond the 30-day window or point-in-time snapshots are needed, a dedicated analytics/summary table populated by a scheduled job would need to be added.

CHAPTER 5:

5.1 Introduction

This chapter provides a comprehensive conclusion to the development of StockSecure POS: Integrating Role-Based Access Control (RBAC) in a Point-of-Sale System with CyberLedger Secure Transaction and Audit Monitoring, the integrated point-of-sale and inventory management system developed for Dulci Cafe in Mandurriao, Iloilo City. It synthesizes the results presented in the preceding chapters, assesses the extent to which the general and specific objectives of the study were achieved, and offers recommendations for future enhancement and implementation strategies that can further maximize the system's utility for Dulci Cafe and for other small and medium-sized food-service enterprises (SMEs) facing similar operational and security challenges.

5.2 Summary of Findings

Reviewing the system that was actually built against the general and specific objectives set out in Chapter 1 shows that the core research objectives were substantially achieved, while several of the more expansive, forward-looking features described in the specific objectives remain as opportunities for future development rather than as delivered functionality. The findings for each major objective are summarized below.

Role-Based Access Control

The first objective, to design and develop a role-based access control module, was achieved. StockSecure POS enforces access restrictions at the server level through Flask decorators (**@login_required**, **@role_required**, and **@manager_required**) that are applied to every protected route so that permissions cannot be bypassed by navigating directly to a restricted page. The delivered system organizes staff into three account types, administrator, manager, and cashier, rather than the four-tier structure (owner, manager, cashier, and barista) originally envisioned in Chapter 1. In practice, the barista-level functions were folded into the cashier role because Dulci Cafe's actual staffing does not separate order-taking from beverage preparation as distinct system users. Every user must authenticate individually before performing any action, satisfying the study's goal of tagging each action to the specific person who performed it.

Real-Time Inventory Management

The second objective, to create a real-time inventory management system, was likewise achieved in its core form. Each product record carries a live stock count and a configurable minimum-stock threshold, and the system deducts inventory automatically as each sale is completed rather than requiring a manual end-of-day reconciliation. Administrators and managers can view low-stock items directly on the dashboard.

However, several of the more advanced inventory capabilities described among the specific objectives in Chapter 1, such as barcode or RFID-driven stocktaking, demand-based forecasting algorithms, multi-location stock tracking, and a separate ledger for waste and spoilage distinct from sales deductions, were not part of the delivered system. The current design assumes a single storage location and a single point of sale, which is appropriate for Dulci Cafe's present scale but would need to be extended before the system could serve a multi-branch operation.

CyberLedger

The third objective, to integrate CyberLedger as a secure, tamper-evident ledger, was achieved through the **simple_ledger** table, an append-only structure protected by SQLite triggers that reject UPDATE and DELETE operations, with each entry cryptographically linked to the previous one through an HMAC-SHA256 hash chain and any detected inconsistency written to a companion **tamper_log** table. This satisfies the study's requirement that financial and operational records cannot be altered retroactively without the alteration being detectable. At the same time, some of the more ambitious audit-related objectives described in Chapter 1, including automated real-time anomaly detection for unusual void patterns or after-hours discounts, a formal thirty-day active-retention and archival policy, and encrypted off-site backup for long-term audit storage, were not implemented in the version of the system presented in Chapter 4. The ledger reliably captures and protects the audit trail, but the proactive analysis of that trail currently depends on a manager or owner actively reviewing the records rather than on the system surfacing anomalies on its own.

Supporting Modules and Objectives

Beyond the three core modules, the study also achieved a functioning sales and transaction-processing workflow that supports cash, card, and digital payment methods, applies Senior Citizen and PWD discounts in accordance with RA 9994 and RA 10754, and produces BIR-formatted receipt numbers and printable, ReportLab-generated reports. The reporting and analytics dashboard delivers real-time figures for daily sales, low-stock counts, top-selling products, and category performance, computed directly from the sales tables rather than from a separate analytics store, which keeps the figures accurate but limits historical trend reporting to the underlying transaction data rather than to stored point-in-time snapshots.

Two objectives described in Chapter 1 were only partially realized. A supplier management capability was described in the system overview, but no dedicated supplier or delivery table appears in the final database design presented in Chapter 4, so procurement tracking remains an area for completion rather than a finished module. Similarly, the customer-relationship and loyalty features envisioned in the general objective, including purchase history and loyalty points, were not carried into the final schema; StockSecure POS captures only an optional walk-in customer name on each sale and does not maintain a dedicated customers table. These gaps do not undermine the three core objectives that defined the study's central contribution, but they do mark the clearest boundary between what the research set out to explore in its broadest framing and what was ultimately delivered as a working, testable system.

5.3 Conclusions

Based on the findings summarized above, the study concludes that the general objective of developing and implementing an integrated point-of-sale system for Dulci Cafe that combines role-based access control, real-time inventory management, and CyberLedger secure transaction monitoring was substantially achieved. The delivered system directly addresses the operational problems identified in Chapter 1, replacing undifferentiated staff access with permission-based roles, replacing manual stock counting with automatic real-time deduction and low-stock alerts, and replacing an absent or informal record of transactions with a cryptographically protected, tamper-evident ledger.

The study further concludes that the three specific technical objectives, the RBAC module, the real-time inventory module, and CyberLedger, were each implemented in a form appropriately scaled to a single-terminal, single-branch café operation rather than in the more expansive form originally sketched in the study's broader aspirational objectives. This scaling down was not a failure of the research process but a reasonable and defensible outcome of grounding the system in Dulci Cafe's actual operational context, consistent with the developmental research design and the Rapid Application Development methodology adopted in Chapter 3, both of which prioritize a working artifact validated against real conditions over a feature list validated only on paper.

The study also concludes that the features left unimplemented, supplier and procurement tracking, customer relationship and loyalty management, automated anomaly detection, multi-location or multi-branch support, and formal data lifecycle management, are not evidence that RBAC and audit-ledger integration are unsuitable for small Philippine cafés. On the contrary, the successful implementation of the core security and inventory functions demonstrates that the approach is technically sound and operationally practical at this scale, and the unimplemented features represent a natural and well-defined roadmap for continued development rather than unresolved problems with the underlying design.

Finally, the study concludes that StockSecure POS and CyberLedger, taken together, provide Dulci Cafe with a meaningfully more secure and accountable operating environment than the manual and loosely supervised processes it previously relied on, and that the system offers a transferable reference architecture, encompassing its role structure, its append-only ledger design, and its single-terminal deployment model, that other small and medium-sized café businesses in the Philippines can adapt to their own operations.

5.4 Recommendations

On the basis of the findings and conclusions presented above, the following recommendations are offered to Dulci Cafe, to future researchers and developers who may continue this work, and to BSIT students who may undertake similar projects.

For Dulci Cafe and Similar Café Operators

- Conduct a formal round of user acceptance testing with recorded task completion times, error rates, and structured satisfaction ratings for each role, together with an Apache JMeter load test under simulated peak-hour conditions, so that the functionality, usability, efficiency, reliability, and security criteria defined in Chapter 3 are validated with actual measurements rather than assessed only qualitatively.
- Adopt a documented, offsite, or encrypted cloud backup routine in addition to the local backup.bat schedule already in place, since a single on-premises terminal represents a single point of hardware failure for both transaction data and the CyberLedger audit trail.
- Establish a recurring schedule for reviewing CyberLedger audit logs, for example weekly, rather than relying solely on ad hoc review, until automated anomaly alerts described below are implemented.
- Continue the role-based training program established during deployment for every new hire, using the reference guides already produced for each role, to sustain the employee preparedness that the literature identifies as critical to long-term POS adoption.

For Future Development of StockSecure POS

- Introduce a dedicated suppliers and deliveries table so that the supplier management capability described in the system overview is fully reflected in the database design and available through the administrator interface.
- Add a customers table with a foreign key from the sales record to support purchase history, loyalty points, and targeted promotions, fulfilling the customer-relationship objective set out in Chapter 1.
- Build automated anomaly detection on top of the existing ledger data, for example, flagging an unusual concentration of voids from a single cashier or discounts applied outside authorized hours, so that CyberLedger moves from a passive, reviewable record to an active, alert-generating monitoring tool.
- Implement a scheduled data lifecycle policy, including archival of older ledger and transaction records and a dedicated analytics or summary table, to support trend reporting beyond the current thirty-day query window without slowing down the live database.
- Extend authentication to include multi-factor login for administrator and manager accounts, not only the existing TOTP-based password-recovery flow, and evaluate barcode or RFID integration to speed up stocktaking and receiving.
- Plan a migration path toward multi-terminal or multi-branch deployment, potentially with secure synchronization between branch databases, should Dulci Cafe or an adopting business expand beyond a single location.

For Future Researchers and BSIT Students

- Use StockSecure POS as a baseline for comparative studies that apply the same RBAC-plus-audit-ledger approach to other MSME contexts, such as retail stores, pharmacies, or small restaurants, to test how well the architecture generalizes beyond the café setting.
- Conduct a longitudinal evaluation after several months of live deployment to measure actual reductions in inventory discrepancy and any change in reported or detected fraud incidents, since the present study evaluates the system's design and construction rather than its long-run operational impact.
- Explore the addition of predictive analytics or simple machine-learning models over the sales and inventory data already being collected to support demand forecasting and staff scheduling as later extensions of the reporting module.

5.5 Impact of the Study

The completion of StockSecure POS and CyberLedger carries implications that extend beyond the immediate technical deliverable, touching the café that served as the study's setting, the wider community of small food-service operators, and the academic and professional development of the researchers themselves.

Impact on Dulci Cafe

For Dulci Cafe, the most direct impact of the study is operational. Order processing, payment recording, and receipt generation are now handled through a single, automated workflow rather than through separate manual steps, which reduces the time cashiers spend on each transaction and lowers the likelihood of pricing or computation errors reaching the customer. Inventory levels are updated the moment a sale is completed, giving the owner and managers a live picture of stock on hand instead of a count that is only as current as the last manual audit. Perhaps most significantly, the RBAC framework and the CyberLedger audit trail together give the owner a level of accountability that the café did not previously have: every sale, void, refund, and administrative change is now attributable to a specific authenticated user and cannot be quietly altered afterward. This shifts internal control from something the owner had to enforce through direct supervision to something the system itself enforces continuously, even when the owner is not physically present.

Impact on the Local SME Sector

Beyond Dulci Cafe itself, the study contributes a documented, working reference architecture that other small cafés and similarly scaled food-service businesses in Iloilo and elsewhere in the Philippines can study and adapt. Because the system was deliberately built around a single, modest terminal and a local SQLite database rather than around expensive cloud infrastructure, the approach demonstrated here is realistic for owners who cannot commit to large recurring technology costs. In this sense, the study advances the broader goal, discussed in the review of related literature in Chapter 2, of closing the gap between the security and accountability tools available to large retail chains and the tools realistically available to independently owned, resource-constrained MSMEs.

Impact on the Researchers and the Academic Community

For the researchers, the project provided direct, hands-on experience in translating a real business problem into a working, security-conscious software system, from stakeholder interviews and requirements analysis through RAD-based iterative development, testing, and deployment. This experience strengthened practical competencies in role-based access control design, secure audit-log architecture, and web application development that extend well beyond the specific requirements of this study. For the wider BSIT program and future capstone researchers, the study leaves behind a concrete example of how developmental research design and design-science principles can be applied to produce an artifact that is simultaneously an academic contribution and a genuinely usable tool for a real small business partner.

5.6 Summary

This study set out to design, develop, and evaluate an integrated point-of-sale system for Dulci Cafe, a small café in Mandurriao, Iloilo City, capable of addressing the operational inefficiencies, weak inventory control, and internal fraud exposure that commonly affect small Philippine food-service businesses. The resulting system, StockSecure POS, combines a role-based access control module that limits each staff member to the functions their position requires, a real-time inventory management module that automatically updates stock as sales occur, and CyberLedger, an append-only, cryptographically chained audit ledger that records every significant transaction and system action in a form that cannot be silently altered.

Guided by a developmental research design and the Rapid Application Development methodology, the study proceeded through requirements gathering, iterative prototyping with direct stakeholder feedback, full system development, and structured testing before deployment. The findings presented in this chapter show that the study's three central technical objectives were achieved in a form appropriately scaled to a single-terminal café operation, while several broader, more ambitious features described early in the study, including supplier management, customer loyalty tracking, automated anomaly detection, and multi-branch support, remain unrealized and are recommended as directions for continued development rather than as shortcomings of the core research.

On the whole, the study concludes that StockSecure POS and CyberLedger meaningfully strengthen Dulci Cafe's operational efficiency and internal accountability and that the system offers a practical, affordable, and transferable model for other small and medium-sized café businesses seeking to modernize their operations without the cost and complexity of enterprise-grade retail systems.

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